INCH-POUND

MIL-P-24764/2
16 December 1992

MILITARY SPECIFICATION

POWER SUPPLIES, SHIPBOARD, ELECTRONIC

5 VOLTS AND ± 15 VOLTS, KEY CODE BB3A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this associated detail specification and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-P-24764.

1. SCOPE

1.1 Scope. This document, along with MIL-P-24764, establishes the requirements and test procedures for an electronic switching mode power supply with 5 volt and ± 15 volt outputs.

1.2 Classification. This specification covers family S1 power supplies in accordance with MIL-P-24764.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Code 603, Bldg. 2917, NAVSURFWARCENDIV, 300 Highway 361, Crane, IN 47522-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards, and handbook. The following specification, standards, and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-P-24764 - Power Supplies, Shipboard, Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-241 - Design Guide for Electromagnetic Interference (EMI) Reduction in Power Supplies.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. Power supplies shall comply with the requirements of MIL-P-24764 except as specified herein.

3.2 Correlation requirements. All testers and test equipment shall be correlated in accordance with MIL-P-24764.

3.3 Reliability requirements. Power supplies shall have a minimum mean time between failures (MTBF) of 100,000 hours when predicted in accordance with MIL-P-24764. The prediction shall be made with a mounting surface temperature of 50 degrees Celsius (°C).

3.4 Performance requirements. Power supplies shall conform to the electrical requirements specified in table I and shall operate within the environment specified in MIL-P-24764.

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TABLE I. Power supply electrical requirements.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1	IV	Isolation-input to chassis/output, 800 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A2	IV	Isolation-output to chassis, 50 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A3	IV	Isolation-output to output, 20 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A4	IV	Isolation-excessive temperature, 100 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A5	IV	Isolation-unsafe temperature, 100 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A6	IV	Isolation-output status, 100 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A7	IV	Isolation-input connector used to unused contacts, 100 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A8	IV	Isolation-output connector used to unused contacts, 100 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A9	IV	Isolation-input to chassis/output, 800 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm
A10	IV	Isolation-output to chassis, 50 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm
A11	IV	Isolation-excessive temperature, 100 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm
A12	IV	Isolation-unsafe temperature, 100 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm
A13	IV	Isolation-output status, 100 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm
A14	IV	Isolation-input connector used to unused contacts, 100 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm
A15	IV	Isolation-output connector used to unused contacts, 100 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm

1/ See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
B1	V	Chassis contact to chassis continuity	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B2	V	15V return to remote ON/OFF return continuity	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B3	V	5V RTN sense to voltage program return continuity	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B4-B47	V	Parallel connector contact continuity	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B48-B52	V	Chassis contact to chassis continuity	Q	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	Ohm
C1	VI	Load regulation, 5.05V, 5A, (±15V, 0A)	X	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.15	VDC
C2	VI	Load regulation, 5.05V, 5A, (±15V, 1A)	X	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.15	VDC
C3	VI	Load regulation, 5.05V, 50A, (±15V, 0A)	X	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.15	VDC
C4	VI	Load regulation, 5.05V, 50A, (±15V, 1A)	X	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.15	VDC
C5	VI	Load regulation, 5.25V, 5A	X	5.20	5.30	5.17	5.33	5.14	5.36	5.12	5.36	VDC

- See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
C6	VI	Load regulation, 5.25V, 50A	X	5.20	5.30	5.17	5.33	5.14	5.36	5.12	5.38	VDC
C7	VI	Load regulation, +15V, 0A (5V, 5A, -15V, 11A)	X	14.85	15.15	14.77	15.23	14.70	15.30	14.62	15.38	VDC
C8	VI	Load regulation, +15V, 0A (5V, 50A, -15V, 11A)	X	14.85	15.15	14.77	15.23	14.70	15.30	14.62	15.38	VDC
C9	VI	Load regulation, +15V, 11A (5V, 5A, -15V, 0A)	X	14.85	15.15	14.77	15.23	14.70	15.30	14.62	15.38	VDC
C10	VI	Load regulation, +15V, 11A (5V, 50A, -15V, 0A)	X	14.85	15.15	14.77	15.23	14.70	15.30	14.62	15.38	VDC
C11	VI	Load regulation, -15V, 0A (5V, 5A, +15V, 11A)	X	-15.15	-14.85	-15.23	-14.77	-15.30	-14.70	-15.38	-14.62	VDC
C12	VI	Load regulation, -15V, 0A (5V, 50A, +15V, 11A)	X	-15.15	-14.85	-15.23	-14.77	-15.30	-14.70	-15.38	-14.62	VDC
C13	VI	Load regulation, -15V, 11A (5V, 5A, +15V, 0A)	X	-15.15	-14.85	-15.23	-14.77	-15.30	-14.70	-15.38	-14.62	VDC
C14	VI	Load regulation, -15V, 11A (5V, 50A, +15V, 0A)	X	-15.15	-14.85	-15.23	-14.77	-15.30	-14.70	-15.38	-14.62	VDC
C15	VI	Load regulation, 5.05V, 0A	X	4.95	5.15	4.89	5.21	4.85	5.15	4.89	5.21	VDC
C16	VI	Load regulation, 5.05V, 0A, output status	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	VDC
C17	VI	Load regulation, -5.25V, 5A, (±15V,0A)	X	-5.30	-5.20	-5.33	-5.17	-5.36	-5.14	-5.38	-5.12	VDC

- See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 80°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
D1	VII	Line regulation, 86 VRMS, 5A, 0A	S	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
D2	VII	Line regulation, 86 VRMS, 50A, 11A	S	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
D3	VII	Line regulation, 144 VRMS, 5A, 0A	S	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
D4	VII	Line regulation, 144 VRMS, 50A, 11A	S	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
D5	VII	Line regulation, 86 VRMS, 5A, 0A	S	14.85	15.15	14.77	15.23	14.70	15.30	14.62	15.38	VDC
D6	VII	Line regulation, 86 VRMS, 50A, 11A	S	14.85	15.15	14.77	15.23	14.70	15.30	14.62	15.38	VDC
D7	VII	Line regulation, 144 VRMS, 5A, 0A	S	14.85	15.15	14.77	15.23	14.70	15.30	14.62	15.38	VDC
D8	VII	Line regulation, 144 VRMS, 50A, 11A	S	14.85	15.15	14.77	15.23	14.70	15.30	14.62	15.38	VDC
D9	VII	Line regulation, 86 VRMS, 5A, 0A	S	-15.15	-14.85	-15.23	-14.77	-15.30	-14.70	-15.38	-14.62	VDC
D10	VII	Line regulation, 86 VRMS, 50A, 11A	S	-15.15	-14.85	-15.23	-14.77	-15.30	-14.70	-15.38	-14.62	VDC
D11	VII	Line regulation, 144 VRMS, 5A, 0A	S	-15.15	-14.85	-15.23	-14.77	-15.30	-14.70	-15.38	-14.62	VDC
D12	VII	Line regulation, 144 VRMS, 50A, 11A	S	-15.15	-14.85	-15.23	-14.77	-15.30	-14.70	-15.38	-14.62	VDC
D13	VII	Line regulation, 108 VDC, 5A, 0A	X	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
D14	VII	Line regulation, 108 VDC, 50A, 11A	X	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
D15	VII	Line regulation, 202 VDC, 5A, 0A	X	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
D16	VII	Line regulation, 202 VDC, 50A, 11A	X	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
D17	VII	Line regulation, auxiliary input 108 VDC, 50A, 11A	X	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC

- See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
D18	VII	Line regulation, auxiliary input 202 VDC, 50A, 11A	X	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
D19	VII	Line regulation, 50A, 11A 47 Hz, 115 VRMS	Q	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
D20	VII	Line regulation, 50A, 11A 63 Hz, 115 VRMS	Q	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
D21	VII	Line regulation, 108 VDC, 5A, 0A	X	14.85	15.15	14.77	15.23	14.70	15.30	14.62	15.38	VDC
D22	VII	Line regulation, 108 VDC, 50A, 11A	X	14.85	15.15	14.77	15.23	14.70	15.30	14.62	15.38	VDC
D23	VII	Line regulation, 202 VDC, 5A, 0A	X	14.85	15.15	14.77	15.23	14.70	15.30	14.62	15.38	VDC
D24	VII	Line regulation, 202 VDC, 50A, 11A	X	14.85	15.15	14.77	15.23	14.70	15.30	14.62	15.38	VDC
D25	VII	Line regulation, 108 VDC, 5A, 0A	X	-15.15	-14.85	-15.23	-14.77	-15.30	-14.70	-15.38	-14.62	VDC
D26	VII	Line regulation, 108 VDC, 50A, 11A	X	-15.15	-14.85	-15.23	-14.77	-15.30	-14.70	-15.38	-14.62	VDC
D27	VII	Line regulation, 202 VDC, 5A, 0A	X	-15.15	-14.85	-15.23	-14.77	-15.30	-14.70	-15.38	-14.62	VDC
D28	VII	Line regulation, 202 VDC, 50A, 11A	X	-15.15	-14.85	-15.23	-14.77	-15.30	-14.70	-15.38	-14.62	VDC
E1	VIII	Low input voltage protection	Q	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
F1-F3	VIII	Loss of phase protection	Q	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
G1	IX	Efficiency, 5.05V, 50A/±15V, 11A	X	65	100	65	100	65	100	65	100	%
G2	IX	Efficiency, 5.05V, 5A/±15V, 1A	Q	58	100	58	100	58	100	58	100	%
G3	IX	Efficiency, 5.25V, 50A/±15V, 11A	Q	65	100	65	100	65	100	65	100	%

See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
H	IX	Power factor, 50A, 11A	Q	0.8	1.0	0.8	1.0	0.8	1.0	0.8	1.0	-
J1-J3	IX	Input current balance, 50A, 11A	Q	-5	5	-5	5	-5	5	-5	5	%
K1, K2	X	Inrush current, 50A, 11A	Q	--	55	--	55	--	55	--	55	Apk
L1, L2	XI	Remote sense compensation, 5.05V	Q	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
		Remote sense compensation, 5.25V	Q	5.20	5.30	5.17	5.33	5.14	5.36	5.12	5.38	VDC
M1	XII	Open + sense lead transient amplitude	Q	-400	100	-400	100	-400	100	-400	100	mV
M2	XII	Open + sense lead transient time	Q	0	100	0	100	0	100	0	100	ms
M3	XII	Open + sense lead voltage	S	4.80	5.56	4.80	5.56	4.80	5.56	4.80	5.56	VDC
M4	XII	Open - sense lead transient amplitude	Q	-400	100	-400	100	-400	100	-400	100	mV
M5	XII	Open - sense lead transient time	Q	0	100	0	100	0	100	0	100	ms
M6	XII	Open - sense lead voltage	S	4.80	5.56	4.80	5.56	4.80	5.56	4.80	5.56	VDC
M7	XII	Open + and - sense lead voltage	S	4.80	5.56	4.80	5.56	4.80	5.56	4.80	5.56	VDC
M8	XII	Sense lead reversal protection	Q	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
N1	XIII	Overload protection, 5V	X	60.0	62.5	60.0	63.5	60.0	62.5	60.0	63.5	ADC
N2	XIII	Overload protection, +15V	X	11.0	13.8	11.0	14.0	11.0	13.8	11.0	14.0	ADC

- See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N3	XIII	Overload protection, -15V	X	11.0	13.8	11.0	14.0	11.0	13.8	11.0	14.0	ADC
N4	XIII	Short circuit recovery, 5V	Q	0	100	0	100	0	100	0	100	ms
N5	XIII	Short circuit overshoot, 5V	Q	-250	250	-250	250	-250	250	-250	250	mV
N6	XIII	Short circuit recovery, +15V	Q	0	100	0	100	0	100	0	100	ms
N7	XIII	Short circuit overshoot, +15V	Q	-750	750	-750	750	-750	750	-750	750	mV
N8	XIII	Short circuit recovery, -15V	Q	0	100	0	100	0	100	0	100	ms
N9	XIII	Short circuit overshoot, -15V	Q	-750	750	-750	750	-750	750	-750	750	mV
N10	XIII	Short circuit current, 5V	S	0	62.5	0	63.5	0	62.5	0	63.5	ADC
N11	XIII	Short circuit current, +15V	S	0	13.8	0	14.0	0	13.8	0	14.0	ADC
N12	XIII	Short circuit current, -15V	S	0	13.8	0	14.0	0	13.8	0	14.0	ADC
P1	XIV	Turn-on transient, 5V, 50A	Q	-250	250	-250	250	-250	250	-250	250	mV
P2	XIV	Turn-on time, 5V, 50A	S	15	250	15	250	15	250	15	250	ms
P3	XIV	Turn-off transient, 5V, 50A	Q	-250	250	-250	250	-250	250	-250	250	mV
P4	XIV	Turn-on transient, +15V, 11A	Q	-750	750	-750	750	-750	750	-750	750	mV
P5	XIV	Turn-on time, +15V, 11A	S	15	250	15	250	15	250	15	250	ms
P6	XIV	Turn-off transient, +15V, 11A	Q	-750	750	-750	750	-750	750	-750	750	mV
P7	XIV	Turn-on transient, -15V, 11A	Q	-750	750	-750	750	-750	750	-750	750	mV

See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P8	XIV	Turn-on time, -15V, 11A	S	15	250	15	250	15	250	15	250	ms
P9	XIV	Turn-off transient, -15V, 11A	Q	-750	750	-750	750	-750	750	-750	750	mV
P10	XIV	Control turn-on transient, 5V, 50A	Q	-253	253	-253	253	-253	253	-253	253	mV
P11	XIV	Control turn-on time, 5V, 50A	S	15	250	15	250	15	250	15	250	ms
P12	XIV	Control turn-off transient, 5V, 50A	Q	-253	253	-253	253	-253	253	-253	253	mV
Q1	XV	Ripple, total, mVp-p, 5V (115 VAC, 50A, 10 Hz - 1 MHz)	X	0	65	0	75	0	65	0	75	mVp-p
Q2	XV	Ripple, total, mVp-p, +15V (115 VAC, 11A, 10 Hz - 1 MHz)	X	0	25	0	29	0	25	0	29	mVp-p
Q3	XV	Ripple, total, mVp-p, -15V (115 VAC, 11A, 10Hz - 1 MHz)	X	0	25	0	29	0	25	0	29	mVp-p
Q4	XV	Ripple, low frequency, 5V (115 VAC, 50A, 10 Hz - 10 kHz)	S	0	50	0	60	0	50	0	60	mVp-p
Q5	XV	Ripple, high frequency, 5V (115 VAC, 50A, 10 kHz - 1 MHz)	S	0	65	0	75	0	65	0	75	mVp-p
Q6	XV	Spectral line, +15V (115 VAC, 11A, 50 Hz - 10 kHz) 6/	S	0	0.056	0	0.065	0	0.056	0	0.065	mVRMS $\sqrt{\text{Hz}}$
Q7	XV	Spectral line, +15V (115 VAC, 11A, 10 kHz - 10 MHz) 6/	S	0	0.10 7/	0	0.12 7/	0	0.10 7/	0	0.12 7/	mVRMS $\sqrt{\text{Hz}}$
Q8	XV	Spectral line, -15V (115 VAC, 11A, 50 Hz - 10 kHz) 6/	S	0	0.056	0	0.065	0	0.056	0	0.065	mVRMS $\sqrt{\text{Hz}}$

See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q9	XV	Spectral line, -15V (115 VAC, 11A, 10 kHz - 10 MHz) 6/	S	0	0.10 7/	0	0.12 7/	0	0.10 7/	0	0.12 7/	mVRMS / √Hz
Q10	XV	Ripple, total, 5V (115 VAC, 50A, 10 Hz - 20 MHz)	Q	0	300	0	300	0	300	0	300	mVp-p
Q11	XV	Ripple, total, +15V (115 VAC, 11A, 10 Hz-20 MHz)	Q	0	100	0	100	0	100	0	100	mVp-p
Q12	XV	Ripple, total, -15V (115 VAC, 11A, 10 Hz-20 MHz)	Q	0	100	0	100	0	100	0	100	mVp-p
Q13	XV	Ripple, total, 5V, >50A, (10 Hz - 5 MHz)	Q	0	200	0	200	0	200	0	200	mVp-p
Q14	XV	Ripple, total, 5V, no load,(10 HZ - 5MHz)	Q	0	130	0	150	0	130	0	150	mVp-p
Q15	XV	Ripple, total, +15V, >11A, (10 Hz - 5MHz)	Q	0	65	0	65	0	65	0	65	mVp-p
Q16	XV	Ripple, total, -15V, >11A, (10 HZ - 5MHz)	Q	0	65	0	65	0	65	0	65	mVp-p
Q17	XV	Ripple, total, 5V, 50A, 86 Vrms (10 HZ - 5 MHz)	Q	0	130	0	150	0	130	0	150	mVp-p
Q18	XV	Ripple, total, +15V, 11A, 86 Vrms (10 HZ - 5 MHz)	Q	0	25	0	29	0	25	0	29	mVp-p
Q19	XV	Ripple, total, -15V, 11A, 86 Vrms (10 HZ - 5 MHz)	Q	0	25	0	29	0	25	0	29	mVp-p
Q20	XV	Ripple, total, 5V, 50A, 144 Vrms (10 HZ - 5 MHz)	Q	0	130	0	150	0	130	0	150	mVp-p

- See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q21	XV	Ripple, total, +15V, 11A, 144 VRMS (10 HZ - 5 MHz)	Q	0	25	0	29	0	25	0	29	mVp-p
Q22	XV	Ripple, total, -15V, 11A, 144 VRMS (10 HZ - 5 MHz)	Q	0	25	0	29	0	25	0	29	mVp-p
Q23	XV	Ripple, total, -5.05V.50A (115 VAC; ±15V, 11A; 10Hz - 5 MHz)	Q	0	130	0	150	0	130	0	150	mVp-p
Q24	XV	Spectral line, 5V (115 VAC, 50A, 50Hz - 10 MHz) 8/	S	0	10	0	10	0	10	0	10	mVRMS $\sqrt{Hz}$
R1	XVI	Dynamic line regulation, 5V, +21%	Q	0	152	0	152	0	152	0	152	mV
R2	XVI	Dynamic line ringing, 5V, +21% (2nd half cycle)	Q	0	76	0	76	0	76	0	76	mV
R3	XVI	Dynamic line regulation, 5V, -21%	Q	0	152	0	152	0	152	0	152	mV
R4	XVI	Dynamic line ringing, 5V, -21% (2nd half cycle)	Q	0	76	0	76	0	76	0	76	mV
R5	XVI	Dynamic line regulation, +15V, +21%	Q	0	450	0	450	0	450	0	450	mV
R6	XVI	Dynamic line ringing, +15V, +21% (2nd half cycle)	Q	0	225	0	225	0	225	0	225	mV
R7	XVI	Dynamic line regulation, +15V, -21%	Q	0	450	0	450	0	450	0	450	mV
R8	XVI	Dynamic line ringing, +15V, -21% (2nd half cycle)	Q	0	225	0	225	0	225	0	225	mV

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R9	XVI	Dynamic line regulation, -15V, +21%	Q	0	450	0	450	0	450	0	450	mV
R10	XVI	Dynamic line ringing, -15V, +21%, (2nd half cycle)	Q	0	225	0	225	0	225	0	225	mV
R11	XVI	Dynamic line regulation, -15V, -21%	Q	0	450	0	450	0	450	0	450	mV
R12	XVI	Dynamic line ringing, -15V, -21%, (2nd half cycle)	Q	0	225	0	225	0	225	0	225	mV
S1	XVII	Dynamic load regulation, 80% to 90%, 5.05V	X	0	152	0	152	0	152	0	152	mV
S2	XVII	Dynamic load ringing (2nd half cycle)	S	0	76	0	76	0	76	0	76	mV
S3	XVII	Dynamic load regulation, 90% to 80%, 5.05V	X	0	152	0	152	0	152	0	152	mV
S4	XVII	Dynamic load ringing (2nd half cycle)	S	0	76	0	76	0	76	0	76	mV
S5	XVII	Dynamic load regulation, 80% to 90%, full capacitor load, 5.05V	Q	0	152	0	152	0	152	0	152	mV
S6	XVII	Dynamic load ringing (2nd half cycle)	Q	0	76	0	76	0	76	0	76	mV
S7	XVII	Dynamic load regulation, 90% to 80%, full capacitor load, 5.05V	Q	0	152	0	152	0	152	0	152	mV
S8	XVII	Dynamic load ringing (2nd half cycle)	Q	0	76	0	76	0	76	0	76	mV
S9	XVII	Dynamic load regulation, 80% to 90%, 5.25V (max sense lead)	X	0	162	0	162	0	162	0	162	mV

- See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
S10	XVII	Dynamic load ringing (2nd half cycle)	S	0	81	0	81	0	81	0	81	mV
S11	XVII	Dynamic load regulation, 90% to 80%, 5.25V (max sense lead)	X	0	162	0	162	0	162	0	162	mV
S12	XVII	Dynamic load ringing (2nd half cycle)	S	0	81	0	81	0	81	0	81	mV
S13	XVII	Dynamic load regulation, 80% to 90%, +15V	X	0	450	0	450	0	450	0	450	mV
S14	XVII	Dynamic load ringing (2nd half cycle)	S	0	225	0	225	0	225	0	225	mV
S15	XVII	Dynamic load regulation, 90% to 80%, +15V	X	0	450	0	450	0	450	0	450	mV
S16	XVII	Dynamic load ringing (2nd half cycle)	S	0	225	0	225	0	225	0	225	mV
S17	XVII	Dynamic load regulation, 80% to 90%, -15V	X	0	450	0	450	0	450	0	450	mV
S18	XVII	Dynamic load ringing (2nd half cycle)	S	0	225	0	225	0	225	0	225	mV
S19	XVII	Dynamic load regulation, 90% to 80%, -15V	X	0	450	0	450	0	450	0	450	mV
S20	XVII	Dynamic load ringing (2nd half cycle)	S	0	225	0	225	0	225	0	225	mV
S21	XVII	Dynamic load regulation, 60% to 80%,5.05V	S	0	500	0	500	0	500	0	500	mV
S22	XVII	Dynamic load ringing, 60% to 90%, 5.05V	S	0	250	0	250	0	250	0	250	mV
S23	XVII	Dynamic load regulation, 90% to 60%,5.05V	S	0	500	0	500	0	500	0	500	mV
S24	XVII	Dynamic load ringing, 90% to 60%, 5.05V	S	0	250	0	250	0	250	0	250	mV
S25	XVII	Dynamic load regulation, 40% to 90%, +15V	S	0	1200	0	1200	0	1200	0	1200	mV
S26	XVII	Dynamic load ringing, 40% to 90%, +15V	S	0	600	0	600	0	600	0	600	mV

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
S27	XVII	Dynamic load regulation, 90% to 40%, +15V	S	0	1200	0	1200	0	1200	0	1200	mV
S28	XVII	Dynamic load ringing, 90% to 40%, +15V	S	0	600	0	600	0	600	0	600	mV
S29	XVII	Dynamic load regulation, 40% to 90%, -15V	S	0	1200	0	1200	0	1200	0	1200	mV
S30	XVII	Dynamic load ringing, 40% to 90%, -15V	S	0	600	0	600	0	600	0	600	mV
S31	XVII	Dynamic load regulation, 90% to 40%, -15V	S	0	1200	0	1200	0	1200	0	1200	mV
S32	XVII	Dynamic load ringing, 90% to 40%, -15V	S	0	600	0	600	0	600	0	600	mV
T1	XVII	Turn-on time into capacitive load	Q	15	250	15	250	15	250	15	250	ms
T2	XVII	Turn-off into capacitive load	Q	14.85	15.15	14.77	15.23	14.70	15.30	14.62	15.38	VDC
T3	XVII	Turn-off into capacitive load	Q	-15.15	-14.85	-15.23	-14.77	-15.30	-14.70	-15.38	14.62	VDC
U1	XVIII	5V Undervoltage - output status LED off	X	4.39	4.80	4.39	4.80	4.39	4.80	4.39	4.80	VDC
U2	XVIII	5V Overvoltage - output status LED off	X (4/)	5.57	6.45	5.57	6.55	5.57	6.45	5.57	6.55	VDC
U3	XVIII	+15V Undervoltage - output status LED off	X	13.05	14.25	13.05	14.25	13.05	14.25	13.05	14 25	VDC
U4	XVIII	+15V Overvoltage - output status LED off	X (4/)	15.90	17.25	15.90	17.40	15.90	17.25	15.90	17 40	VDC
U5	XVIII	-15V Undervoltage - output status LED off	X	-14.25	-13.05	-14.25	-13.05	-14.25	-13.05	-14.25	-13.05	VDC
U6	XVIII	-15V Overvoltage - output status LED off	X (4/)	-17.25	-15.90	-17.40	-15.90	-17.25	-15.90	-17.40	-15.90	VDC

- See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
U7	XVIII	Status LED on	X	----------			Status LED on	----------				VDC
V1	XVIII	Output status signal, logic low	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	VDC
V2	XVIII	Output status signal, logic high	X	29.0	30.5	29.0	30.5	29.0	30.5	28.0	30.5	VDC
V3	XVIII	5V Undervoltage- output status signal	X	4.39	4.80	4.39	4.80	4.39	4.80	4.39	4.80	VDC
V4	XVIII	5V Overvoltage- output status signal	X (4/)	5.57	6.45	5.57	6.55	5.57	6.45	5.57	6.55	VDC
V5	XVIII	+15V Undervoltage, "output status signal"	X	13.05	14.25	13.05	14.25	13.05	14.25	13.05	14.25	VDC
V6	XVIII	+15V Overvoltage, "output status signal"	X (4/)	15.90	17.25	15.90	17.40	15.90	17.25	15.90	17.40	VDC
V7	XVIII	-15V Undervoltage, "output status signal"	X	-14.25	-13.05	-14.25	-13.05	-14.25	-13.05	-14.25	-13.05	VDC
V8	XVIII	-15V Overvoltage, "output status signal"	X (4/)	-17.25	-15.90	-17.40	-15.90	17.25	-15.90	-17.40	-15.90	VDC
V9	XVIII	Output status signal leakage current	Q	0	4.4	0	4.4	0	50	0	200	uADC
V10	XVIII	Output status signal settling time (high to low)	Q	0	2.0	0	2.0	0	2.0	0	2.0	ms
V11	XVIII	Output status signal settling time (low to high)	Q	0	2.0	0	2.0	0	2.0	0	2.0	ms
V12	XVIII	Output status signal, logic low output	Q	0	0.4	0	0.4	0	0.4	0	0.4	VDC

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
W1	XIX	Excessive temp indication (5 mA current)	Q	-	-	-	-	0	0.4	-	-	VDC
W2	XIX	Excessive temp indication (500 mA current)	Q	-	-	-	-	0	0.4	-	-	VDC
W3	XIX	Unsafe temp indication (5 mA current)	Q	-	-	-	-	0	0.4	-	-	VDC
W4	XIX	Unsafe temp indication (500 mA current)	Q	-	-	-	-	0	0.4	-	-	VDC
W5	XIX	Unsafe - excessive differential (5mA)	Q	-	-	-	-	1.0	25	-	-	C
W6	XIX	Indicator reset (5mA)	Q	-	-	-	-	60		-		C
W7	XIX	Unsafe - excessive differential (500mA)	Q	-	-	-	-	1.0	25	-	-	C
W8	XIX	Indicator reset (500mA)	Q	-	-	-	-	60	-	-	-	C
X	XX	Output voltage hold-up time (Input volt: 108Vdc)	Q	50	-	50	-	50	-	50	-	us
Y1	XXI	AC input to chassis capacitance	S	0	0.03	0	0.03	0	0.03	0	0.03	uF
Y2	XXI	Auxiliary input to chassis capacitance	S	0	0.05	0	0.05	0	0.05	0	0.05	uF
Z1-Z3	XXII	Output impedance, 10Hz to 25 kHz 75% rated load	Q	0	30	0	30	0	30	0	30	mOhm
AA1	XXIII	Open 5.05V load lead protection	Q	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
AA2	XXIII	Open 5.05V rtn load lead protection	Q	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC

See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
BB1	XXIV	Spike voltage protection, AC input, initial transient	Q	-253	253	-253	253	-253	253	-253	253	mV
BB2	XXIV	Spike voltage protection, AC input, secondary transient	Q	-126	126	-126	126	-126	126	-126	126	mV
BB3	XXIV	Spike voltage protection, auxiliary input, initial transient	Q	-253	253	-253	253	-253	253	-253	253	mV
BB4	XXIV	Spike voltage protection, auxiliary input, secondary transient	Q	-126	126	-126	126	-126	126	-126	126	mV
CC	XXV	Reverse voltage protection on auxiliary input	Q	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
DD1	XXV	Reverse voltage protection on ouput, 5V	Q	-1.0	0	-1.0	0	-1.0	0	-1.0	0	VDC
DD2	XXV	Reverse voltage bias, turn-on, 5V	Q	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
DD3, DD4	XXV	Reverse voltage protection on output, ± 15V	Q	-1.0	0	-1.0	0	-1.0	0	-1.0	0	VDC
DD5	XXV	ON/OFF parallel outputs, 5V off	Q	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
DD6	XXV	ON/OFF parallel outputs, +15V off	Q	14.85	15.15	14.77	15.23	14.70	15.30	14.62	15.38	VDC
DD7	XXV	ON/OFF parallel outputs, -15V off	Q	-15.15	-14.85	-15.23	-14.77	-15.30	-14.70	-15.38	-14.62	VDC
DD8	XXV	ON/OFF parallel outputs, 05 high	Q	2.4	5.5	2.4	5.5	2.4	5.5	2.4	5.5	VDC
DD9	XXV	ON/OFF parallel outputs, 05 LED off	Q		Output	status	LED	is	off.			---

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
DD10	XXV	Positive voltage bias, turn-on, 5V	Q	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
EE1, EE2	XXVI	Common mode current, broadband, 5V	Q	0	200	0	200	0	200	0	200	mAp-p
EE3, EE4	XXVI	Common mode current, broadband, ±15V	Q	0	200	0	200	0	200	0	200	mAp-p
EE5	XXVI	Common mode current, 5V, (40Hz - 40 kHz)	Q	-INF	-60	-INF	-60	-INF	-60	-INF	-60	dBA / √Hz
EE6	XXVI	Common mode current, 5V, (40kHz-350 kHz)	Q	-INF	-36.5	-INF	-36.5	-INF	-36.5	-INF	-36.5	dBA / √Hz
EE7	XXVI	Common mode current, 5V, (350kHz - 5MHz)	Q	See paragraph 3.18 and figure 11								dBA / √Hz
EE8	XXVI	Common mode current, 5V, (5MHz - 10 MHz)	Q	-INF	-60	-INF	-60	-INF	-60	-INF	-60	dBA / √Hz
EE9	XXVI	Common mode current, ±15V, (40 Hz-40kHz)	Q	-INF	-60	-INF	-60	-INF	-60	-INF	-60	dBA / √Hz
EE10	XXVI	Common mode current, ±15V, (40kHz-350kHz)	Q	-INF	-36.5	-INF	-36.5	-INF	-36.5	-INF	-36.5	dBA / √Hz

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
EE11	XXVI	Common mode current, ±15V, (350kHz-5MHz)	Q	See paragraph 3.18 and figure 11								dBA / √Hz
EE12	XXVI	Common mode current, ±15V, (5MHz - 10 MHz)	Q	-INF	-60	-INF	-60	-INF	-60	-INF	-60	dBA / √Hz
FF1-FF3	XXVII	Voltage programming contact signal rejection	Q	15	-	15	-	15	-	15	-	dBV
GG1	XXVIII	Input voltage turn-on threshold	S	100	135	100	135	100	135	100	135	VDC
GG2	XXVIII	Input voltage turn-off threshold	S	-	100	-	100	-	100	-	100	VDC
GG3	XXVIII	Input voltage turn-on/turn-off hysteresis	S	18	-	18	-	18	-	18	-	VDC
GG4	XXVIII	Input voltage turn-on threshold, auxiliary input	S	100	135	100	135	100	135	100	135	VDC
GG5	XXVIII	Input voltage turn-off threshold, auxiliary input	S	-	100	-	100	-	100	-	100	VDC
GG6	XXVIII	Input voltage turn-on/turn-off hysteresis, auxiliary input	S	18	-	18	-	18	-	18	-	VDC
GG7	XXVIII	Maximum steady state input current	Q	-	11.9	-	11.9	-	12.1	-	12.1	ADC
GG8	XXVIII	Slowly rising input peak current	Q	-	11.9	-	11.9	-	12.1	-	12.1	Apk
GG9	XXVIII	Slowly rising input rate of current rise	Q	-	9.4	-	9.4	-	9.4	-	9.4	A/ms
GG10	XXVIII	Input peak current, remote turn-on	Q	-	11.9	-	11.9	-	12.1	-	12.1	Apk

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
GG11	XXVIII	Input rate of current rise, remote turn-on	Q	-	9.4	-	9.4	-	9.4	-	9.4	A/ms
HH1	XXIX	Emergency condition input voltage protection, 5.25 V	Q	4.99	5.51	4.99	5.51	4.99	5.51	4.99	5.51	VDC
HH2	XXIX	Emergency condition input voltage protection, 5.25V	Q	5.20	5.30	5.17	5.33	5.14	5.36	5.12	5.38	VDC
HH3	XXIX	Emergency condition input voltage protection, +15V	Q	14.25	15.75	14.25	15.75	14.25	15.75	14.25	15.75	VDC
HH4	XXIX	Emergency condition input voltage protection, +15V	Q	14.85	15.15	14.77	15.23	14.70	15.30	14.62	15.38	VDC
HH5	XXIX	Emergency condition input voltage protection, -15V	Q	-15.75	-14.25	-15.75	-14.25	-15.75	-14.25	-15.75	-14.25	VDC
HH6	XXIX	Emergency condition input voltage protection, -15V	Q	-15.15	-14.85	-15.23	-14.77	-15.30	-14.70	-15.38	-14.62	VDC
JJ1	XXX	Output status noise, positive	Q	0	500	0	500	0	500	0	500	mVpk
JJ2	XXX	Output status noise, negative	Q	-500	0	-500	0	-500	0	-500	0	mVpk

· See footnotes on next page.

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NOTES:

1/ "X" parameters shall be 100 percent tested during power supply quality conformance testing.

2/ "S" parameters shall be sample tested (one percent or two power supplies from each inspection lot, whichever is greater) during quality conformance testing.

3/ "Q" parameters shall be tested during qualification testing. It is not required that these parameters be tested during quality conformance testing, however, the vendor is responsible for ensuring that the power supplies meet these parameters.

4/ Proper operation of the overvoltage protection circuitry shall be verified as an "X" test by the power supply vendor during manufacture of the supplies.

5/ The symbols used in this table and not defined elsewhere are as follows.

A - amperes
ADC - amperes direct current
Apk - amperes peak
cond - condition
INF - infinity
mAp-p - milliamperes peak-to-peak
max - maximum
min - minimum
mOhm - milliohm
MOhm - megohm
mV - millivolts
mVpk - millivolts peak
mVp-p - millivolts peak-to-peak
% - percent
uADC - microamperes direct current
V - volts
VDC - volts direct current
VRMS - volts root-mean-square
mVRMS - millivolts root-mean-square
dBA - decibel amperes

6/ Test conditions Q6-Q9 and Q24 limits apply to single spectral lines with test bandwidths as specified. Do not apply bandwidth correction to the limits.

7/ Test conditions Q7 and Q9 limits will be as specified except between the frequencies of 1 MHZ to 3 MHZ where the maximum limit shall be 0.18 mVRMS/\/HZ.

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3.5 Power supply configuration. The configuration of the power supplies shall be in accordance with MIL-P-24764 and as specified herein.

3.5.1 Size. Power supplies shall be style 3.

3.5.2 Weight. Power supplies shall have a maximum weight of 14 pounds.

3.5.3 Visual indicator. The visual indicator shall be located as specified on figure 1. A green light emitting diode (LED) shall indicate status of output voltage.

3.5.4 Marking. The power supply marking shall contain the information and shall be located in the relative location and orientation shown on figure 1. The marking "OUTPUT STATUS" shall be immediately adjacent to the output status LED. When authorization has been granted by the Shipboard Power Supply Products Quality Assurance Activity (SPSP-QAA), the qualification certification mark "QUAL" shall be marked as shown. The size of the letters, information per line, and spacing shall, within the limits of MIL-P-24764, be allowed to vary.

3.5.5 Keying. Keying shall be BB3A.

3.5.6 Mounting fastener torque. The power supply shall operate as specified when it is mounted to a cooling plate with the mounting fasteners torqued to 32 ± 3 inch-pounds.

3.6 Function diagram and contact assignments. Power supplies shall conform to the function diagram shown on figure 2 and contact assignments shall be as shown in table II. When multiple connector contacts are specified for an interface function (such as output) for current capacity, redundancy, and so forth, all multiple contacts for each function shall be connected together before being used to interface the power supply to external sources and loads. A summary of power supply functions is shown in table III.

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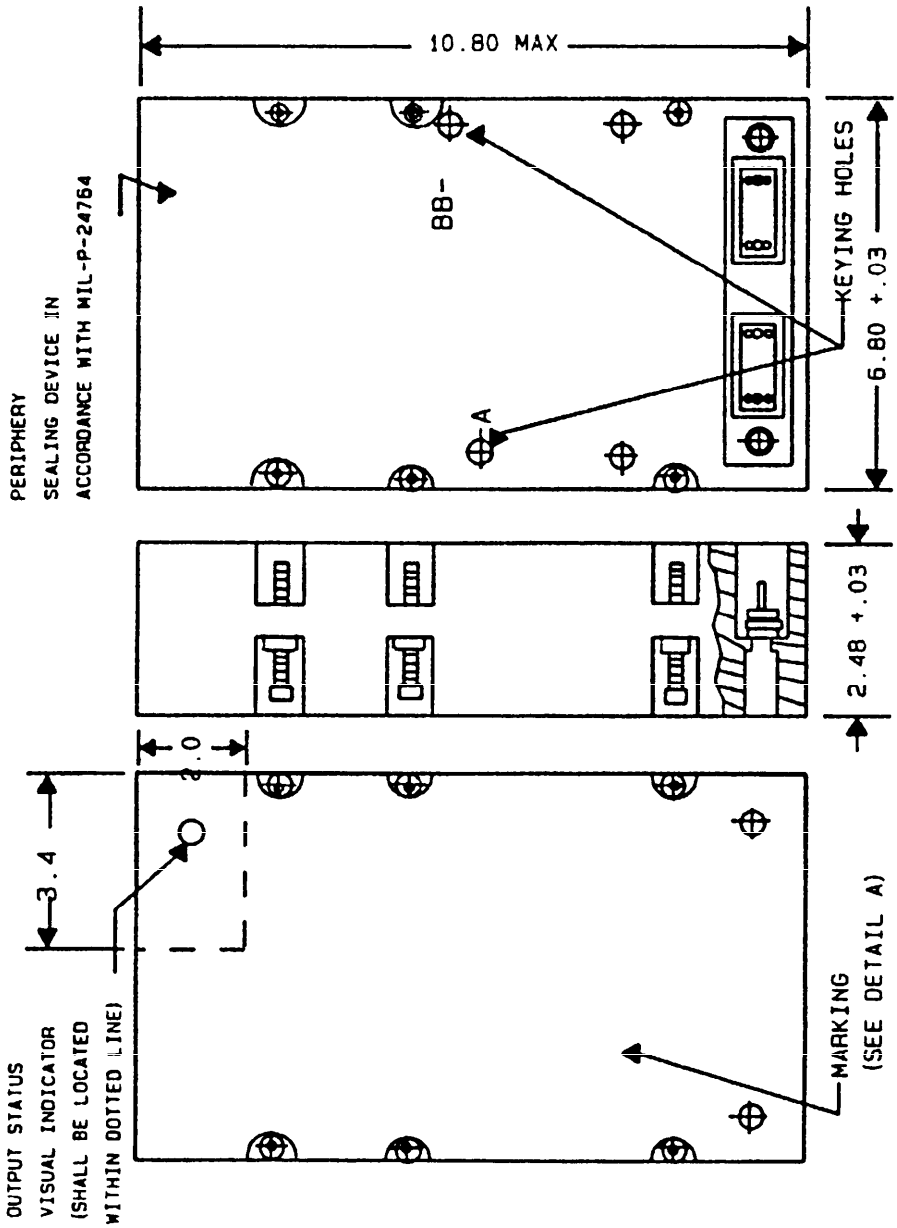
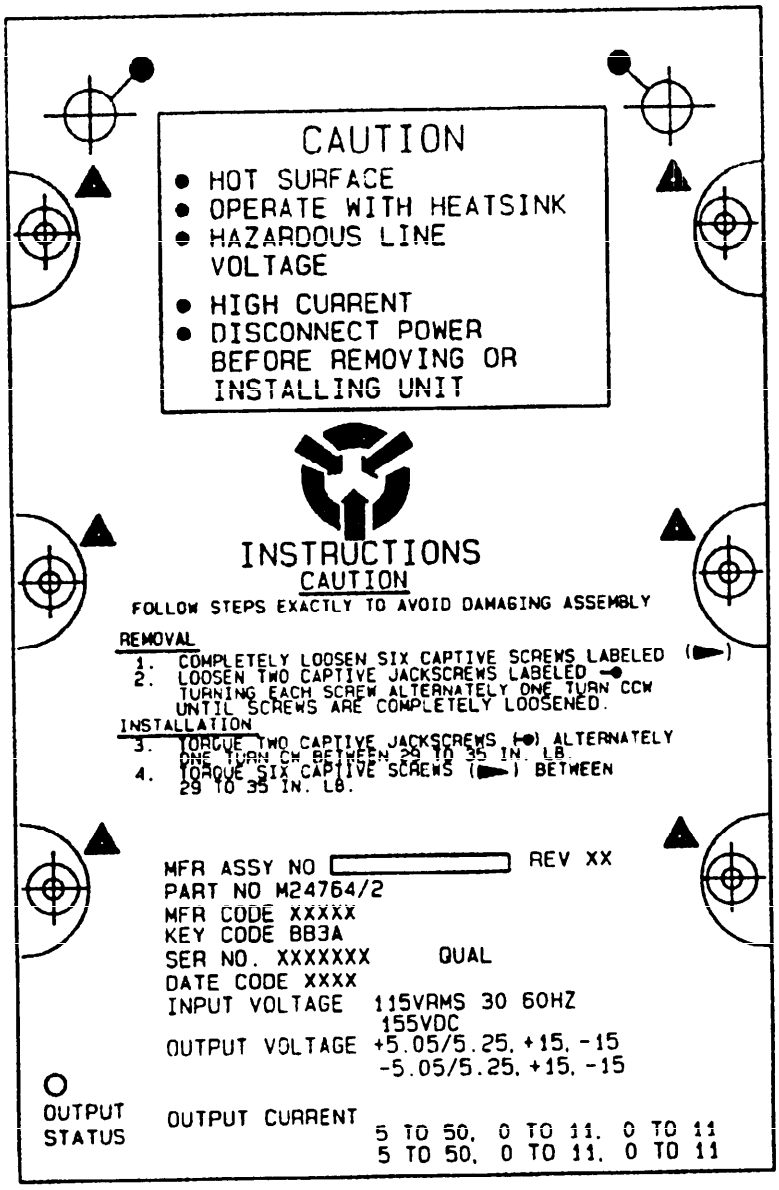


FIGURE 1. Power supply marking and visual indicator location.

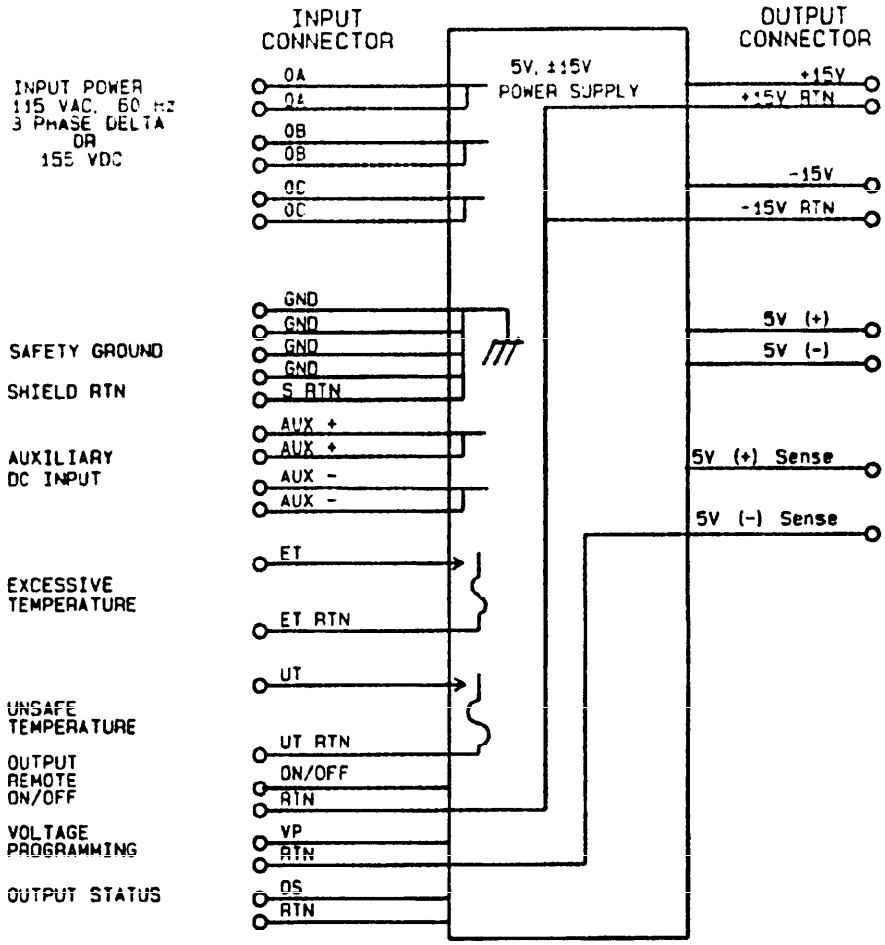
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DETAIL A

FIGURE 1. Power supply marking and visual indicator location - Continued.

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NOTE:

1. See table II for contact assignments.

FIGURE 2. Power supply function diagram.

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TABLE II. Connector contact assignments.

INPUT CONNECTOR		OUTPUT CONNECTOR	
A	Phase A	A	5 V Sense
B	Unused	B	5 V RTN Sense
C	Phase C	C	5 V (+)
D	Phase B	D	5 V (-)
E	Phase A	E	5 V (+)
F	Unused	F	5 V (-)
H	Phase C	H	5 V (+)
J	Phase B	J	5 V (-)
K	Auxiliary DC (+)	K	5 V (+)
L	Unused	L	5 V (-)
M	Auxiliary DC (-)	M	5 V (+)
N	Shield RTN	N	5 V (-)
P	Auxiliary DC (+)	P	5 V (+)
R	Safety Ground	R	5 V (-)
S	Auxiliary DC (-)	S	5 V (+)
T	Safety Ground	T	5 V (-)
U	Safety Ground	U	5 V (+)
V	Unused	V	5 V (-)
W	Safety Ground	W	5 V (+)
X	Unused	X	5 V (-)
Y	Excessive Temperature RTN	Y	5 V (+)
Z	Remote ON/OFF	Z	5 V (-)
a	Excessive Temperature	a	5 V (+)
b	Remote ON/OFF RTN	b	5 V (-)
c	Unsafe Temperature RTN	c	5 V (+)
d	Voltage Program-1	d	5 V (-)
f	Unsafe Temperature	e	Unused
g	VP-1 RTN	f	Unused
h	Output Status RTN	h	Unused
i	Unused	j	Unused
j	Output Status	k	+15 V Output
k	Unused	m	Unused
m	Unused	n	+15 V Output
n	Unused	p	Unused
		r	+15 V Output
		s	+15 V Common RTN
		t	+15 V Output
		u	+15 V Common RTN
		v	Unused
		w	+15 V Common RTN
		x	Unused
		y	+15 V Common RTN
		z	-15 V Output

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TABLE II. Connector contact assignments - Continued.

INPUT CONNECTOR		OUTPUT CONNECTOR
	AA	-15 V Common RTN
	BB	-15 V Output
	CC	-15 V Common RTN
	DD	-15 V Output
	EE	-15 V Common RTN
	FF	-15 V Output
	HH	-15 V Common RTN

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TABLE III. Power supply functions.

Function	Logic "0" or closed circuit	Logic "1" or open circuit	Reference
Remote ON/OFF (Input contacts b and Z)	All outputs ON	All outputs OFF	MIL-P-24764

Function	Open circuit	Closed circuit	Reference
Voltage programming (Input contacts d and g)	5.05V	5.25V	MIL-P-24764

Function	Logic "0"	Logic "1"	Reference
Output status signal (Input contacts j and h)	Voltage within rating and remote ON/OFF = ON	Voltage out of tolerance or remote ON/OFF = OFF	3.11

Function	On	Off	Reference
Output status LED	Voltage within rating and remote ON/OFF = ON	Voltage out of tolerance or remote ON/OFF = OFF	3.11

Function	Open 1/	Closure 1/	Reference
Excessive temperature (Input contacts a and Y)	Any Tj prior to an excessive closure.	When Tj is from 1 to 25 C prior to the unsafe closure	MIL-P-24764
Unsafe temperature (Input contacts c and f)	Any Tj prior to an unsafe closure.	When Tj exceeds 80 C Tj by a minimum of 5 C	MIL-P-24764

1/ The symbols in this table and not defined elsewhere are as follows: Tj - semiconductor junction temperature.

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3.7 Quality conformance tests. All production power supplies shall be tested in accordance with footnotes 1 and 2 in table I at a mounting surface temperature of 25°C ± 5°C.

3.8 Qualification. Initial qualification and periodic check shall be in accordance with MIL-P-24764 (see 4.5).

3.9 End-of-life limits. Power supplies which have been plugged into an operating system, or exceed 2 years from the date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.3).

3.10 Output voltage polarity. The 5 volt output of the power supply shall be capable of being utilized as a positive output or as a negative output with respect to the chassis or the return of the 15 volt outputs.

3.11 Output status signal. The output status signal shall operate as defined in MIL-P-24764 for output voltage status. Any output out of tolerance shall cause an appropriate indication on the status signal output. The output status signal shall be at a logic "1" whenever the remote ON/OFF equals OFF.

3.12 Remote sense leads. Power supplies shall have remote sense capability for the 5 volt output with sense and load lengths up to 10 inches in length. The voltage between the 5V(+) and 5V(-) output terminals shall be capable of reaching a value of at least 5.408 volts, regardless of the voltage programming level, in order to compensate for load lead voltage drop. The power supply shall not be operated without the sense leads connected to the correct output or load point.

3.13 Parallel power supplies. All outputs of a BB3A power supply shall be capable of being connected in parallel with the respective outputs of one additional BB3A power supply. In the parallel configuration, the outputs of only one power supply shall be energized at a time. The remote ON/OFF control of the power supply with unenergized outputs shall be programmed for the OFF condition.

3.13.1 Output capacitive loading. When two BB3A power supplies are paralleled, the energized output shall be capable of providing specified output voltage and current into the maximum allowable capacitance of the external load and the output capacitance of the paralleled, unenergized power supply. The capacitance of the external load shall not exceed 30 microfarads (uF) per rated output ampere. The output capacitance of the

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paralleled, unenergized power supply shall not exceed 50 uF per rated output ampere for the 5 volt output and 30 uF per rated output ampere for each of the 15 volt outputs. Therefore, the energized power supply shall drive a maximum capacitance of 80 uF per rated output ampere connected to the 5 volt output and 60 uF per rated output ampere connected to each of the 15 volt outputs.

3.14 Electromagnetic interference requirements. The power supply shall meet the following electromagnetic interference (EMI) requirements of MIL-STD-461, class A4 and class A5, with modifications to the limits where indicated (see 4.5.7):

(a) CE01 direct current (DC) power leads - see figure 3.

(b) CE01 alternating current (AC) power leads- see figure 3.

(c) CE03 narrowband, DC power leads- see figure 4.

(d) CE03 broadband, DC power leads- see figure 5.

(e) CE03 narrowband, AC power leads- see figure 6.

(f) CE03 broadband, AC power leads- see figure 7.

(g) CS01.

(h) CS02.

(i) RE01- see figure 8.

(j) RE02, installed configuration, see 4.5.7.2. The maximum test frequency shall be 1 gigahertz (GHz).

(k) RE02, connector opening, see 4.5.7.2. The maximum test frequency shall be 1 GHz.

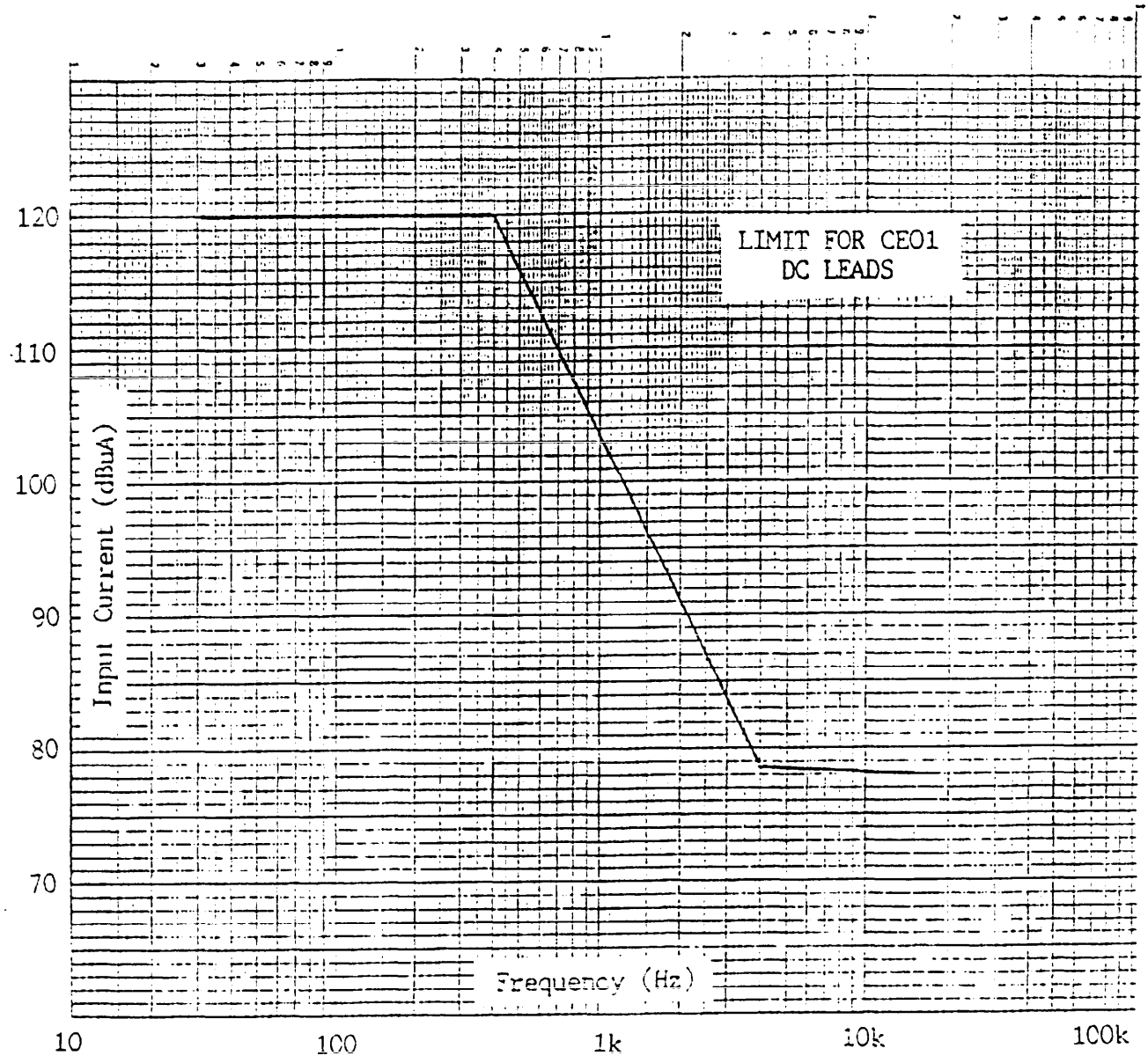
(l) RS01.

(m) RS02.

(n) RS03- class A4 for below deck metallic hull.

3.14.1 Criteria for susceptibility. When the power supply is being tested for susceptibility to a conducted or radiated signal, the output voltage shall be monitored for compliance to the table I output regulation limits.

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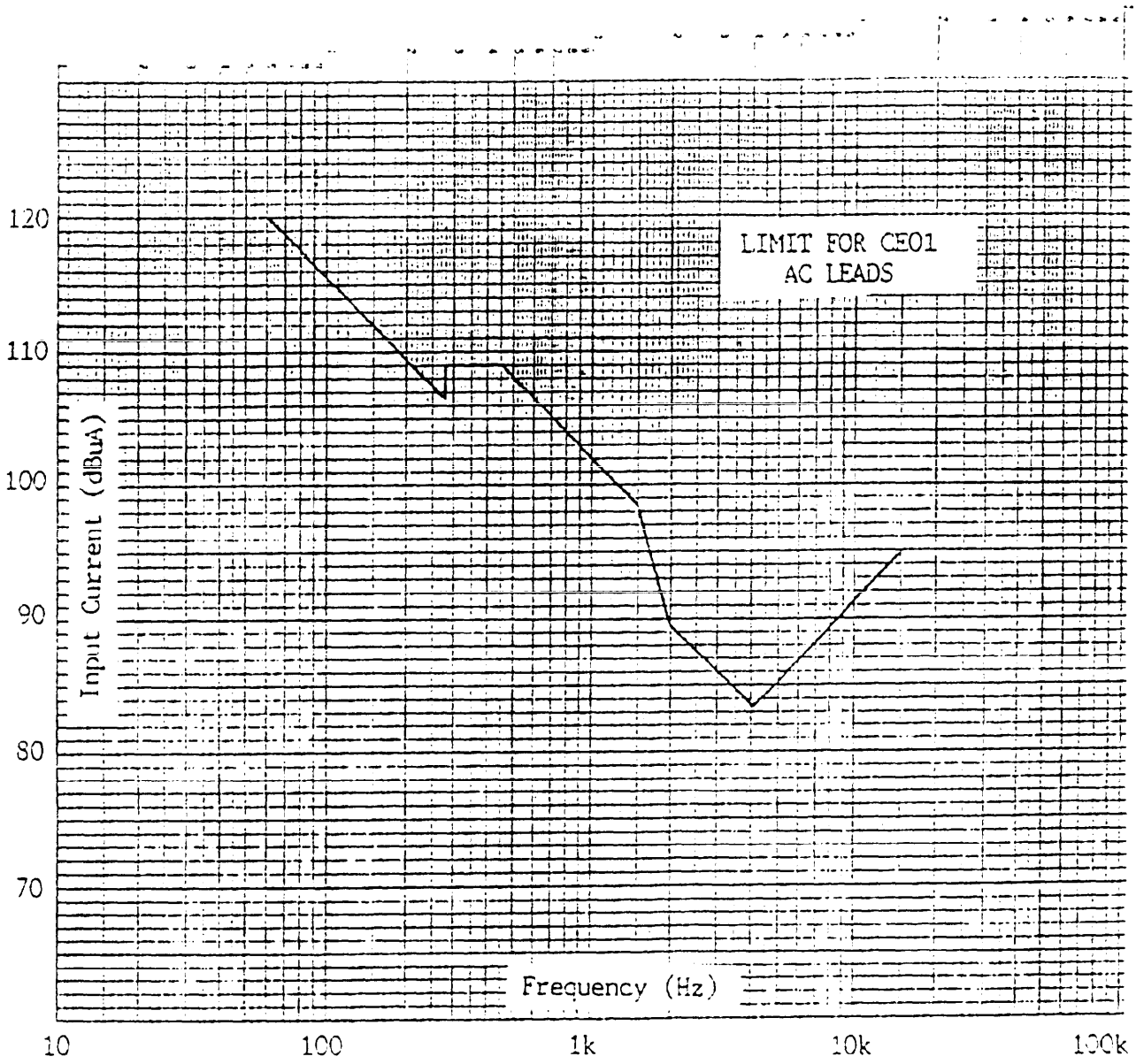
For power supplies with input current less than 1 ampere, the limit shall be as shown. For power supplies with input current greater than 1 ampere, the limit shall be relaxed as follows:

dB relaxation = 20 log(input current),

where input current is in amperes.
The unit dBuA is decibels with respect to 1 microampere.

FIGURE 2. Powerline harmonic current limitation (CE01).

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NOTE:

For power supplies with input current less than 1 ampere, the limit shall be as shown. For power supplies with input current greater than 1 ampere, the limit shall be relaxed as follows:

dB relaxation = 20 log(input current),

where, input current is the wideband, RMS value in amperes.

FIGURE 5. Powerline harmonic current limitation (CE01) - Continued.

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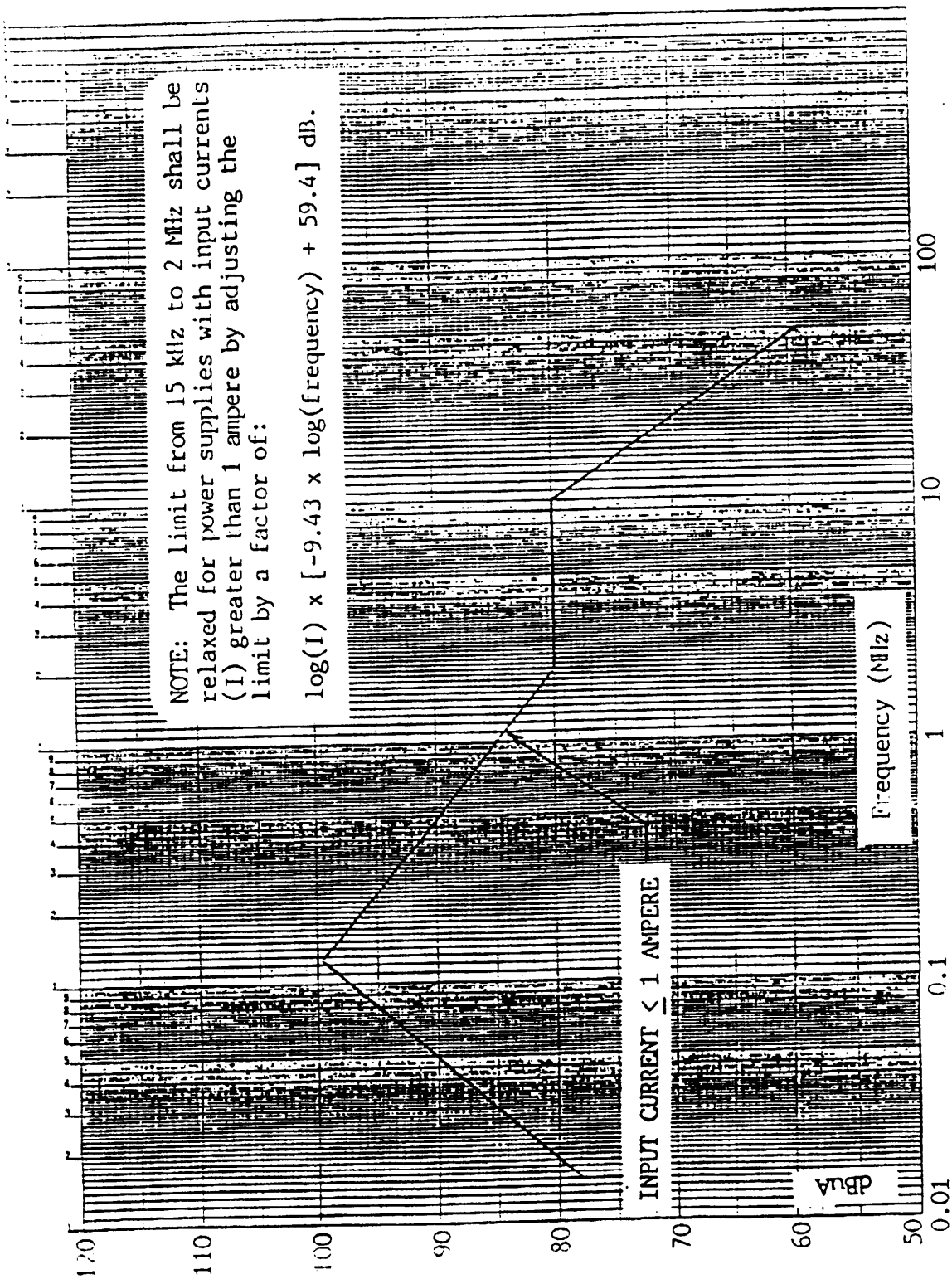


FIGURE 4. Narrowband conducted emissions on DC power leads (CE03).

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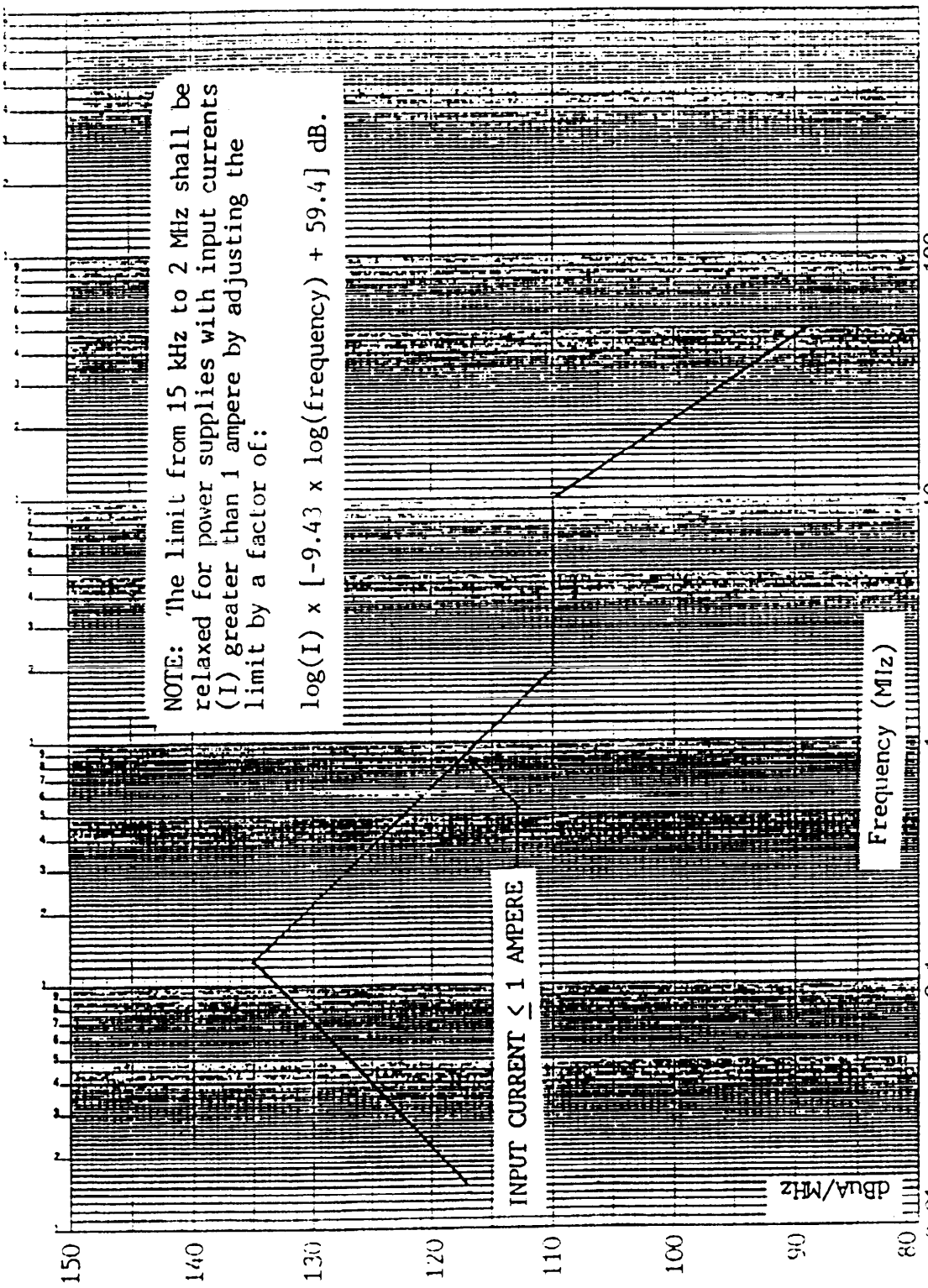
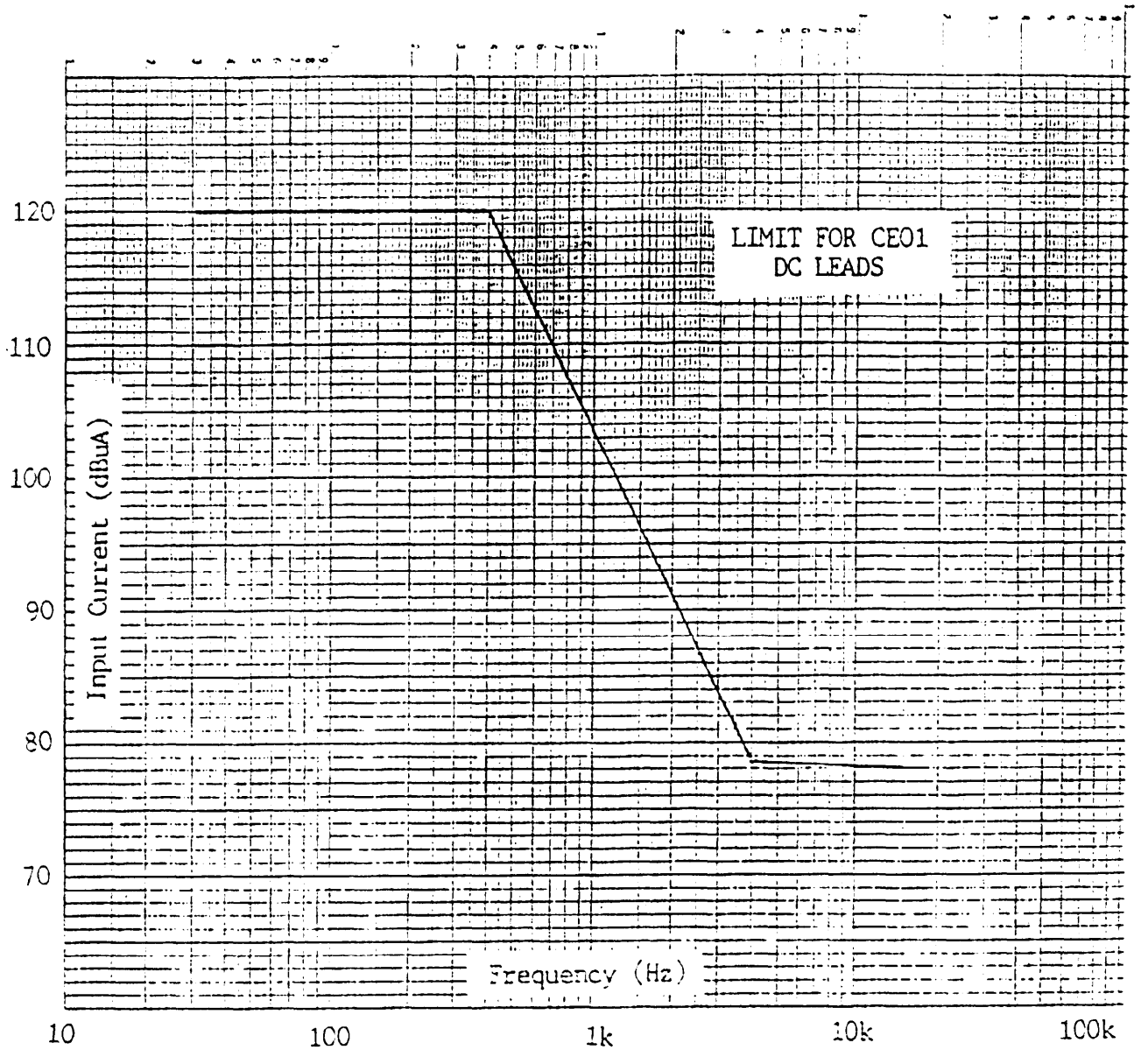


FIGURE 5. Broadband conducted emissions on DC power leads (CE03).

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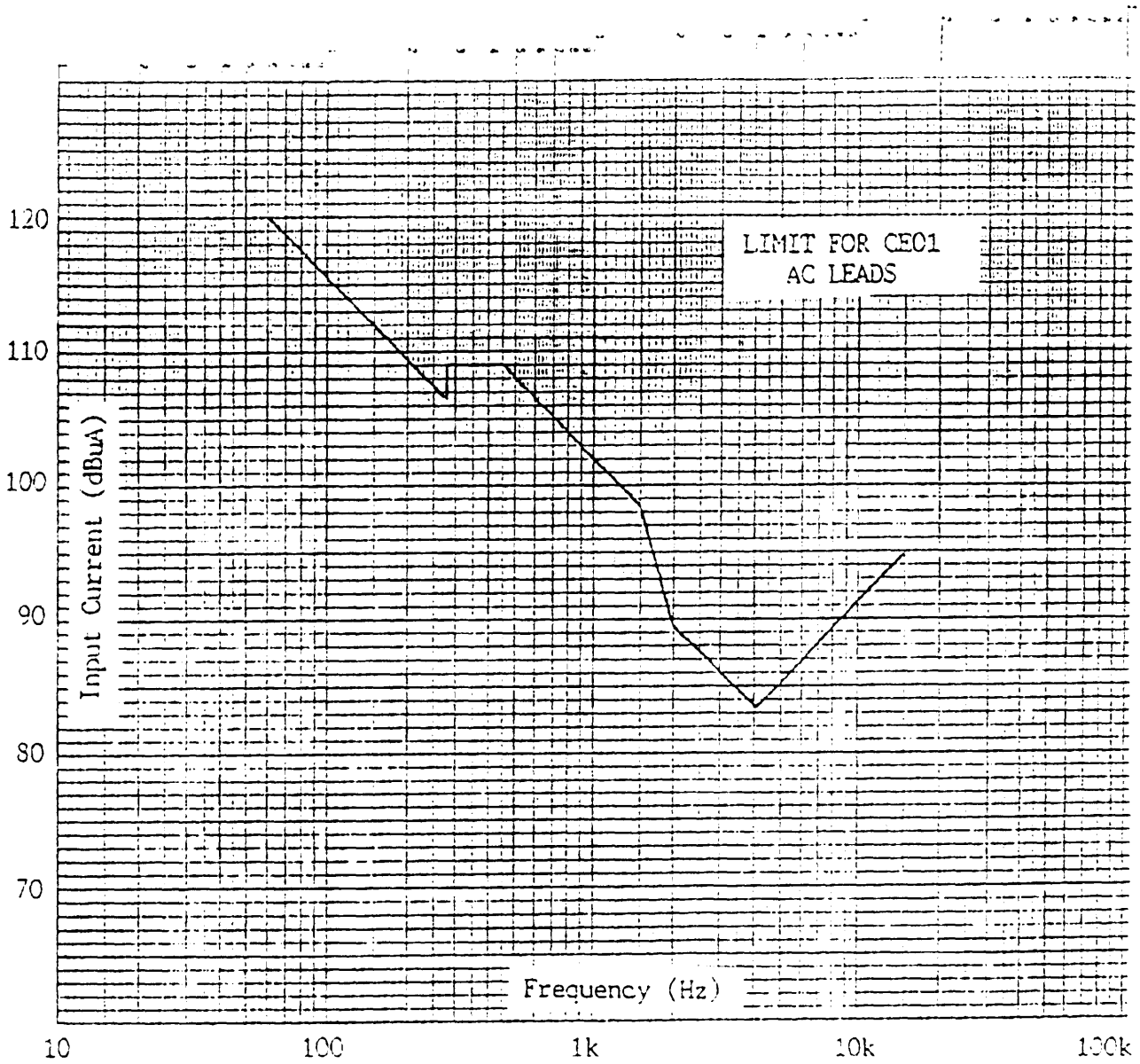
For power supplies with input current less than 1 ampere, the limit shall be as shown. For power supplies with input current greater than 1 ampere, the limit shall be relaxed as follows:

dB relaxation = 20 log(input current),

where input current is in amperes.
The unit dBuA is decibels with respect to 1 microampere.

FIGURE 2. Powerline harmonic current limitation (CE01).

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NOTE:

For power supplies with input current less than 1 ampere, the limit shall be as shown. For power supplies with input current greater than 1 ampere, the limit shall be relaxed as follows:

dB relaxation = 20 log(input current).

where input current is the wideband, RMS value in amperes.

FIGURE 3. Powerline harmonic current limitation (CE01) - Continued.

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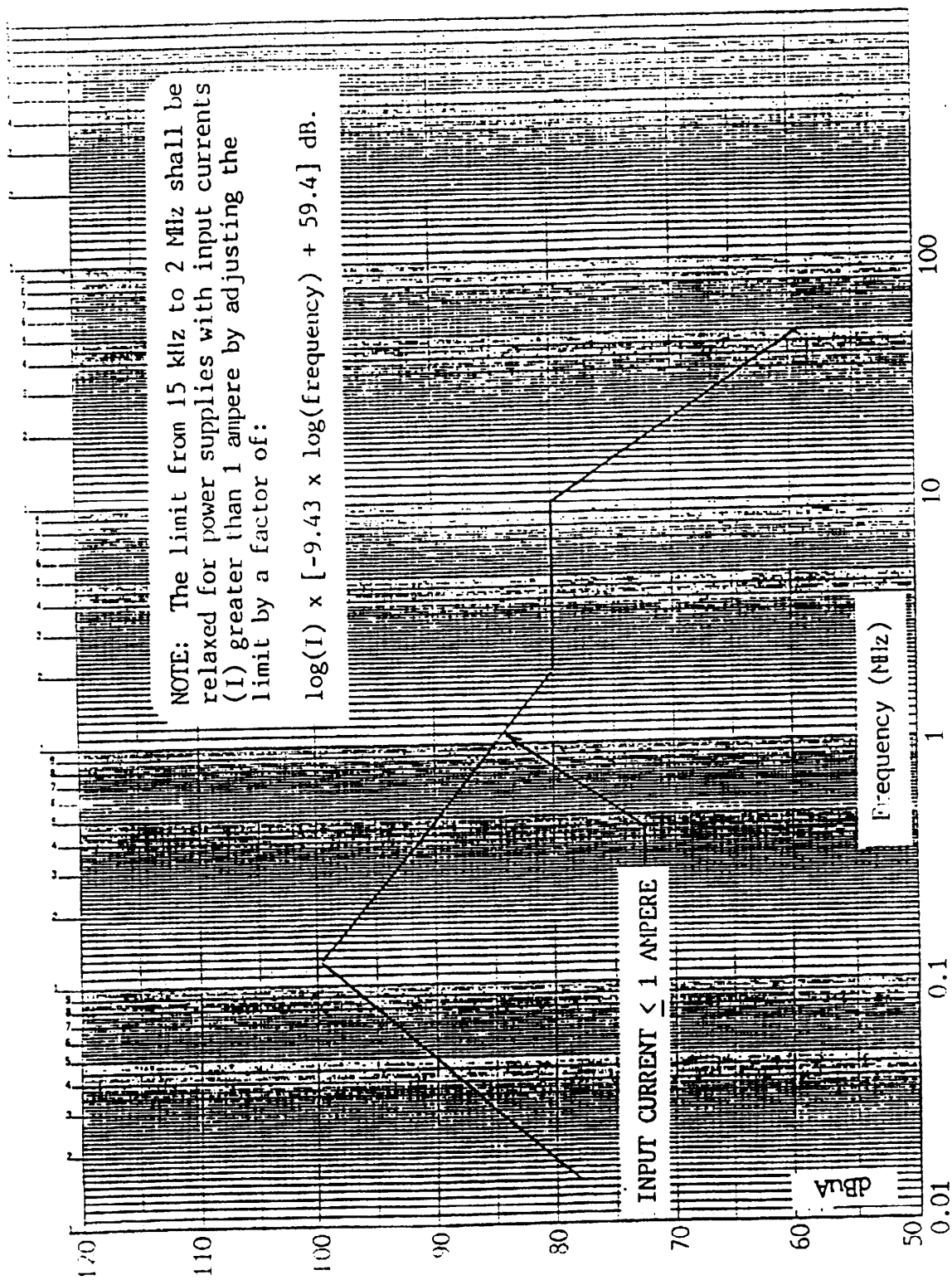


FIGURE 4. Narrowband conducted emissions on DC power leads (CE03).

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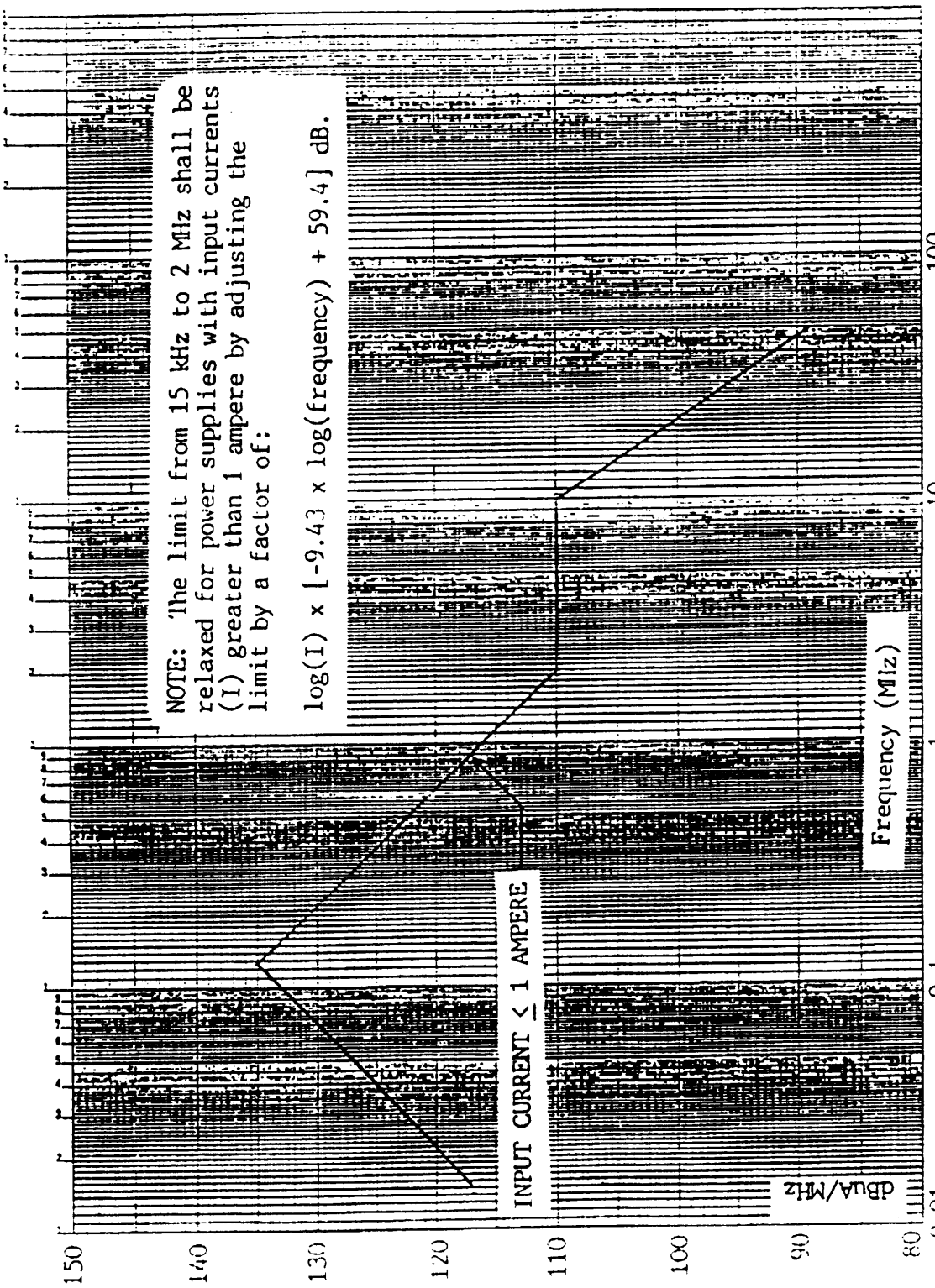


FIGURE 5. Broadband conducted emissions on DC power leads (CE03).

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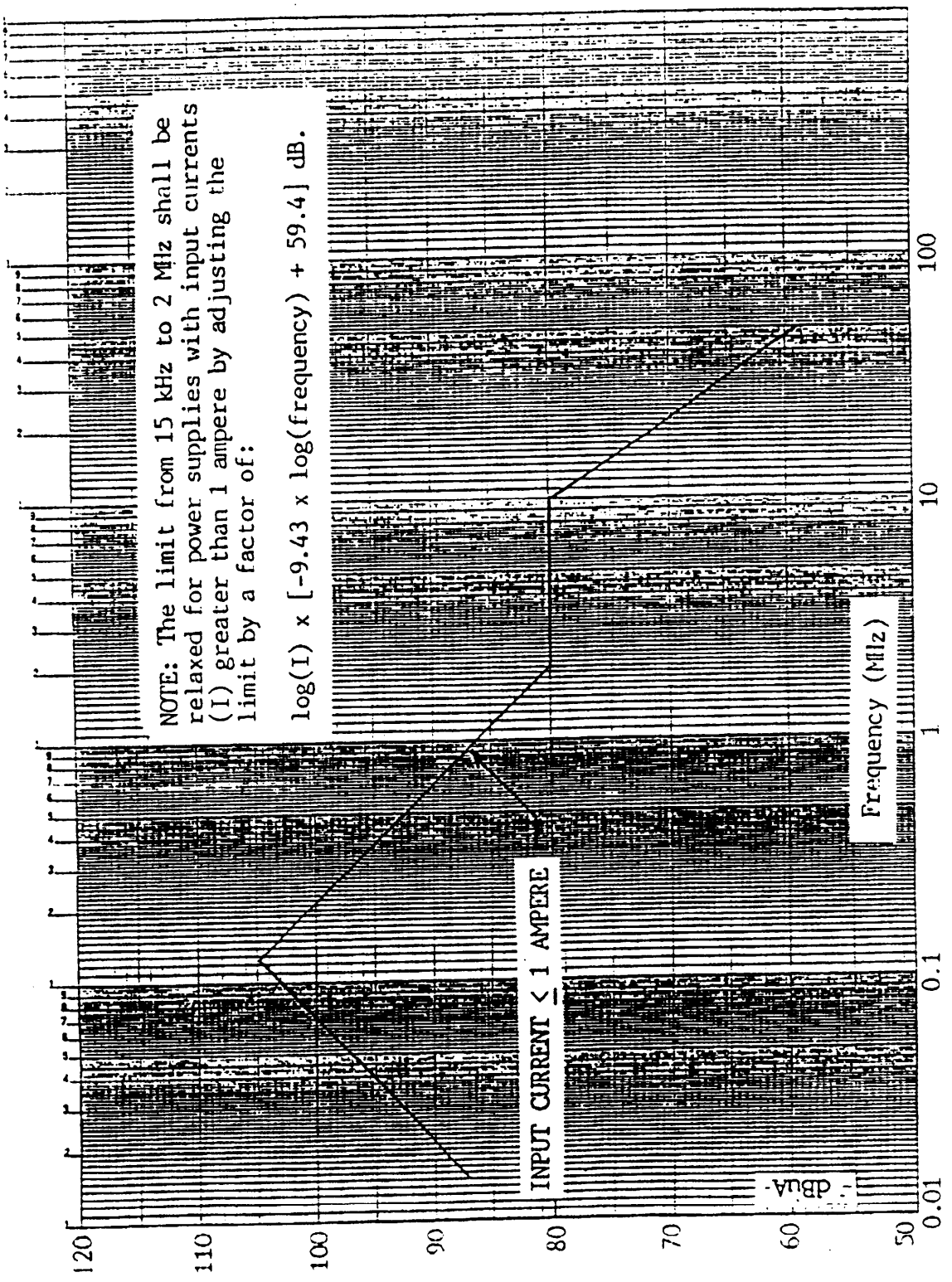


FIGURE 6. Narrowband conducted emissions on AC power leads (CE03).

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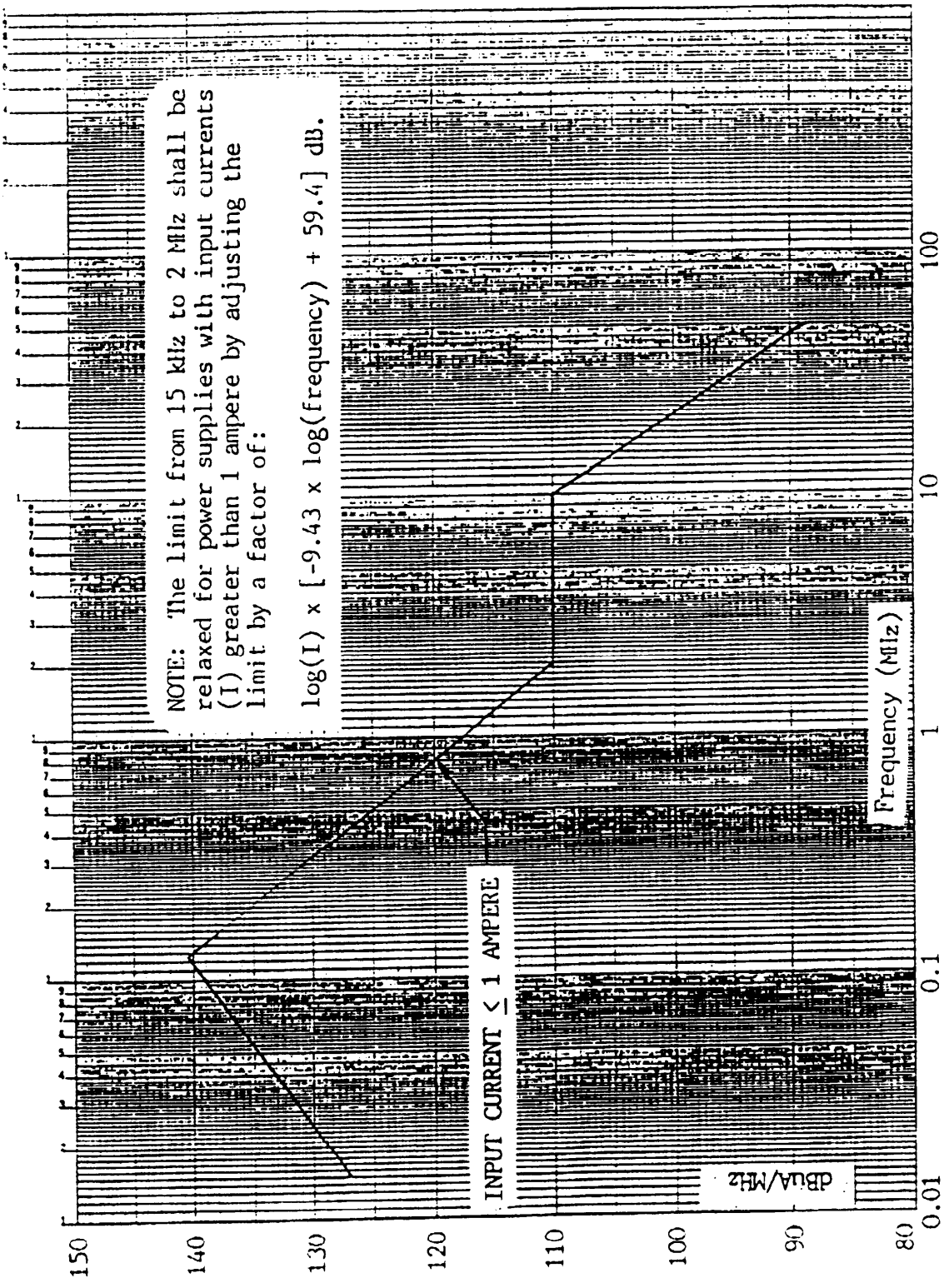


FIGURE 7. Broadband conducted emissions on AC input power leads (CE03).

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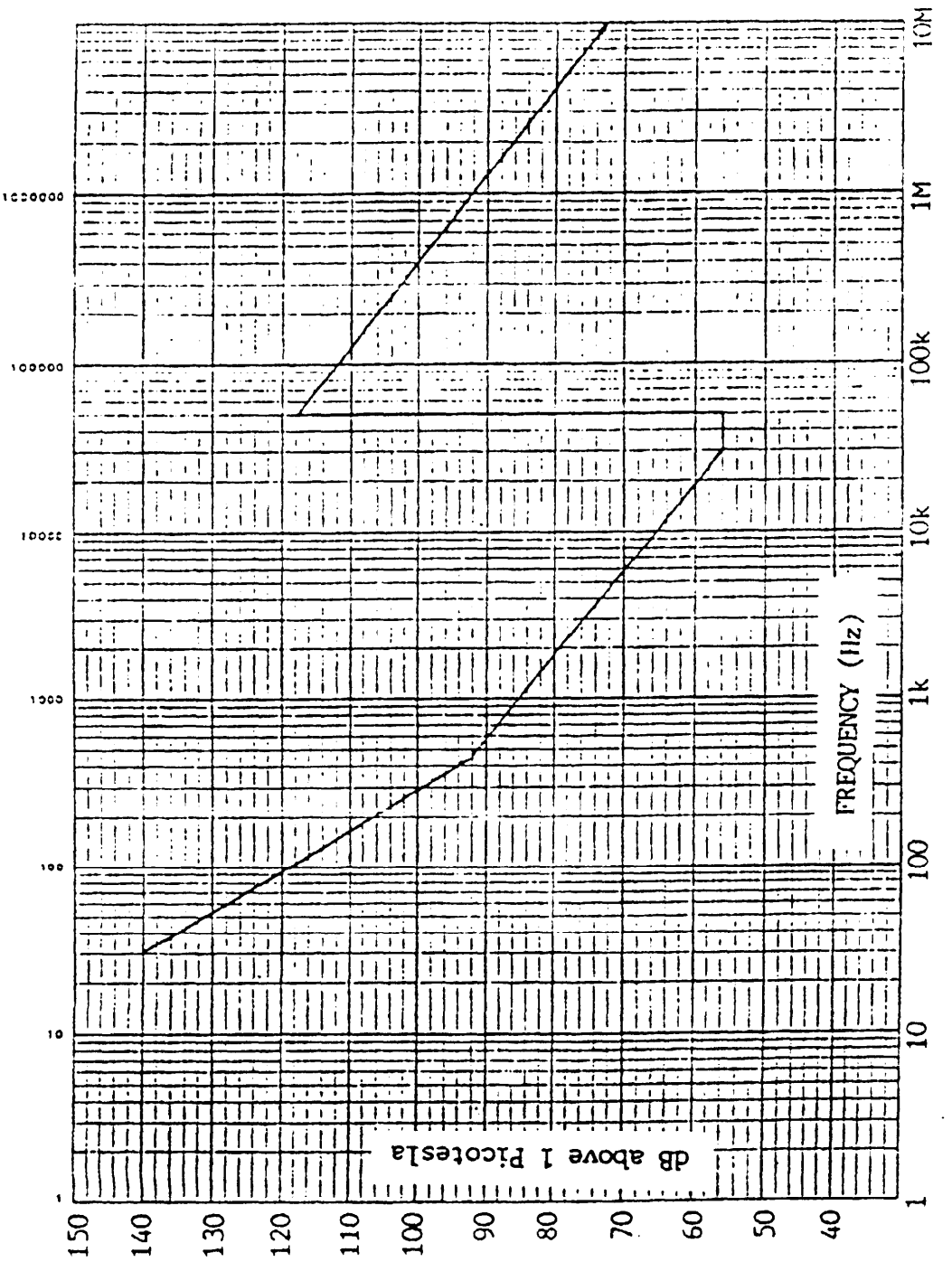


FIGURE 8. Magnetic field emissions (RE01).

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3.15 EMI filter design. The power supply is not required to have an internal input EMI filter. Therefore, for a power system to meet MIL-STD-461 conducted emissions requirement CE03, the system must have an EMI filter between the power source output and the power supply input. For one power supply operating at full load, the necessary EMI filter attenuation can be found by taking the difference between the allowed applicable power supply conducted emissions curves (see figures 3 through 7) and the system required conducted emissions. For reduced power supply loading and more than one power supply attached to an EMI filter, the determination of the required EMI filter attenuation becomes more complex. The power supply input impedance characteristics as specified in 3.15.1 are also needed in order to design an EMI filter. MIL-HDBK-241 should be referenced for precautions to be taken in the design of an EMI filter to prevent the filter/power supply combination from forming a negative impedance oscillator. The design of the power source system must ensure that damage cannot occur under the system's worst case minimum input voltage including brownout conditions. The power supply output voltage may continue within regulation as the input voltage drops to half of the nominal 155 VDC (115 volts alternating current (VAC)); however, the power source voltage may not be specified to go this low.

3.15.1 Input impedance requirement. The power supply input impedance magnitude, at full load output, shall be greater than the figure 9 curve for the input voltages specified. The input impedance phase shall be more positive than the figure 10 curve for the input voltages specified. These curves are composites of the minimum input impedances from several different power supply designs. The actual model for each vendor's design may be obtained from the SPSP-QAA. During initial qualification testing, the SPSP-QAA will verify that each vendor's design meets the above impedance requirements.

3.15.2 Input impedance computer model. Each manufacturer of power supplies shall generate a computer model of the power supply input impedance (Zi) as observed from the input power contacts. The model shall diagram the input network and show both the AC and auxiliary inputs. It shall include elements simulating the negative impedance of the converter. Nominal component values together with their tolerances shall be given. To verify the accuracy of the model, each manufacturer shall measure the input impedance (magnitude and phase) of their power supply under the following conditions:

(a) From any two of the AC, phase A, B, or C input contacts. Also from the auxiliary input contacts if the data taken would differ significantly from that taken at the AC input contacts.

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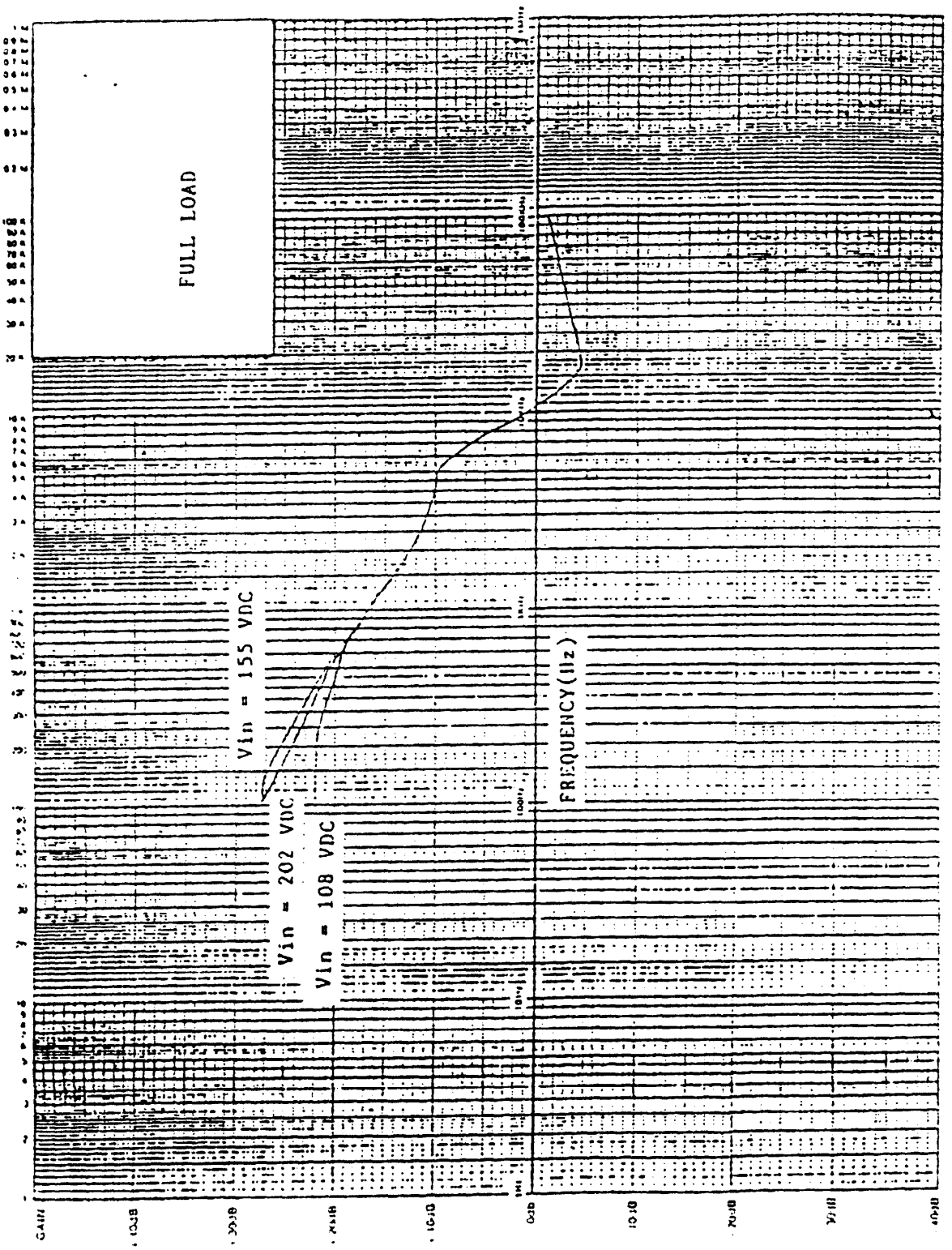


FIGURE 9. Minimum input impedance magnitude.

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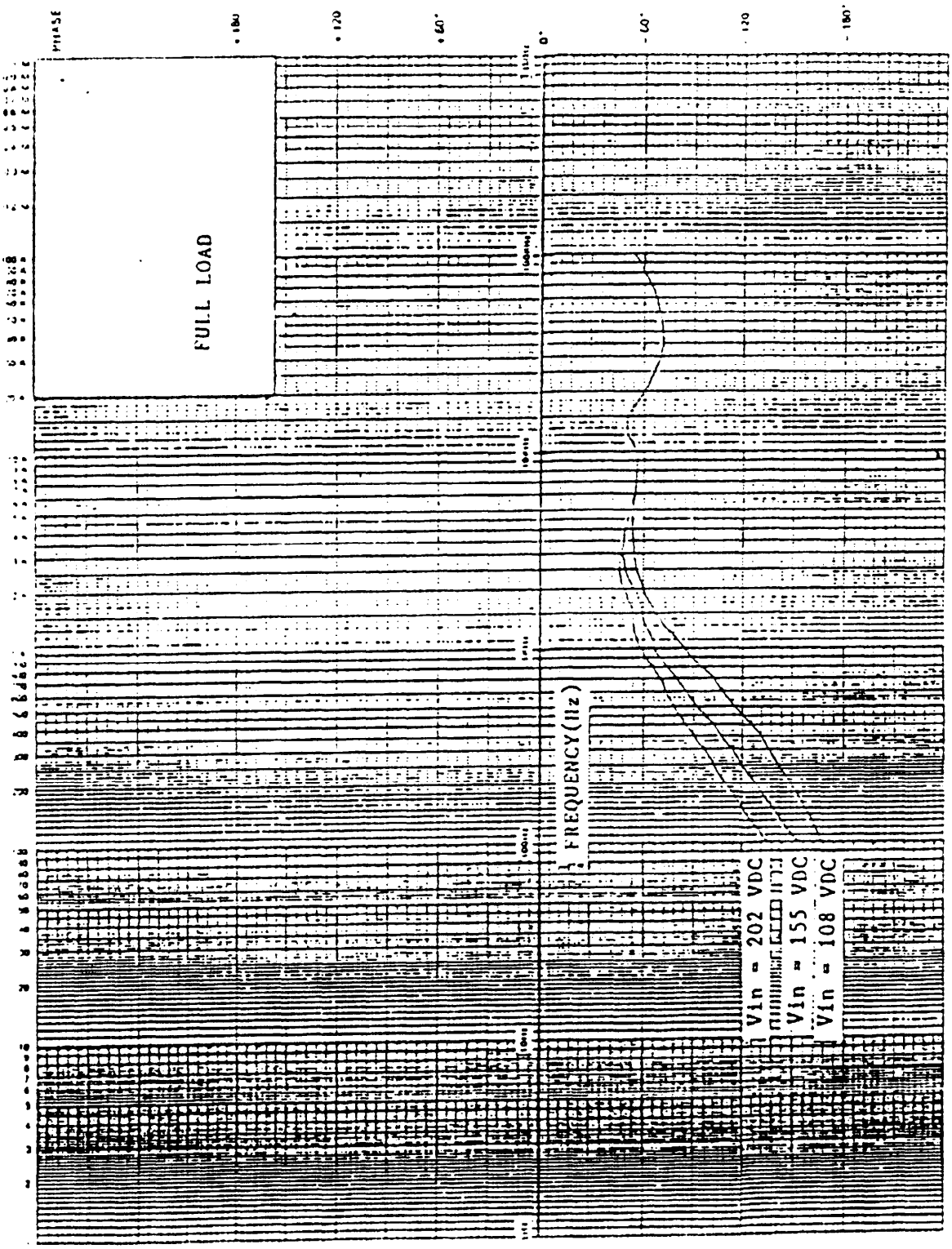


FIGURE 10. Minimum input impedance phase.

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(b) Over the frequency range of 100 hertz (Hz) to 100 kilohertz (kHz) or up to the voltage feedback loop gain crossover frequency, whichever is greater.

(c) With input voltages of 108, 155, and 202 VDC. Also measurement at a minimum input voltage, just prior to the output voltage going out of regulation or turning off.

(d) At full load for all input voltages. For 108 VDC input voltage, also measure at 10 percent and 50 percent full load.

The Zi computer model shall be made to match the Zi measured data within ± 1.0 dB for gain and ± 6° for phase at frequencies up to 10 kHz. Above 10 kHz, the required match shall be ± 2.0 dB for gain and ± 24° for phase. The measured data, the computer model, and curves obtained from the computer model shall be submitted to the SPSP-QAA with no restrictions on use by system manufacturers needing to ensure system stability.

3.16 Spike voltage. The power supply shall be designed to withstand, without damage, a 2500 volt spike clipped to a 250 volt amplitude by external system or equipment devices. The duration of the spike shall be 162 ±16 microseconds (us) at the 250 volt level.

3.17 +15 V and -15 V output protection. The +15 V and -15 V outputs shall not be damaged by repeated turn-off into a purely capacitive load.

3.18 Common mode current. The spectral line common mode current (per square root hertz basis) shall be less than the figure 11 curve over the frequency range of 40 Hz to 10 megahertz (MHz).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-P-24764 shall apply.

4.2 Quality conformance tests. Quality conformance tests of all power supplies shall be in accordance with the procedures and equipment specified in MIL-P-24764.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as

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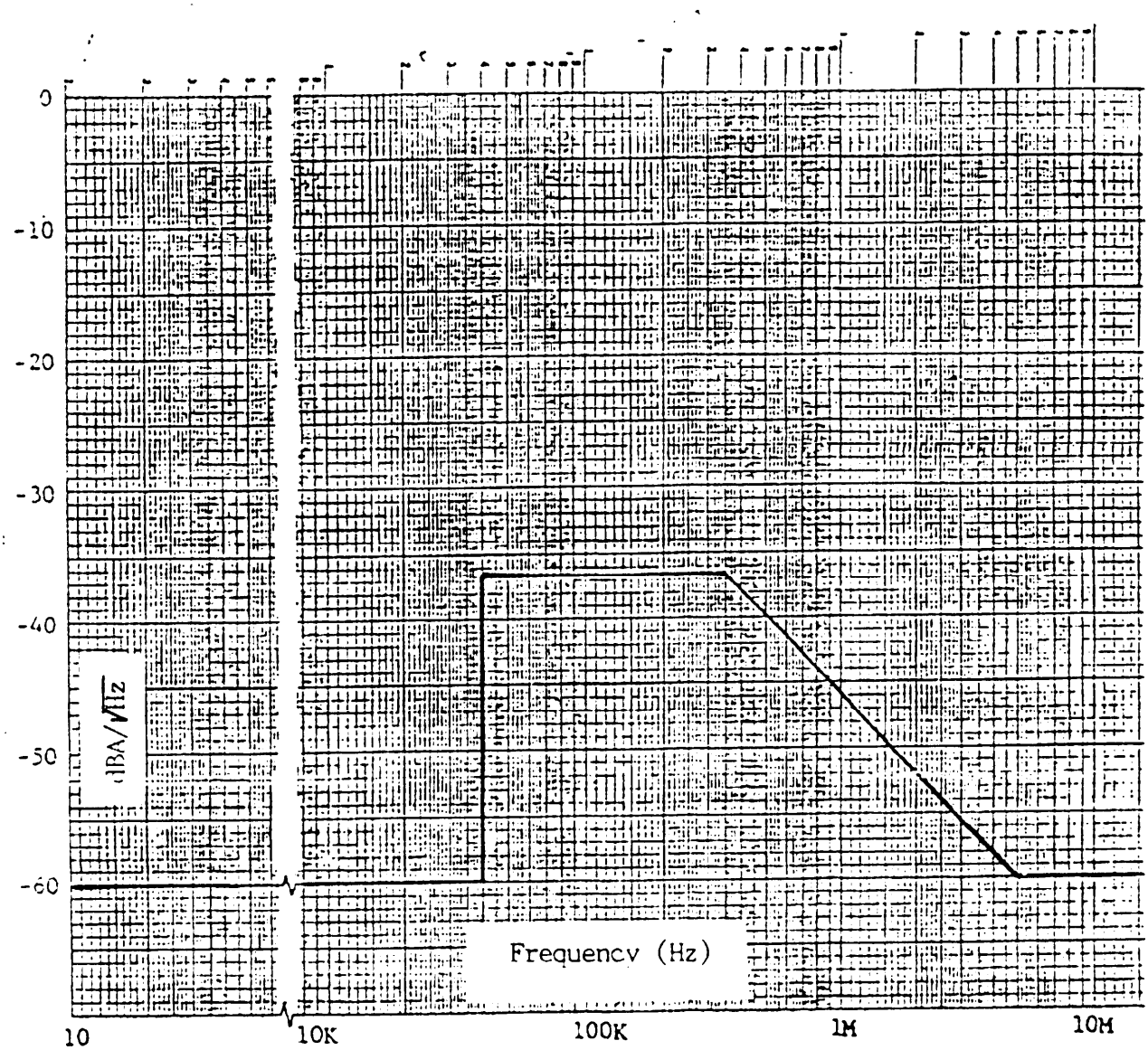


FIGURE 11. Spectral line common mode current limit (per square root hertz basis).

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specified on the figures. Equivalent equipment may be used for quality conformance and qualification testing provided the correlation requirements of MIL-P-24764 are met.

4.4 Test procedures. The power supply shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the SPSP-QAA and the test methods conform to the requirements of MIL-P-24764.

4.4.2 Order of testing. Test conditions A through JJ shall be performed in any order with the following exceptions. If test conditions A, BB and HH are part of a sequence of tests, the order of testing shall be as follows: A, BB, and HH where applicable, followed by the remaining tests in the sequence. Test condition W shall be performed in accordance with 4.5.3.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-P-24764 with the modifications or additions specified in 4.5.1 through 4.5.7.2.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-P-24764. Before starting these tests, test conditions A through V and X through JJ in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25 ± 5°C mounting surface temperature. At turn on, following one hour stabilization at the low temperature, test conditions C1, C3, C7, C9, C12, C14, D2, D3 and P11 shall be performed. When electrical testing is called for at the end of each temperature stabilization period (at -10°C, 60°C, and 25°C post operational temperature), test conditions C, D, G, H, M7, N1-N3, P1-P3, Q1-Q3, Q6-Q12, S1-S4, S13-S20, T, V1-V9 and GG1-GG3 shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. When electrical testing is called for at 80°C, test conditions C5-C7, C10, C11, C14, D1-D12, N1-N3 and Q1-Q3 shall be performed using the high temperature end-of-life limits. At all other times during the operating temperature tests, the power supply shall be operated in the test circuit shown on figure 12.

4.5.2 Component temperature tests. The component temperatures shall be measured in accordance with MIL-P-24764. The power supply shall be operated as shown on figure 12. These measurements shall be performed on each initial qualification submission that is the result of a new vendor or of a major redesign of a previously qualified vendor.

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4.5.3 Over-temperature indicators. Operation of the excessive temperature indicator and unsafe temperature indicator shall be verified to be within the temperature limits defined in table I. Test conditions W1-W8 are non-operating tests designed to verify that the over-temperature indicators operate. Test condition W shall only be performed after all other electrical and environmental tests are completed.

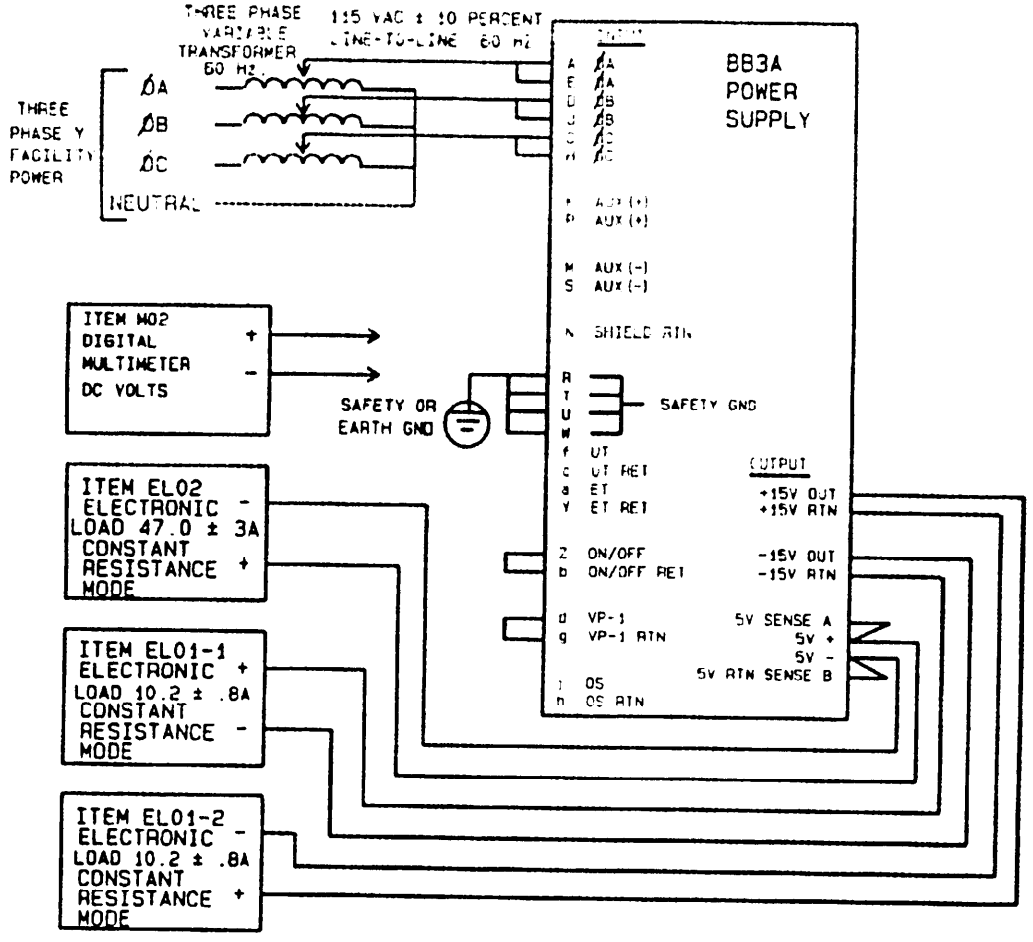
4.5.4 Vibration tests. The vibration tests shall be conducted in accordance with MIL-P-24764. During the vibration tests, the power supply shall be operated in the circuit shown on figure 13. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25 ± 5°C mounting surface temperature.

4.5.5 Shock tests. The mechanical shock tests shall be conducted in accordance with MIL-P-24764. During the shock tests, the power supply shall be operated in the circuit shown on figure 13. Following the shock test, the electrical tests in table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25 ± 5°C mounting surface temperature.

4.5.6 Long term stability test. The long term stability test shall be conducted in accordance with MIL-P-24764. The power supply shall be operated as shown on figure 12. After operating for 20 ± 2 hours, the baseline output voltage readings shall be taken. At the conclusion of the test, prior to disconnecting the power supply or reducing the temperature, the output voltage readings shall again be recorded. The end-of-test voltages shall not change by more than 0.2 percent from the baseline voltages. Following the return of the power supply to 25°C, electrical tests A through V, X, Z, EE, and GG1-GG7 shall be performed in the order specified in 4.4.2 using the 25 ± 5°C mounting surface temperature end-of-life requirements in table I.

4.5.7 EMI test. The EMI tests shall be conducted in accordance with MIL-P-24764, the procedures of MIL-STD-462, and as modified herein. For requirements CE01 and CE03, the power supply shall be tested using 115 VAC ± 2 percent line-to-line, 3-phase, 60 ± 1 Hz input voltage and tested using 155 VDC ± 2 percent input voltage. For requirement RE01, the supply shall be tested using the 115 VAC ± 2 percent line-to-line input voltage. At the option of the testing facility, for the remainder of the requirements, either the nominal AC or DC input voltage may be used. The loading shall be 90 to 100 percent of full load.

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NOTES:

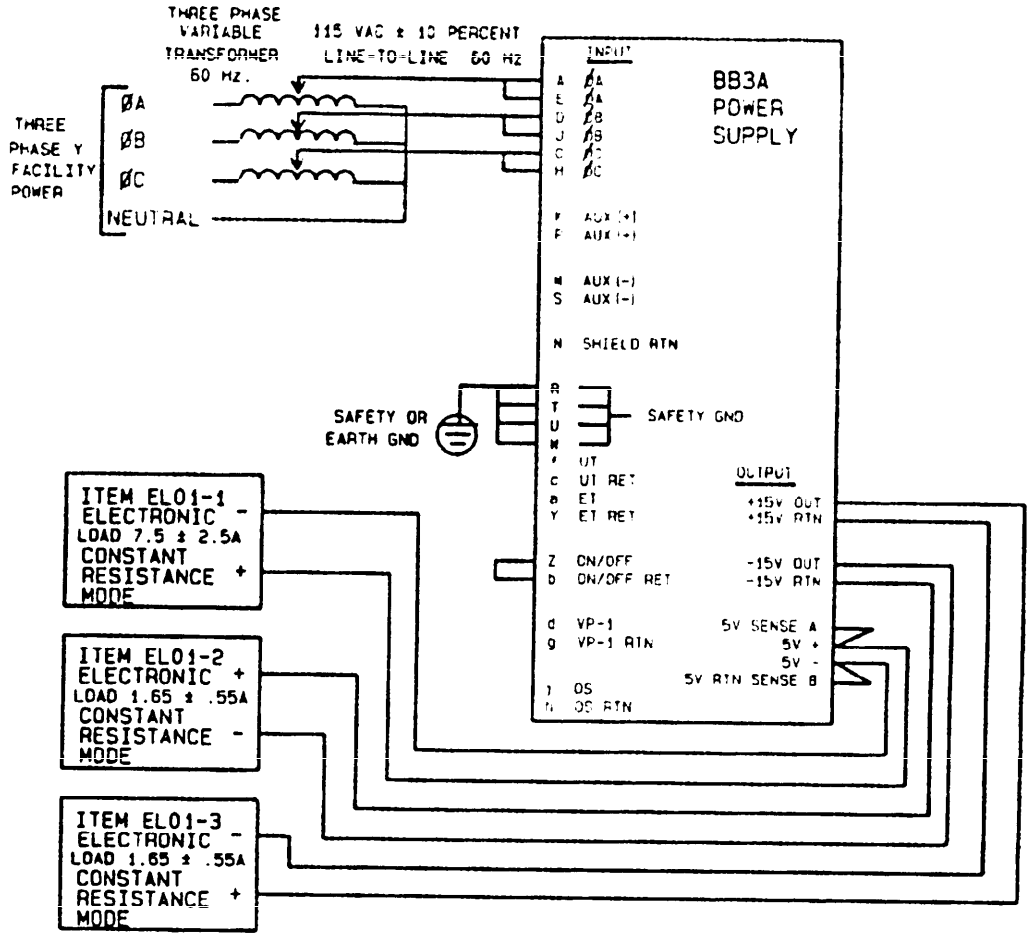
1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. For the long term stability test, the input voltage and output load current shall be within ± 1 percent of nominal when making output voltage measurements. Item M02 shall be used.

FIGURE 12. Operating temperature, component temperature, and long term stability test circuit.

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NOTES:

1. See table II for output connector contact assignments.

2. Output loading is specified at 15 percent of rated load to minimize the requirement for externally supplied cooling. The test may be performed at any load level between 10 percent and 100 percent rated load; however, the power supply mounting surface temperature shall not exceed 60°C.

FIGURE 13. Vibration and shock test circuit.

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4.5.7.1 RE01 test modification. The RE01 upper frequency shall be 10 MHz as shown on figure 8.

4.5.7.2 RE02 test modification. The test shall be performed for two conditions.

(a) The installed configuration shall have the power supply marking (see figure 1) facing the antenna with the connector area and all leads completely shielded. The purpose of this configuration is to ensure that only the radiated emissions through the case and cover of the power supply are being measured.

(b) The connector opening configuration shall have the power supply connector opening facing the antenna and all leads up to the connector opening completely shielded. The purpose of this configuration is to ensure that only the radiated emissions from the power supply connector area are being measured.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-P-24764.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Power supplies specified herein are intended for use in military shipboard and subsurface ship systems and subsystems.

6.2 Acquisition requirements. Acquisition documents should specify the ordering data required in MIL-P-24764.

6.3 End-of-life. When any individual power supply parameter fails to conform to the end-of-life requirements in table I, the power supply is considered to have completed its useful life (see 3.9).

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6.4 Power supply description.

Key code: BB3A

Case size: 2.48 inches (62.99 millimeters (mm)) by 6.80 inches (172.72 mm) by 10.8 inches (274.32 mm)

Mean time between failures (MTBF): 100,000 hours minimum.

Input voltage: Accepts any of the following voltages:

(a) 115 volts root mean square (VRMS) line-to-line, 3-phase, 47-63 Hz, delta.

(b) 155 VDC to any two AC inputs.

(c) 155 VDC to the auxiliary input.

Outputs: Provides either one of the following output voltage and current combinations at one time:

(a) Triple output: +5.05/+5.25 volts at 5 to 50 amperes.
+15 volts at 0 to 11 amperes.
-15 volts at 0 to 11 amperes.

(b) Triple output: -5.05/-5.25 volts at 5 to 50 amperes.
+15 volts at 0 to 11 amperes.
-15 volts at 0 to 11 amperes.

6.5 Power supply function.

6.5.1 Voltage programming. The 5V output is contact programmable for a nominal voltage of 5.05 VDC or 5.25 VDC. The voltage programming return is common with the 5V output sense return. The voltage programming (VP) return will be a negative voltage when the 5V output is used as a negative 5 VDC output.

6.5.2 Remote ON/OFF control. The remote ON/OFF control applies to all three outputs simultaneously.

6.5.3 Remote sense capability. The power supply remote sense capability applies to the 5V output only. The sense leads shall not be left disconnected.

6.5.4 Stability with remote sense. Power supply stability may be adversely affected by certain load lead characteristics

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when the output voltage is remotely sensed. The inductance (L) of the load leads between the power supply output and the remote sense point, together with the load capacitance (C), form an additional LC filter on the output of the power supply. When the corner frequency of this LC filter falls within the unity gain bandwidth of the power supply feedback loop, then the power supply phase margin is reduced. These conditions tend toward and can result in power supply instability. Caution should be exercised when utilizing remote sensing and having significant load lead inductance (L) and load capacitance (C). Power supply stability should be verified by analysis and also by dynamic load testing in the actual circuit configuration.

6.5.5 Output current rating. The 5V output is rated at 5 to 50 ADC. For operation between 0 and 5 ADC, the 25°C initial load regulation and noise ripple tolerances are twice the tolerances for operation within the rated current limits.

6.6 Recovery from overvoltage. The output voltage is not required to recover automatically from an output overvoltage fault and may require the input voltage to be taken to zero and then reapplied to reset the output. Cycling of the remote ON/OFF control will not accomplish this reset function.

6.7 Exceptions to MIL-P-24764. This power supply has the following exceptions to the requirements of MIL-P-24764:

(a) The power interrupt signal and visual indicator are not provided.

(b) The malfunction signal and visual indicator are not provided.

(c) The power input circuit can withstand a spike voltage of only 250 volt peak (see 3.16).

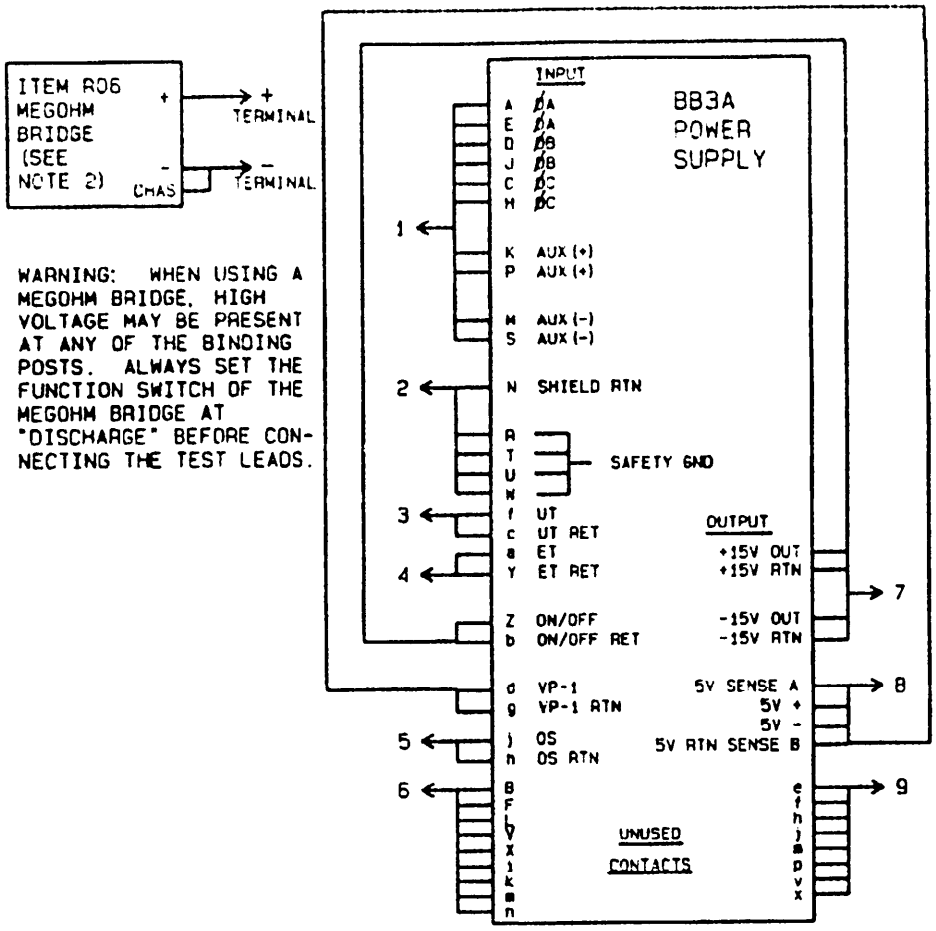
(d) The EMI limits for requirements CE01, CE03, RE01, and RE02 of MIL-STD-461 are modified or extended (see 3.14).

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Test cond	Parameter	Voltage ± 10%	+ Terminal points:	- Terminal points:	Surface temperature Min	Surface temperature Max	Unit	Test fig
A1	Isolation-input to chassis and output	800 VDC	1	2,7,8	100	INF.	kOhm	14
A2	Isolation-output to chassis	50 VDC	7,8	2				
A3	Isolation-output to output	20 VDC	7	8				
A4	Isolation-excessive temp.	100 VDC	4	1-3,5,7,8				
A5	Isolation-unsafe temp.	100 VDC	3	1,2,4,5,7,8				
A6	Isolation-output status	100 VDC	5	1-4,7,8				
A7	Isolation-input connector used to unused contacts	100 VDC	6	1-5				
A8	Isolation-output connector used to unused contacts	100 VDC	9	2,7,8	100		kOhm	
A9	Isolation-input to chassis and output	800 VDC	1	2,7,8	10		MOhm	
A10	Isolation-output to chassis	50 VDC	7,8	2				
A11	Isolation-excessive temp.	100 VDC	4	1-3,5,7,8				
A12	Isolation-unsafe temp.		3	1,2,4,5,7,8				
A13	Isolation-output status		5	1-4,7,8				
A14	Isolation-input connector used to unused contacts		6	1-5				
A15	Isolation-output connector: used to unused contacts	100 VDC	9	2,7,8	10	INF.	MOhm	14

1/ Two or more numbers in one block indicate that these points shall be connected together in addition to being connected to the + terminal or - terminal as applicable (for example, "7,8" under column "+ Terminal points:" indicates point 7, connected to point 8, connected to the + terminal).

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NOTES:

1. See table II for output connector contact assignments.
2. See table IV for item R06 connections and voltage setting.

FIGURE 14. Isolation test circuit for table IV.

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TABLE V. Continuity test.

Test cond	Parameter	Connect to M02 (Ohms mode)		Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		+ to contact	- to contact	Min	Max		
B1	Chassis contact to chassis continuity	Input R	Chassis	0.0	1.0	Ohm	-
B2	15V RTN to remote ON/OFF RTN continuity	Output y	Input b				
B3	5V (-) sense to VP RTN continuity	Output B	Input g				
B4	Safety contact continuity	Input R	Input T				
B5	Safety contact continuity		Input U				
B6	Safety contact continuity		Input W				
B7	Safety GND to shield RTN continuity	Input R	Input N				
B8	Phase A contact continuity	Input A	Input E				
B9	Phase B contact continuity	Input D	Input J				
B10	Phase C contact continuity	Input C	Input H				
B11	Auxiliary DC(+) contact continuity	Input K	Input P				
B12	Auxiliary DC(-) contact continuity	Input M	Input S	0.0	1.0	Ohm	-

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TABLE V. Continuity test - Continued.

Test cond	Parameter	Connect to M02 (Ohms mode)		Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		+ to contact	- to contact	Min	Max		
B13	Output(+15V) contact continuity	Output k	Output n	0.0	1.0	Ohm	-
B14	Output(+15V) contact continuity	Output k	Output r				
B15	Output(+15V) contact continuity	Output k	Output t				
B16	Output(-15V) contact continuity	Output z	Output BB				
B17	Output(-15V) contact continuity	Output z	Output DD				
B18	Output(-15V) contact continuity	Output z	Output FF				
B19	Output(±15V) RTN contact continuity	Output s	Output u				
B20			Output w				
B21			Output y				
B22			Output AA				
B23			Output CC				
B24			Output EE				
B25	Output(±15V) RTN contact continuity	Output s	Output HH	0.0	1.0	Ohm	

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TABLE V. Continuity test - Continued.

Test cond	Parameter	Connect to M02 (Ohms mode)		Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		+ to contact	- to contact	Min	Max		
B26	Output 5V (+) contact continuity	Output C	Output E	0.0	1.0	Ohm	
B27			Output H				
B28			Output K				
B29			Output M				
B30			Output P				
B31			Output S				
B32			Output U				
B33			Output W				
B34			Output Y				
B35			Output a				
B36	Output 5V (+) contact continuity	Output C	Output c	0.0	1.0	Ohm	

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TABLE V. Continuity test - Continued.

Test cond	Parameter	Connect to M02 (Ohms mode)		Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		+ to contact	- to contact	Min	Max		
B37	Output 5V (-) contact continuity	Output D	Output F	0.0	1.0	Ohm	-
B38			Output J				
B39			Output L				
B40			Output N				
B41			Output R				
B42			Output T				
B43			Output V				
B44			Output X				
B45			Output Z				
B46			Output b				
B47	Output 5V (-) contact continuity	Output D	Output d	0.0	1.0	Ohm	-
B48	Chassis contact to chassis continuity	Input N	Chassis	0.0	0.1	Ohm	-
B49	Chassis contact to chassis continuity	Input R	Chassis	0.0	0.1	Ohm	-
B50	Chassis contact to chassis continuity	Input T	Chassis	0.0	0.1	Ohm	-
B51	Chassis contact to chassis continuity	Input U	Chassis	0.0	0.1	Ohm	-
B52	Chassis contact to chassis continuity	Input W	Chassis	0.0	0.1	Ohm	-

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TABLE VI. Static load regulation test.

Test cond	Parameter	Item EL02 Curr. +0, -2%	Item EL01-1 Curr. +0, -2%	Item EL01-2 Curr. +0, -2%	Connect 1 to points	Connect 2 to points	VP-1 Input contact d to g	Initial limits 25 ± 5°C mounting surface temperature Min	Initial limits 25 ± 5°C mounting surface temperature Max	Unit	Test fig
C1	5.05V Static regulation	5A.	0A.	0A.	7	8	Open	5.00	5.10	VDC	15
C2		5A.	11A.	11A.							
C3		50A.	0A.	0A.							
C4	5.05V Static regulation	50A.	11A.	11A.			Open	5.00	5.10		
C5	5.25V Static regulation	5A.	0A.	0A.			Shorted	5.20	5.30		
C6	5.25V Static regulation	50A.	11A.	11A.	7	8	Shorted	5.20	5.30		
C7	+15V Static regulation	5A.	11A.	0A.	3	4	Open	14.85	15.15		
C8		50A.	11A.	0A.							
C9		5A.	0A.	11A.							
C10	+15V Static regulation	50A.	0A.	11A.	3	4	Open	14.85	15.15	VDC	15

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TABLE VI. Static load regulation test - Continued.

Test cond	Parameter	Item EL02 Curr. +0, -2%	Item EL01-1 Curr. +0, -2%	Item EL01-2 Curr. +0, -2%	Connect 1 to points	Connect 2 to points	VP-1 Input contact d to g	Initial limits 25 ± 5°C mounting surface temperature Min	Initial limits 25 ± 5°C mounting surface temperature Max	Unit	Test fig
C11	-15V Static regulation	5A.	0A.	11A.	5	6	Open	-15.15	-14.85	VDC	15
C12		50A.	0A.	11A.							
C13		5A.	11A.	0A.							
C14	-15V Static regulation	50A.	11A.	0A.	5	6		-15.15	-14.85		
C15	5.05V Static regulation No load	0A.	0A.	0A.	7	8		4.95	5.15		
C16	5.05V Static regulation No load	0A.	0A.	0A.	j	h	Open	0.0	0.4	VDC	
C17	-5.25V Static regulation	5A.	0A.	0A.	8	7	Shorted	-5.30	-5.20	VDC	15

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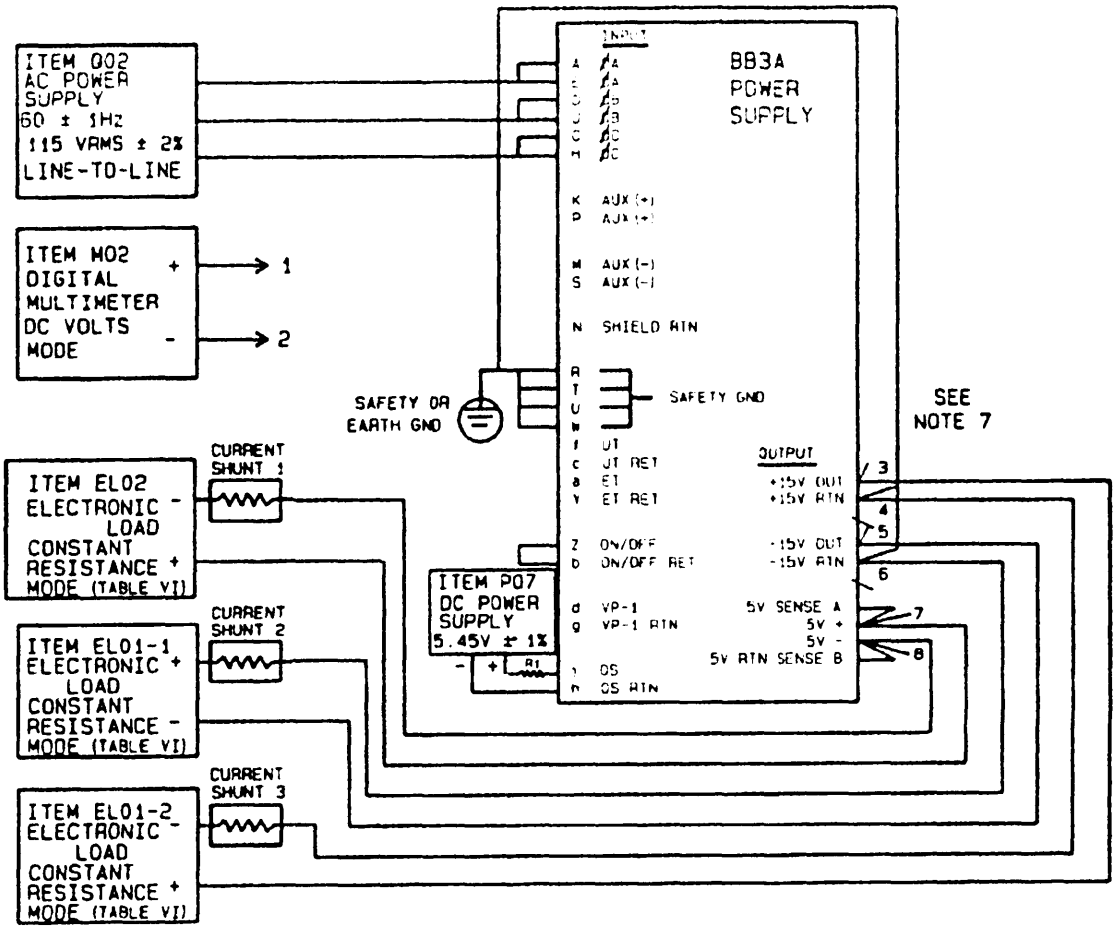


FIGURE 15. Static load regulation test circuit for table VI.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 1=75A/50mV, 2,3=50A/50mV.

4. R1=1.69 kilOhm (kOhm) ±1 percent, 1/8 watt resistor.

5. Monitor shunt voltages with an item VO4 voltmeter to ensure that the load currents are within specification.

6. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

7. For test conditions C1 through C16, output "5V -" shall be connected to the safety ground contacts also. For test condition C17, "5V -" shall be disconnected from safety ground and "5V +" connected to safety ground.

FIGURE 15. Static load regulation test circuit for table VI - Continued.

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TABLE VII. Line regulation test.

Test cond	Parameter	1/ Input voltage ± 1%	Item EL02 Curr. +0, -2%	Item EL01-1 Curr. +0, -2%	Item EL01-2 Curr. +0, -2%	Connect 1 to point	Connect 2 to point	Initial limits 25 ± 5°C mounting surface temperature Min	Initial limits 25 ± 5°C mounting surface temperature Max	Unit	Test fig
D1	86V Line regulation	86VRMS	5A.	0A.	0A.	7	8	5.00	5.10	VDC	16
D2	86V Line regulation maximum load	86VRMS	50A.	11A.	11A.						
D3	144V Line regulation	144VRMS	5A.	0A.	0A.						
D4	144V Line regulation maximum load	144VRMS	50A.	11A.	11A.	7	8	5.00	5.10		
D5	86V Line regulation	86VRMS	5A.	0A.	0A.	3	4	14.85	15.15		
D6	86V Line regulation maximum load	86VRMS	50A.	11A.	11A.						
D7	144V Line regulation	144VRMS	5A.	0A.	0A.						
D8	144V Line regulation maximum load	144VRMS	50A.	11A.	11A.	3	4	14.85	15.15		
D9	86V Line regulation	86VRMS	5A.	0A.	0A.	5	6	-15.15	-14.85		
D10	86V Line regulation maximum load	86VRMS	50A.	11A.	11A.						
D11	144V Line regulation	144VRMS	5A.	0A.	0A.						
D12	144V Line regulation maximum load	144VRMS	50A.	11A.	11A.	5	6	-15.15	-14.85	VDC	16

See footnote on next page.

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TABLE VII. Line regulation test - Continued.

Test cond	Parameter	1/ Input voltage ± 1%	Item EL02 Curr. +0, -2%	Item EL01-1 Curr. +0, -2%	Item EL01-2 Curr. +0, -2%	Connect 1 to point	Connect 2 to point	Initial limits 25 ± 5°C mounting surface temperature Min	Initial limits 25 ± 5°C mounting surface temperature Max	Unit	Test fig
D13	108VDC Line regulation	108VDC	5A.	0A.	0A.	7	8	5.00	5.10	VDC	16
D14	108VDC Line regulation Maximum load	108VDC	50A.	11A.	11A.						
D15	202VDC Line regulation	202VDC	5A.	0A.	0A.						
D16	202VDC Line regulation Maximum load	202VDC	50A.	11A.	11A.						
D17	108VDC Line regulation Maximum load-auxiliary input	108VDC	50A.	11A.	11A.						
D18	202VDC Line regulation Maximum load-auxiliary input	202VDC	50A.	11A.	11A.						
D19	115V Line regulation Maximum load-47Hz	115VRMS	50A.	11A.	11A.						
D20	115V Line regulation Maximum load-63Hz	115VRMS	50A.	11A.	11A.	7	8	5.00	5.10		
D21	108VDC Line regulation	108VDC	5A.	0A.	0A.	3	4	14.85	15.15		
D22	108VDC Line regulation	108VDC	50A.	11A.	11A.						
D23	202VDC Line regulation	202VDC	5A.	0A.	0A.						
D24	202VDC Line regulation	202VDC	50A.	11A.	11A.	3	4	14.85	15.15		
D25	108VDC Line regulation	108VDC	5A.	0A.	0A.	5	6	-15.15	-14.85		
D26	108VDC Line regulation	108VDC	50A.	11A.	11A.						
D27	202VDC Line regulation	202VDC	5A.	0A.	0A.						
D28	202VDC Line regulation	202VDC	50A.	11A.	11A.	5	6	-15.15	-14.85	VDC	16

1/ See figure 16, notes 5 and 6 for connection of input power. The root means square (RMS) voltage is measured line-to-line.

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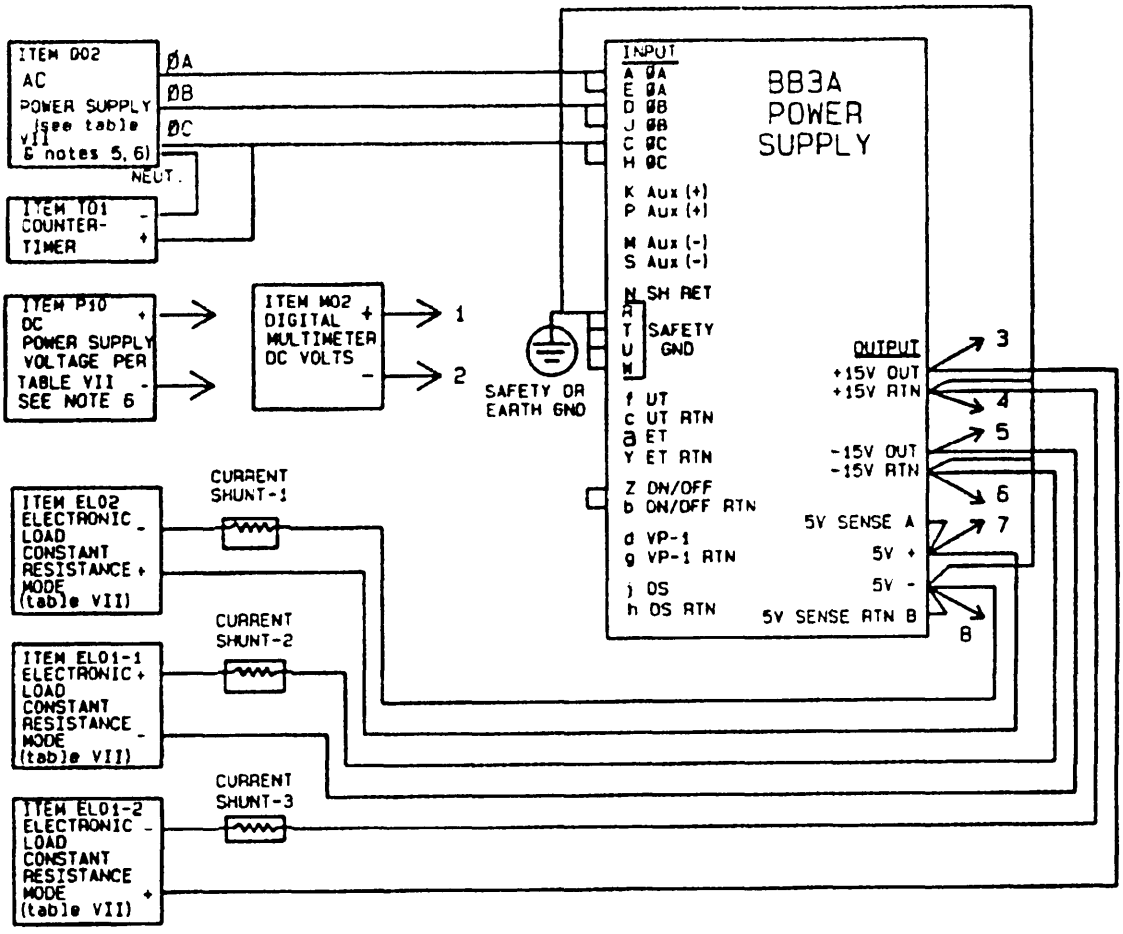


FIGURE 16. Line regulation test circuit for table VII.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 1=75A/50mV; 2,3=50A/50mV.

4. Monitor the shunt voltages with an item V04 voltmeter to ensure that the load currents are within specification.

5. The frequency of the input voltage shall be 60 ± 0.3 Hz for test conditions D1-D12, 47 ± 0.3 Hz for test conditions D19, and 63 ± 0.3 Hz for test condition D20.

6. Where RMS input voltage is specified (test conditions D1-D12, D19, and D20), the AC power supply connected as shown shall be used. For test conditions D13-D16 and D21-D28, disconnect the AC power supply and connect item P10 + to input contacts A and E, and - to input contacts D and J. For test conditions D17 and D18, connect item P10 + to input contacts K and P, and - to input contacts M and S. Only one power source shall be connected at any time.

FIGURE 16. Line regulation test circuit for table VII - Continued.

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TABLE VIII. Low input voltage protection and loss of phase protection test.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
E	Low input voltage protection	5.00	5.10	VDC	17
F1	Loss of phase A protection				
F2	Loss of phase B protection				
F3	Loss of phase C protection	5.00	5.10	VDC	17

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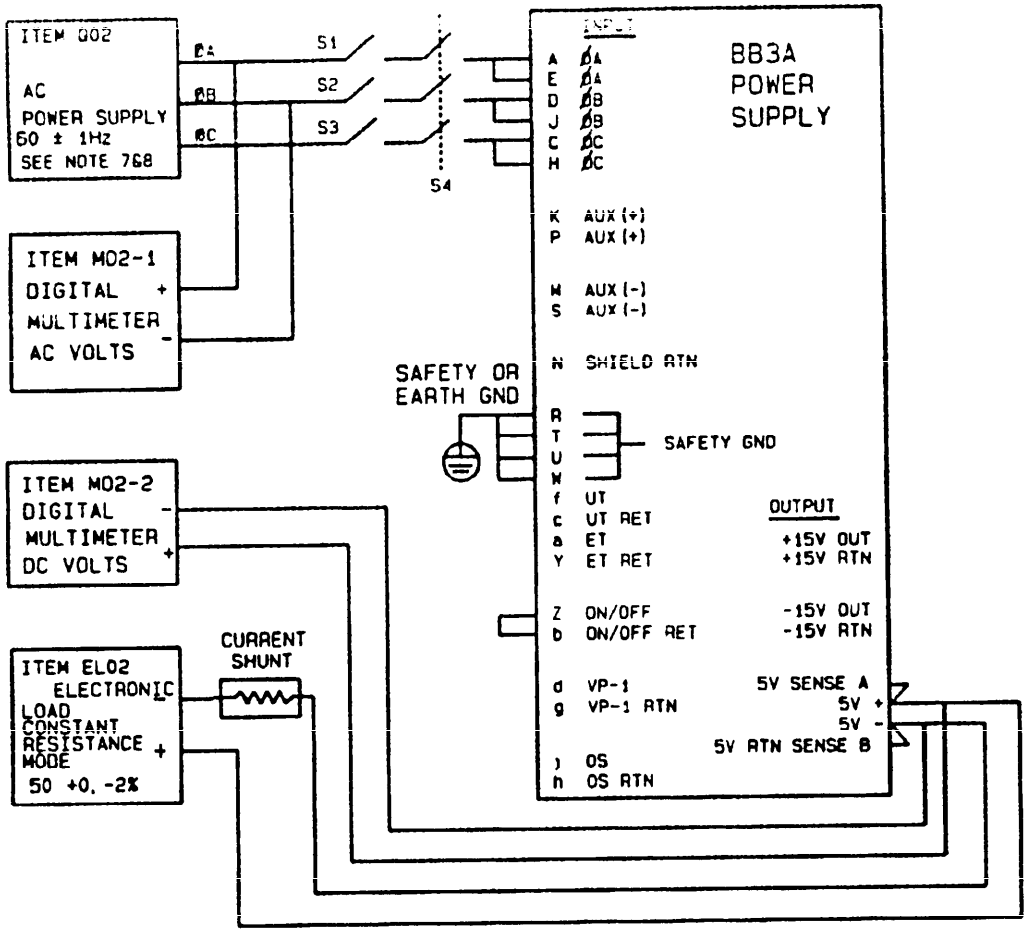


FIGURE 17. Low input voltage protection and loss of phase protection test circuit for table VIII.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 75A/50mV.

4. Monitor the shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. S1 = S2 = S3 = single pole single throw (spst), 160 VRMS, 15 amperes switch.

6. S4 = three pole single throw (3 pst), 160 VRMS, 15 amperes switch.

7. For test condition E, connect the circuit as shown with switches S1, S2, S3, and S4 closed. The item Q02 voltage shall be monitored with item M02-1. Open switch S4 and adjust the item Q02 voltage to 0 VRMS. Close switch S4 and slowly (less than 5 volts per second) increase the item Q02 voltage to 115 VRMS ± 2 percent line-to-line, then decrease it to 0 VRMS at the same rate. Open switch S4, and adjust the item Q02 voltage to 115 VRMS ± 2 percent line-to-line. Close switch S4. The item M02-2 voltage reading shall meet the test condition E limits.

8. Test conditions F1, F2, and F3 shall be performed as follows:

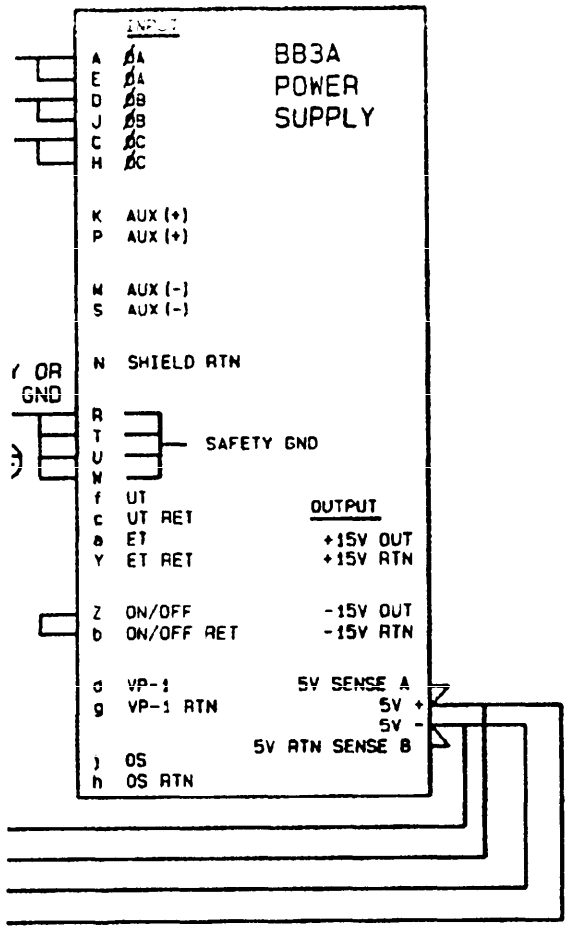
(a) For test condition F1, connect the circuit as shown with item Q02 voltage set to 115 VRMS ± 2 percent and with switches S1, S2, S3, and S4 closed. Open switch S1 for a minimum of 5 seconds, then close switch S1. The item M02-2 voltage reading shall meet the test condition F1 limits. The power supply is not required to meet the regulation limits during the loss of phase period.

(b) For test condition F2, repeat (a), except use switch S2 and test condition F2 limits.

(c) For test condition F3, repeat (a), except use switch S3 and test condition F3 limits.

FIGURE 17. Low input voltage protection and loss of phase protection test circuit for table VIII - Continued.

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age protection and loss of phase
t circuit for table VIII.

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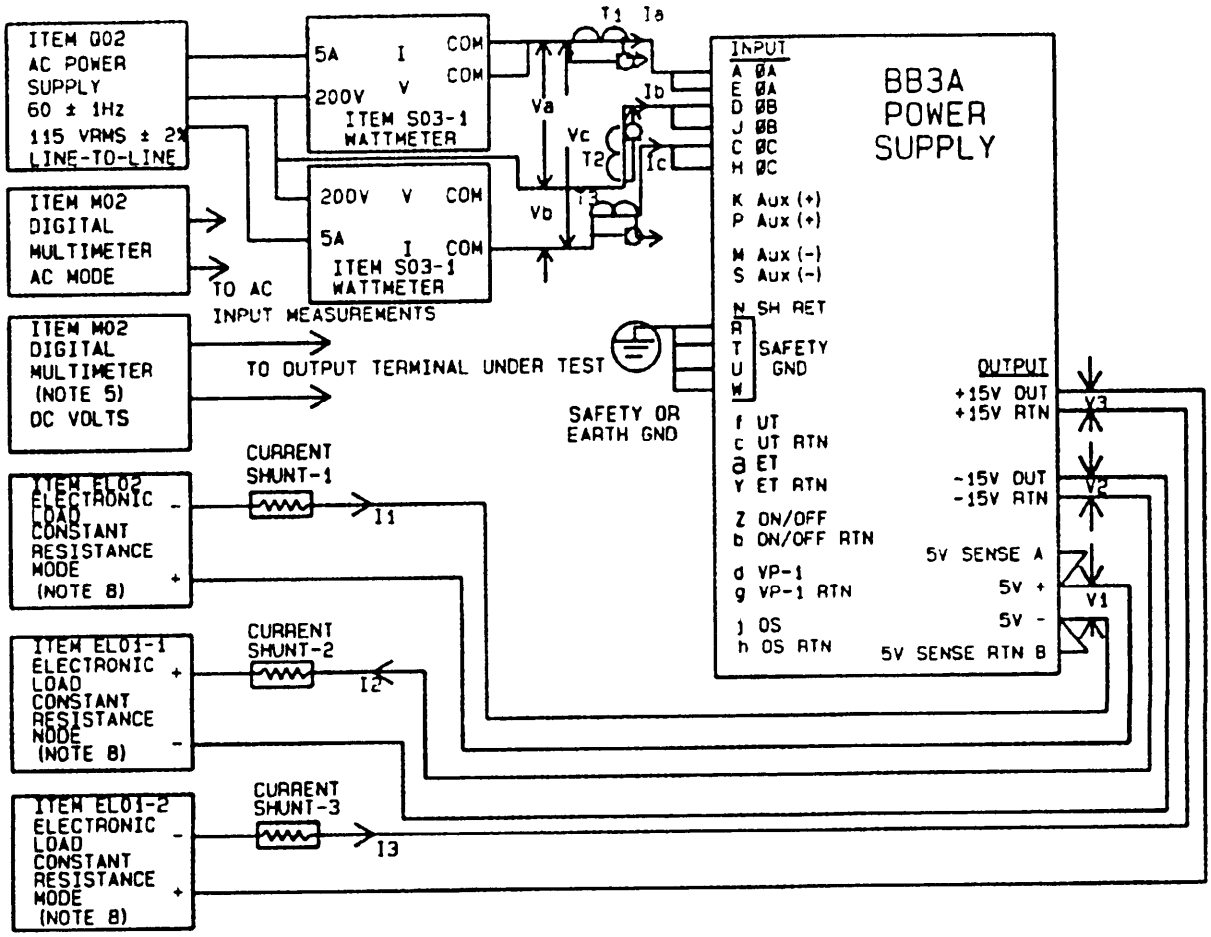


FIGURE 18. Efficiency, power factor, and input current balance test circuit for table IX.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 1=75A/50mV; 2,3=50A/50mV.

4. Monitor the shunt voltages with an item V04 voltmeter to ensure that the load currents are within specification.

5. Only one M02 is required for G1 and G2.

6. Make the M02 measurement directly at the power supply mating connector.

7. T1, T2, and T3 are Pearson Model 411 current transformers.

8. The electronic loads shall be set: EL02 to 50 amperes, EL01-1 to 11 amperes and EL01-2 to 11 amperes; except for test condition G2 where the loads shall be set: EL02 to 5 amperes, EL01-1 to 1 ampere and EL01-2 to 1 ampere. All settings shall have a tolerance of +0, -2 percent.

FIGURE 18. Efficiency, power factor, and input current balance test circuit for table IX - Continued.

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TABLE X. Inrush current test.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
K1	Inrush current-AC inputs	-	55	Apk	19,20
K2	Inrush current-auxiliary input	-	55	Apk	19,20

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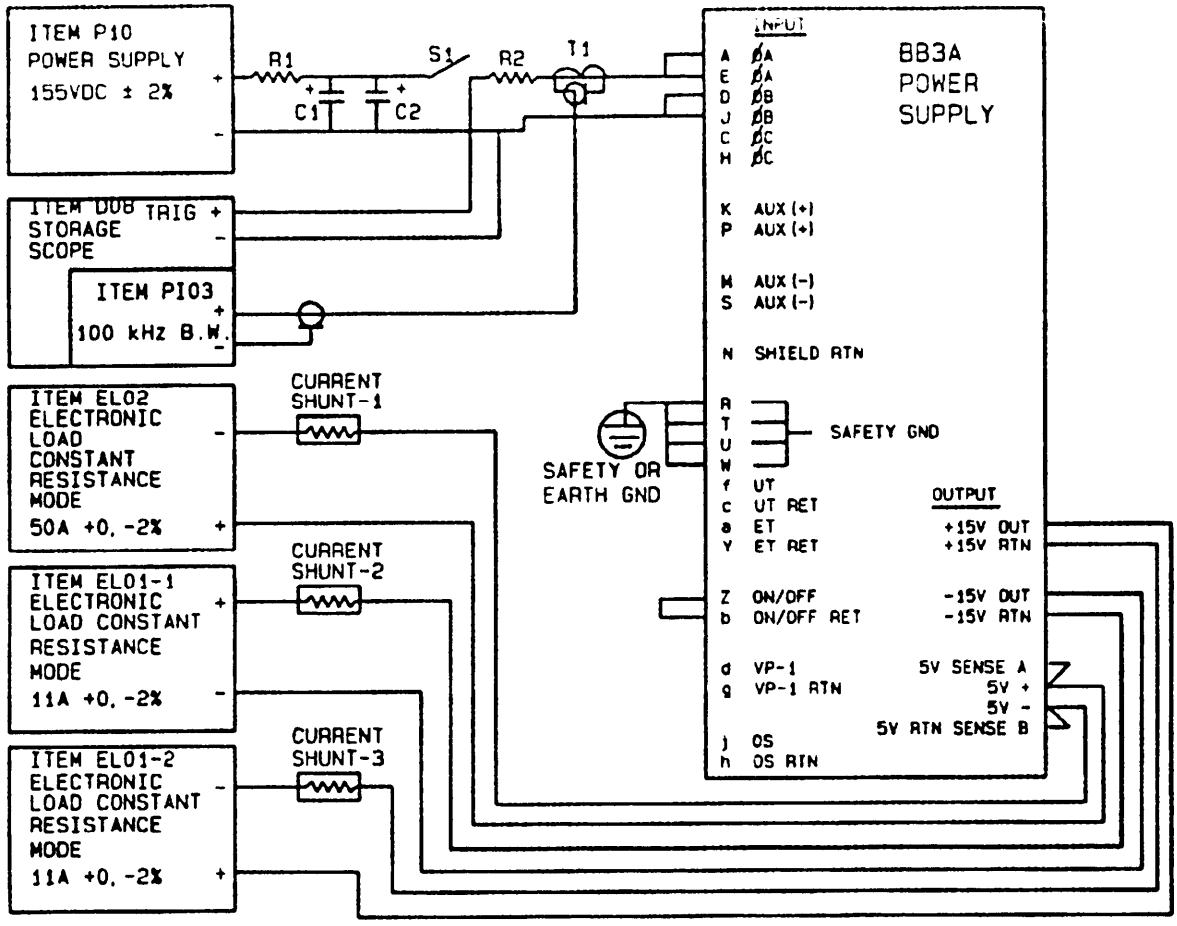


FIGURE 19. Inrush current test circuit for table X.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 1=75A/50mV, 2,3=50A/50mV.

4. Monitor the shunt voltages with an item V04 voltmeter to insure that the load currents are within specification.

5. S1 = spst, 225 VDC, 15 amperes switch.

6. T1 = Pearson Model 411 current transformer.

7. R1 = 1 Ohm ± 5 percent, 40 watt resistor.

8. R2 = 0.1 Ohm ± 5 percent, (represents maximum allowable series resistance).

9. C1 = C2 = 330 uF -15 percent, +50 percent, equivalent series resistance (ESR) of 0.4 Ohm max, 250 VDC (Sprague 80D331P250JE5D or equivalent).

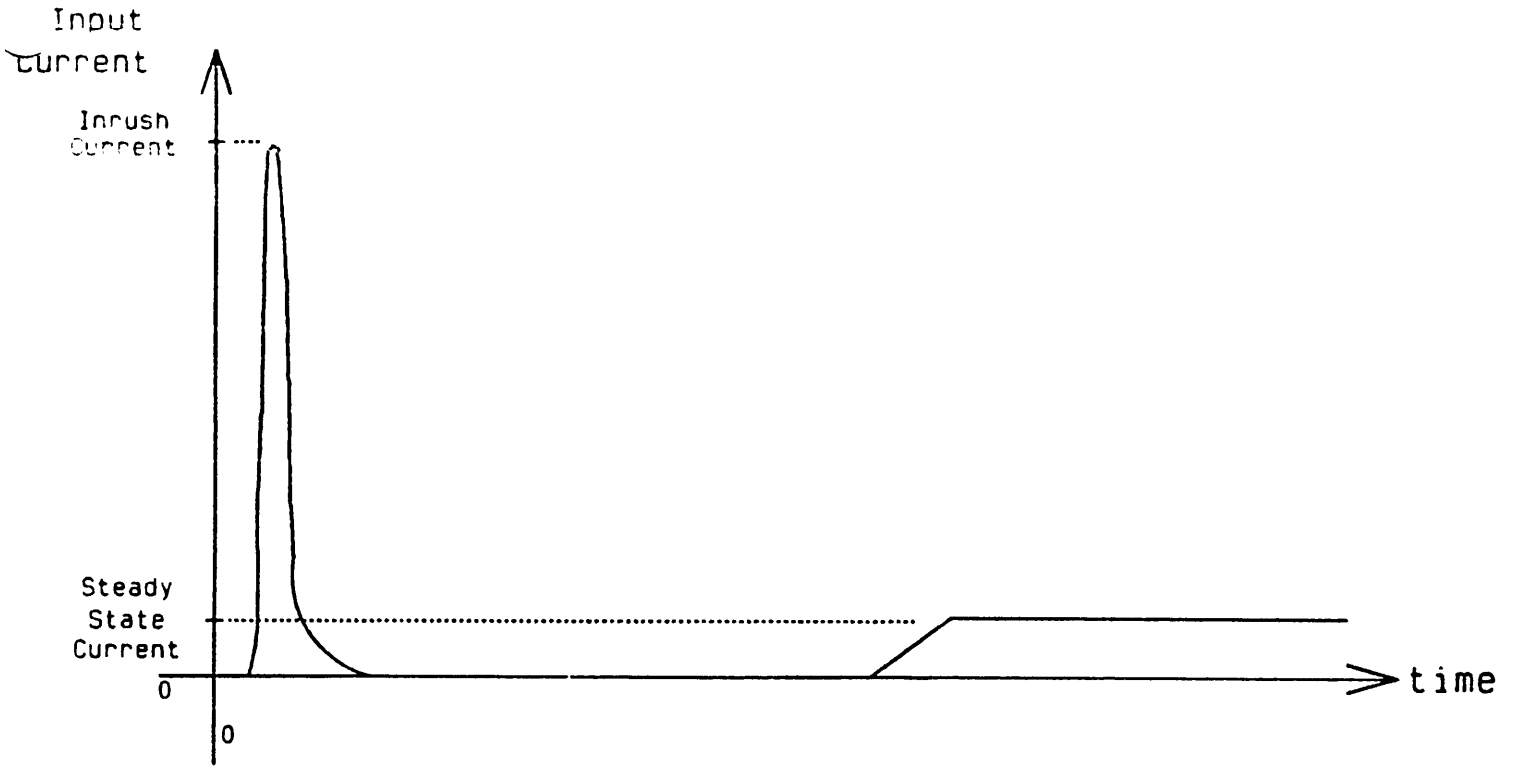
10. For test condition K1, connect item P10 to the AC inputs as shown. For test condition K2, connect item P10 to the auxiliary inputs: + to input contacts K and P, and - to input contacts M and S.

11. Apply power to the power supply under test by closing switch S1 with item P10 energized. Neither the power "OFF" period nor the power "ON" period shall be less than one second.

12. Keep the lead length between C1, C2, S1, and the power supply under test as short as possible.

FIGURE 19. Inrush current test circuit for table X - Continued.

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NOTE:

1. The steady state current is not observable using the figure 19 test circuit, but is shown to indicate its relationship to the inrush current.

FIGURE 20. Inrush current test waveform for table X.

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TABLE XI. Remote sense compensation test.

Test cond	Parameter	VP-1 Input contact d to g	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
			Min	Max		
L1	Remote sense compensation, 5.05V	Open	5.00	5.10	VDC	21
L2	Remote sense compensation, 5.25V	Shorted	5.20	5.30	VDC	21

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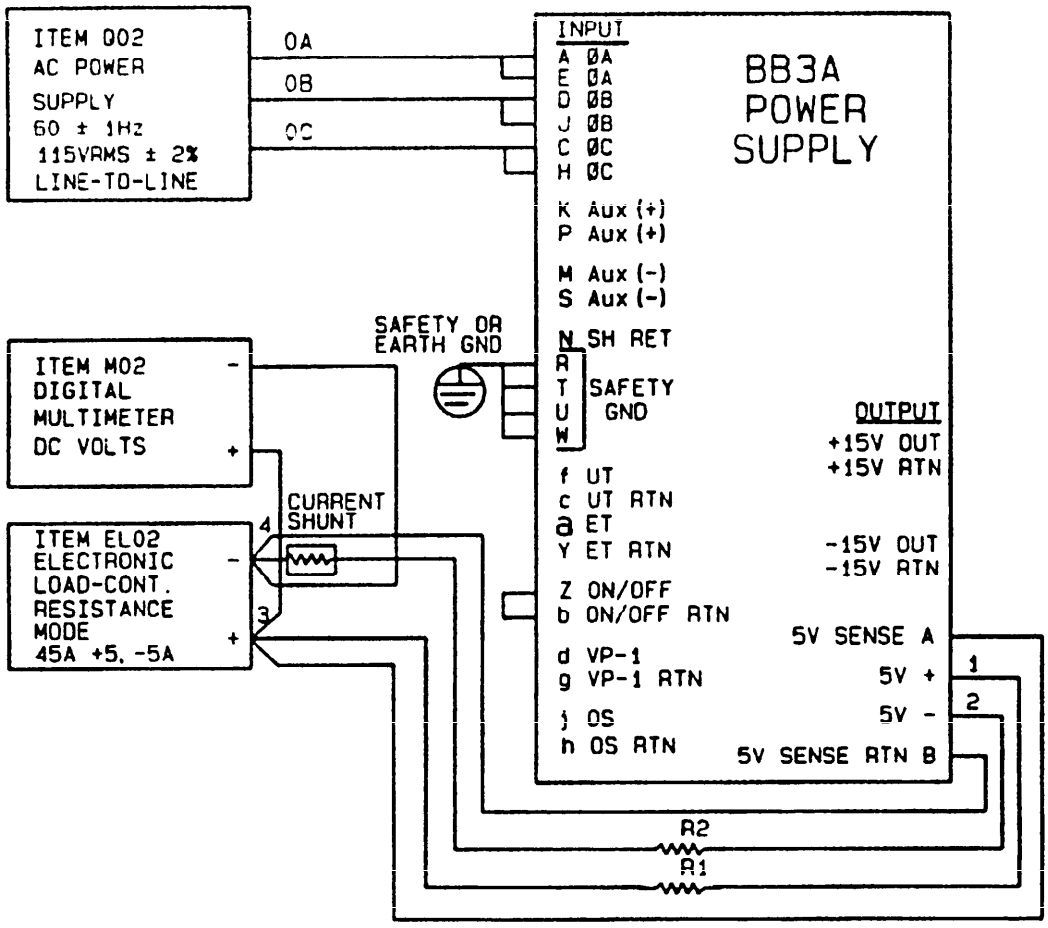


FIGURE 21. Remote sense compensation test circuit for table XI.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 75A/50mV.

4. Monitor the shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

6. The sense leads shall be less than 18 inches long.

7. For test condition L1: R1 and R2, and the current shunt shall be the total resistance (including contact and lead resistance) to cause a 0.358 + .035, -.000 volt drop when voltage between points 1 and 2 is compared to the voltage between points 3 and 4 (V1-2 minus V3-4 = 0.358 + .035, -.000 volts). The load current may be varied between the indicated values to assist in obtaining the correct voltage drop. For test condition L2, the drop across R1, R2, and the current shunt shall be 0.158 +.030, -.000 volts.

FIGURE 21. Remote sense compensation test circuit for table XI - Continued.

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TABLE XII. Open sense lead and reversal test.

Test cond	Parameter	Open switch	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
			Min	Max		
M1	Open "+" sense lead transient amplitude	S1	-400	100	mV	22,23
M2	Open "+" sense lead transient time	S1	0	100	ms	22,23
M3	Open "+" sense lead voltage	S1	4.80	5.56	VDC	22,23
M4	Open "-" sense lead transient amplitude	S2	-400	100	mV	22,23
M5	Open "-" sense lead transient time	S2	0	100	ms	22,23
M6	Open "-" sense lead voltage	S2	4.80	5.56	VDC	22,23
M7	Open "+" and "-" sense lead voltage	S1&S2	4.80	5.56	VDC	22,23
M8	Sense lead reversal protection	note 1/	5.00	5.10	VDC	22,23

1/ Both S1 and S2 are closed.

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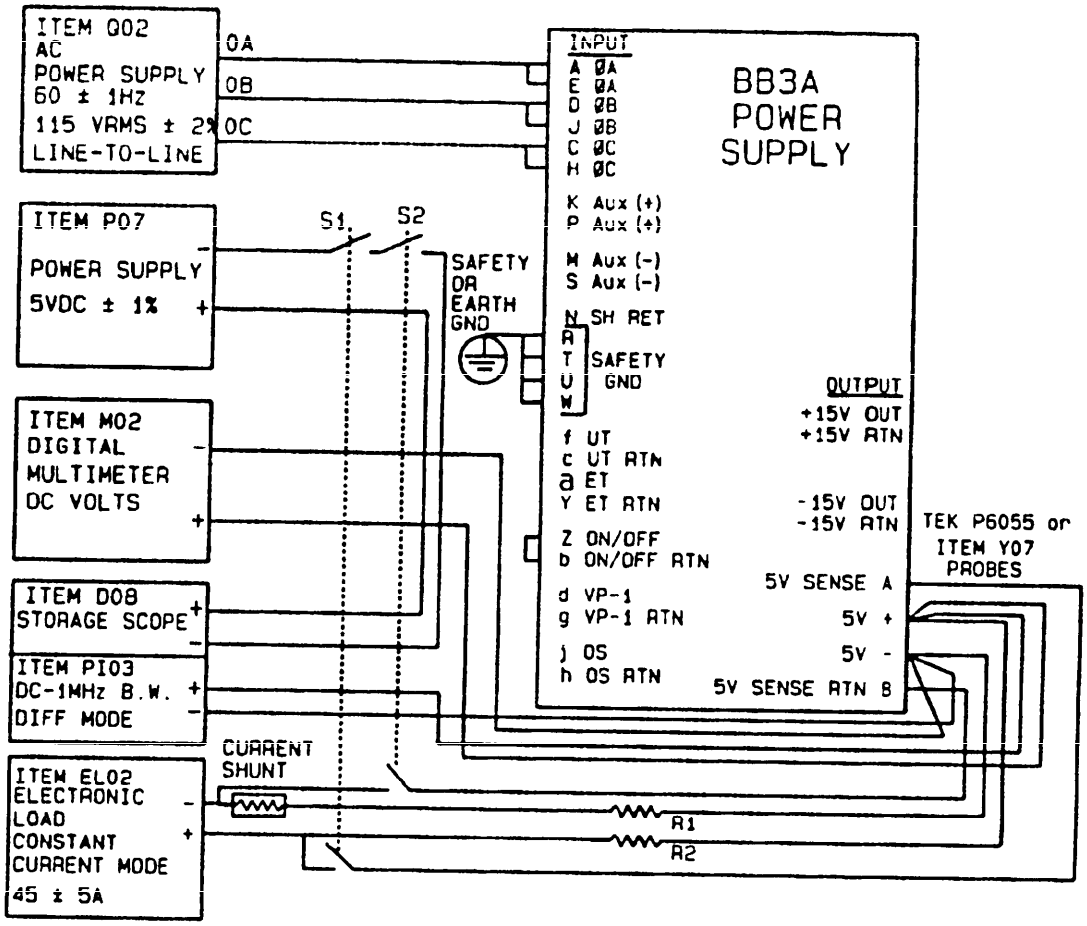


FIGURE 22. Open sense lead and reversal test circuit for table XII.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 75A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. S1 = S2 = double pole single throw (dpst) switch.

6. At the option of the test facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two of the AC inputs maybe used instead of the AC input specified.

7. For test conditions M1 through M8: R1, R2, and the current shunt shall be the total resistance (including lead and contact resistance) to cause a load lead voltage drop of 0.358 + 0.035, -.000 VDC. The load lead voltage drop is the difference between the voltage directly at the BB3A output voltage contacts and at the sense lead connection points. The load current may be varied between the indicated values to assist in obtaining the correct voltage drop.

8. Test conditions M1-M3 shall be performed as follows:

(a) Connect the circuit per figure 22, close switches S1 and S2, and apply power.

(b) Open switch S1.

(c) The maximum transient overshoot or undershoot shall not exceed the limits of M1. The power supply shall completely recover to and remain within the voltage limit of M3 and within the time limit of M2 (see figure 23). Test condition M3 voltage shall be measured with item M02.

9. For test conditions M4-M6, repeat the procedure of note 8: S2 shall be opened while S1 remains closed.

10. For test condition M7, open both S1 and S2 and measure the steady-state voltage with item M02.

FIGURE 22. Open sense lead and reversal test circuit for table XII - Continued.

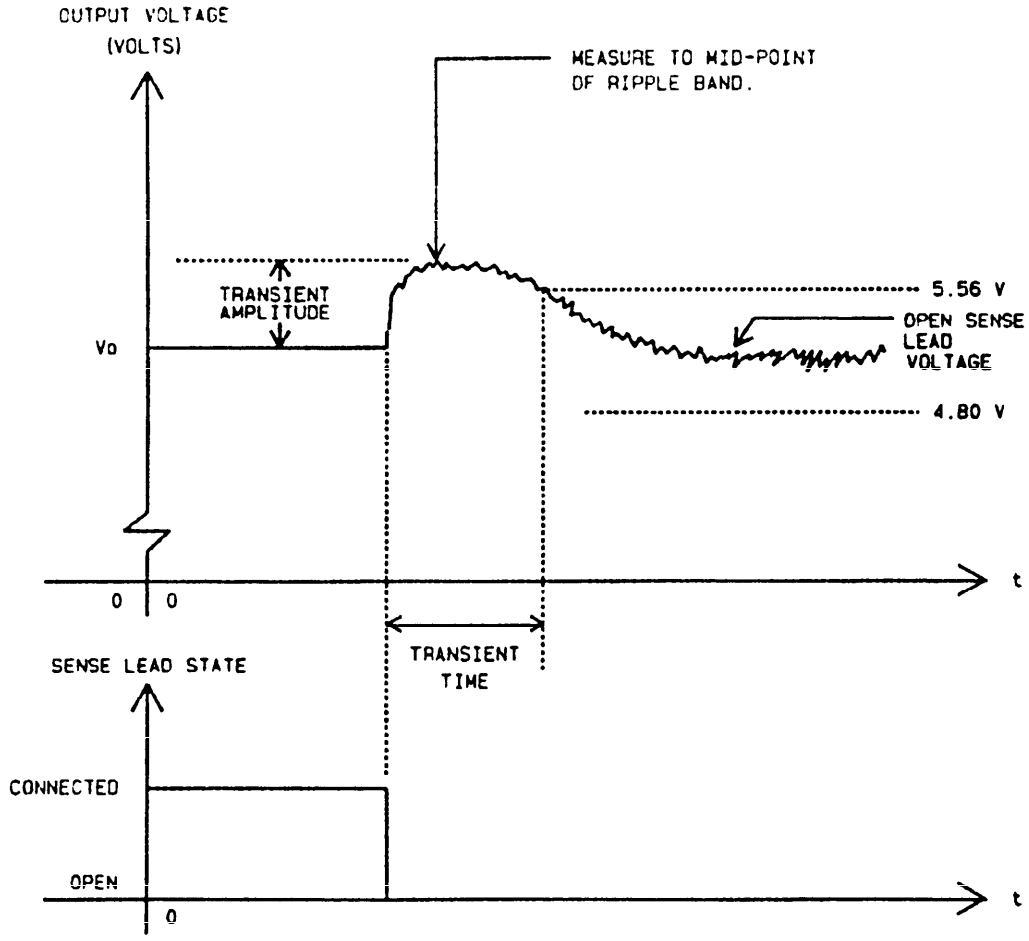
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NOTES: - Continued

11. For test condition M8, remove power reverse the sense leads, (that is, connect 5V sense to 5V- and 5V sense RTN to 5V+). Apply power and leave connected for a minimum of 5 seconds. Then remove power and connect correctly. Finally, apply power and measure the output voltage with item M02 connected across the item ELO2 + and - terminals (at the sense lead connection points).

FIGURE 22. Open sense lead and reversal test circuit for table XII - Continued.

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NOTES:

1. Transient amplitude is defined as the maximum change in voltage after opening the sense lead.

2. Transient time will be zero if the voltage stays within the 4.80 volt to 5.56 volt limits.

3. Transient amplitude may be an overshoot (as shown) or an undershoot.

FIGURE 23. Open sense lead and reversal test definitions for table XII.

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TABLE XIII. Overload and short circuit protection test.

Test cond	Parameter	Measurement device	Connect 1 to	Connect 2 to	Initial limits 25 ± 5°C mounting surface temperature Min	Initial limits 25 ± 5°C mounting surface temperature Max	Unit	Test fig
N1	Overload protection on 5V	shunt & V04	-	-	50.0	62.5	ADC	24,25
N2	Overload protection on +15V	shunt & V04	-	-	11.0	13.8	ADC	24.25
N3	Overload protection on -15V	shunt & V04	-	-	11.0	13.8	ADC	24.25
N4	Short circuit recovery on 5V	Item D08	3	4	0	100	ms	24,25
N5	Short circuit overshoot on 5V	Item D08	3	4	-253	253	mV	24.25
N6	Short circuit recovery on +15V	Item D08	7	8	0	100	ms	24.25
N7	Short circuit overshoot on +15V	Item D08	7	8	-750	750	mV	24.25
N8	Short circuit recovery on -15V	Item D08	5	6	0	100	ms	24.25
N9	Short circuit overshoot on -15V	Item D08	5	6	-750	750	mV	24,25
N10	Short circuit current on 5V	shunt & V04	3	4	0	62.5	ADC	24,25
N11	Short circuit current on +15V	shunt & V04	7	8	0	13.8	ADC	24,25
N12	Short circuit current on -15V	shunt & V04	5	6	0	13.8	ADC	24,25

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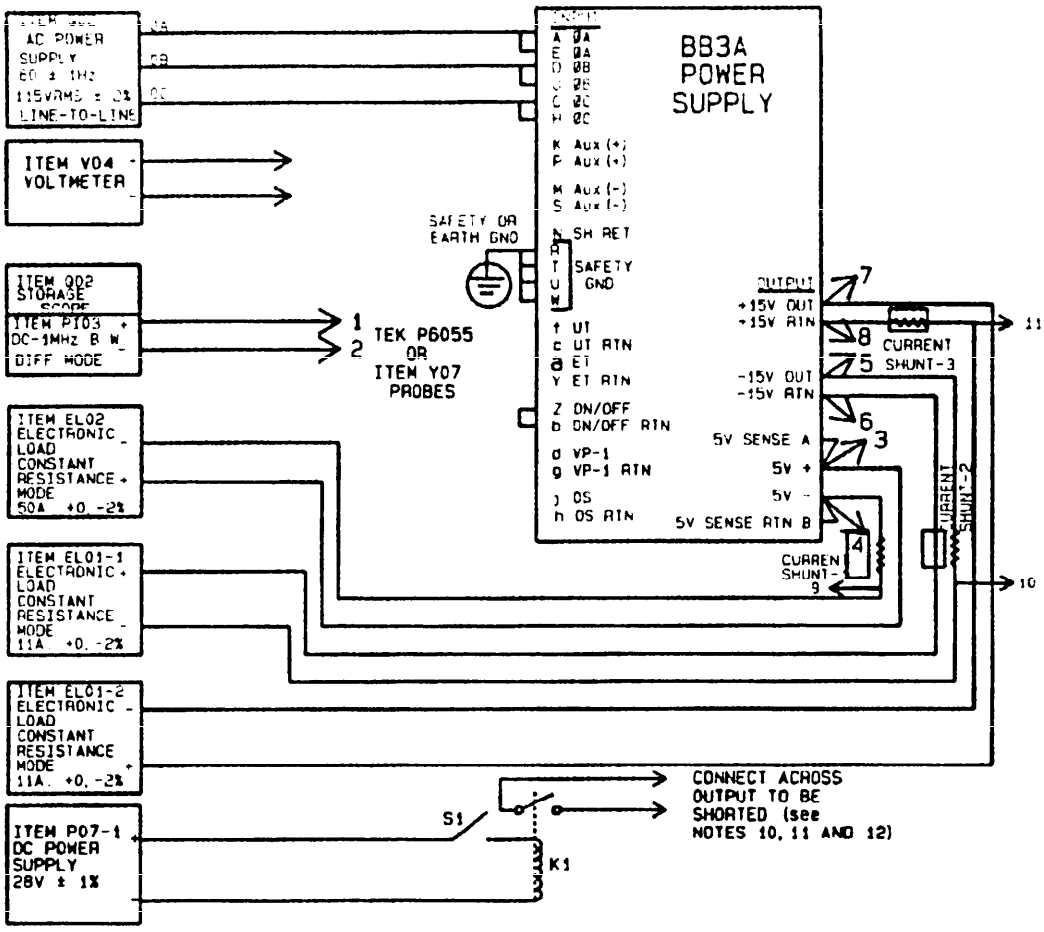


FIGURE 24. Overload and short circuit protection test circuit for table XIII.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 1=75A/50mV, 2,3=50A/50mV.

4. Monitor shunt voltages with an item V04 voltmeter to ensure that the load currents are within specification.

5. S1 = spst switch.

6. K1 = Cutler Hammer MS24185D1 or equivalent contactor.

7. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

8. For test conditions N1-N3: slowly increase the load current, as indicated by the shunt voltage, until the maximum current is obtained (this may be current saturation or prior to current shutdown). This maximum current shall fall within the specified limits. If the current exceeds 64 amperes for test condition N1 or 15 amperes for test conditions N2 and N3, stop this test and do not perform short circuit tests on the failing output.

9. Keep the leads between the power supply under test and the electronic loads as short as possible to minimize lead resistance.

10. Test conditions N4, N5, and N10 shall be performed as follows:

(a) Connect the circuit as shown with the contactor K1 connected between points 3 and 9.

(b) Close the contactor K1 by closing S1 to short the 5V output for a minimum of 10 seconds. The short circuit current, as monitored by current shunt-1, shall meet the limits of test condition N10.

(c) Remove the short and store the resulting transient on the storage scope (see figure 25). The power supply shall automatically recover to and remain within the regulation limits and the recovery time limits of N4. The maximum over- shoot after removal of the short shall meet the limits of N5.

FIGURE 24. Overload and short circuit protection test circuit for table XIII - Continued.

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NOTES: - Continued.

11. For test conditions N6, N7, and N11, repeat the procedure of note 10 for the 15V output with the contactor K1 connected between points 7 and 11.

12. For test conditions N8, N9, and N12, repeat the procedure of note 10 for the -15V output with the contactor K1 connected between points 5 and 10.

FIGURE 24. Overload and short circuit protection test circuit for table XIII - Continued.

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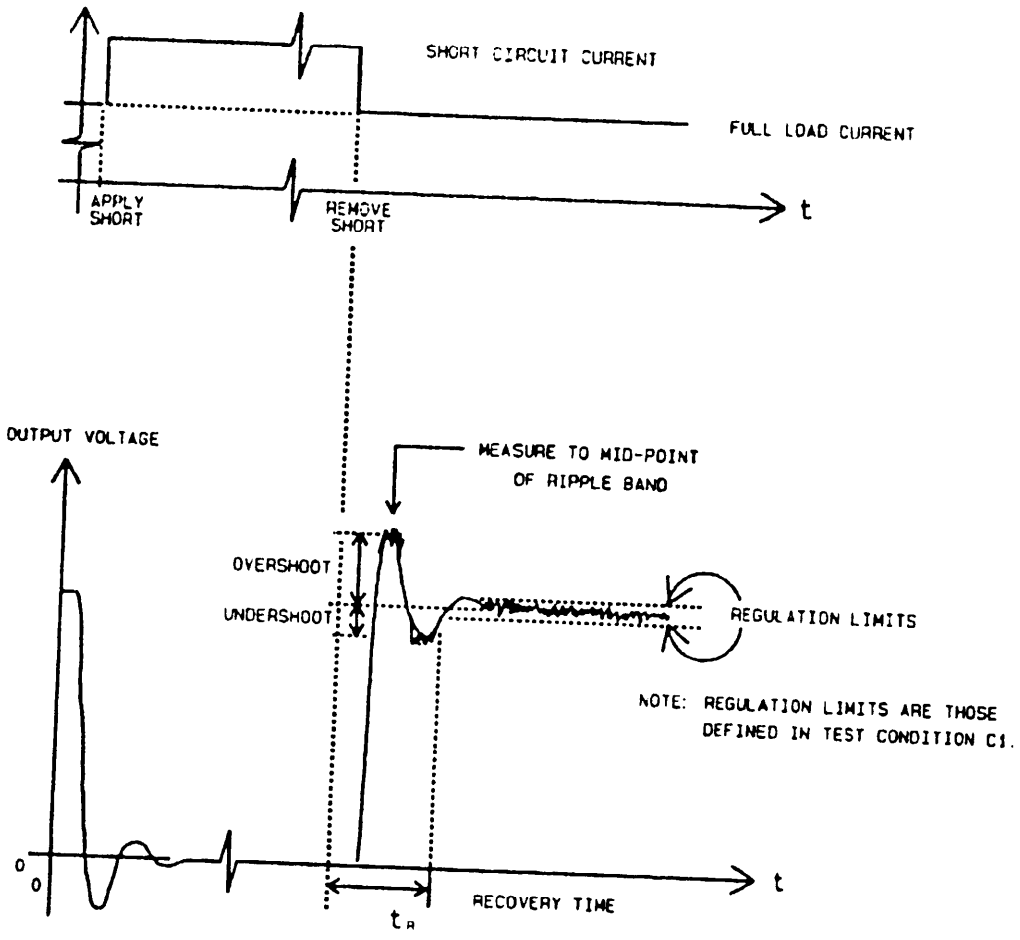


FIGURE 25. Overload and short circuit protection test definitions for table XIII.

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TABLE XIV. Turn on/off transient and turn on time test.

Test cond	Parameter	Connect 1 to	Connect 2 to	Initial limits 25 ± 5°C mounting surface temperature Min	Initial limits 25 ± 5°C mounting surface temperature Max	Unit	Test fig
P1	Turn on transient on 5V	4	3	-253	253	mV	26,28
P2	Turn on time on 5V	4	3	15	250	ms	26,28
P3	Turn off transient on 5V	4	3	-253	253	mV	26,28
P4	Turn on transient on +15V	8	7	-750	750	mV	26,28
P5	Turn on time on +15V	8	7	15	250	ms	26,28
P6	Turn off transient on -15V	8	7	-750	750	mV	26,28
P7	Turn on transient on -15V	5	6	-750	750	mV	26,28
P8	Turn on time on -15V	5	6	15	250	ms	26,28
P9	Turn off transient on -15V	5	6	-750	750	mV	26,28
P10	Control turn on transient on 5V	n/a	n/a	-253	253	mV	27,28
P11	Control turn on time	n/a	n/a	15	250	ms	27,28
P12	Control turn off transient	n/a	n/a	-253	253	mV	27,28

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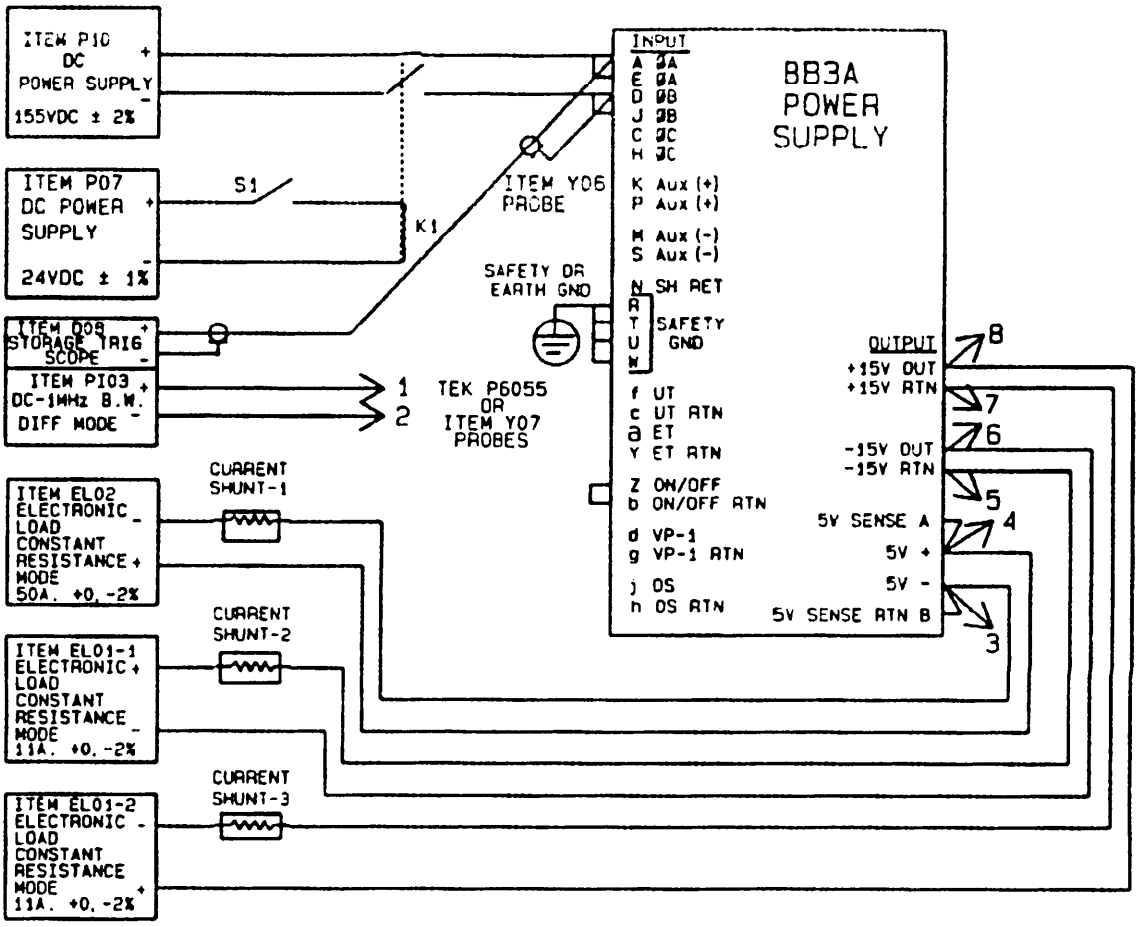


FIGURE 26. Turn on/off transient and turn on time test circuit for table XIV.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 1=75A/50mV; 2,3=50A/50mV.

4. Monitor shunt voltages with an item V04 voltmeter to ensure that the load currents are within specification.

5. S1=spst switch.

6. K1=spst mercury relay, Magnecraft WM35A-24D or equivalent.

7. At the option of the testing facility, an AC input of 115 VRMS ± 2 percent, 60 ± 1 Hz, 3-phase can be applied at the phase A, B, and C inputs.

8. Each test shall be performed several times to ensure measurement of the highest transient and longest times.

FIGURE 26. Turn on/off transients, turn on time test circuit for table XIV - Continued.

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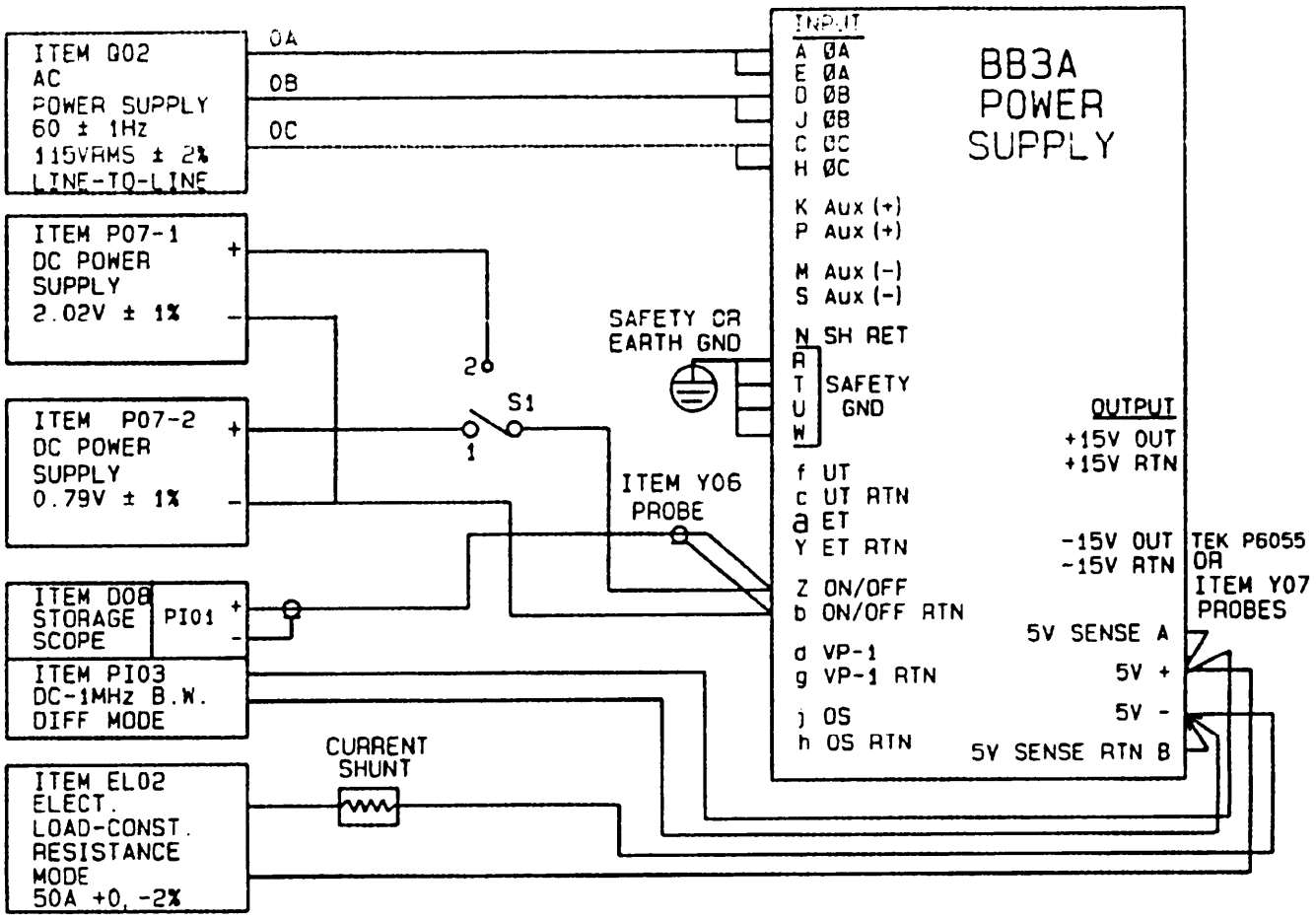


FIGURE 27. Control turn on/off transient test circuit for table XIV.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 75A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. S1=single pole double throw (spdt) switch.

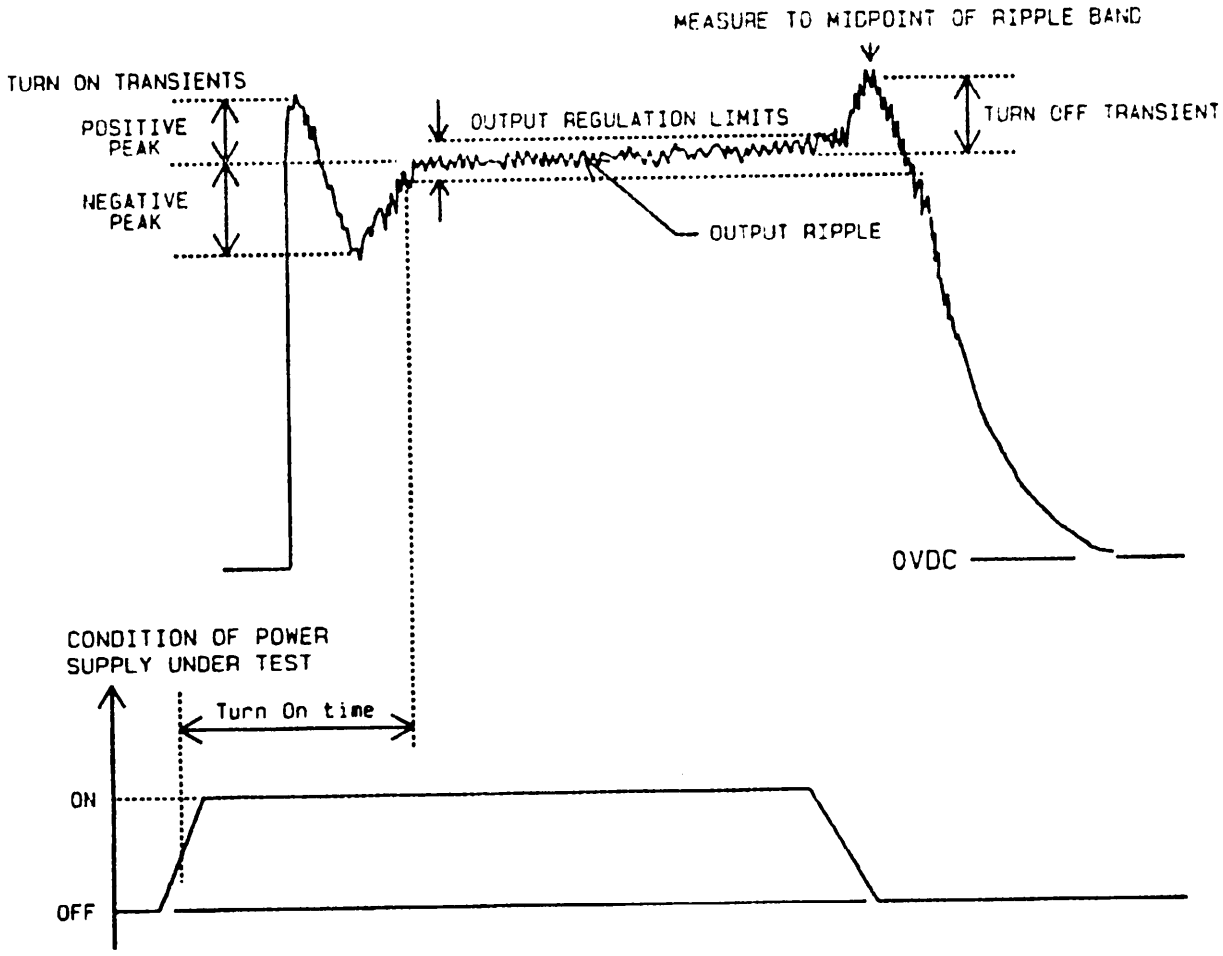
6. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected to any two of the AC inputs can be used instead of the AC input specified.

7. Each test shall be performed several times to ensure measurement of the highest transient and longest time.

8. The power supply shall be turned on when switch S1 is switched from position 2 to 1 and turned off when switch S1 is switched from position 1 to 2.

FIGURE 27. Control turn on/off transient test circuit for table XIV - Continued.

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NOTES:

1. For turn on transient, record the highest amplitude positive or negative peak.

2. The test conditions P1-P9 turn on time is the time from when the input power is applied to the power supply until the output recovers to and remains within the regulation limits. The regulation limits shall be test condition C1 limits for +5V output, C7 limits for the +15V output and C11 for the -15V output (at the applicable temperature).

3. The test conditions P10-P12 control turn on time is defined as the time from when voltage is applied to the ON/OFF input contacts until the output recovers to and remains within the regulation limits (see note 2).

FIGURE 28. Turn on and turn off waveform for table XIV.

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TABLE XV. Ripple and spectral line test.

Test cond	Parameter	Measurement device (Item D08)	AC input line-to line voltage	Measurement frequency range (3dB pt)		Connect		Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
				Low 1/	High	1 to	2 to	Min	Max		
Q1	Ripple, total on 5V	Item P103	115 VRMS ± 2%	10	1 MHz	7	8	0	65	mVpp	29
Q2	Ripple, total on +15V	Item P103	〃	〃	1 MHz	3	4	0	25	〃	〃
Q3	Ripple, total on -15V	Item P103	115 VRMS ± 2%	10	1 MHz	5	6	0	25	〃	〃
Q4	Ripple, low frequency on 5V	Item P103	115 VRMS ± 2%	10	10 kHz	7	8	0	50	〃	〃
Q5	Ripple, high frequency on 5V	Item P103	〃	10 kHz	1 MHz	7	8	0	65	mVpp	29
Q6	Spectral line, +15V	NA	〃	50 Hz	10 kHz	3	4	0	.056	mVRMS $\sqrt{Hz}$	30
Q7	Spectral line, +15V	NA	〃	10 kHz	10 MHz	3	4	0	0.10 2/	〃	〃
Q8	Spectral line, -15V	NA	〃	50 Hz	10 kHz	5	6	0	.056	mVRMS	〃
Q9	Spectral line, -15V	NA	〃	10 kHz	10 MHz	5	6	0	0.10 2/	$\sqrt{Hz}$	〃
Q10	Ripple, total, 20MHz BW	Item P101	〃	10 Hz	20 MHz	7	8	0	300	mVpp	29
Q11	Ripple, total, 20MHz BW	Item P101	〃	10 Hz	20 MHz	3	4	0	100	〃	〃
Q12	Ripple, total, 20MHz BW	Item P101	〃	10 Hz	20 MHz	5	6	0	100	〃	〃
Q13	Ripple, total, 5V, OVL	Item P102	〃	10 Hz	5 MHz	7	8	0	200	〃	〃
Q14	Ripple, total, 5V, no load	Item P102	115 VRMS ± 2%	10 Hz	5 MHz	7	8	0	130	〃	29

- See footnotes at end of table.

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TABLE XV. Ripple and spectral line test - Continued.

Test cond	Parameter	Measurement device (Item D08)	AC input line-to line voltage	Measurement frequency range (3dB pt)		Connect		Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
				Low 1/	High	1 to	2 to	Min	Max		
Q15	Ripple, total on +15V, OVL	Item P102	115 VRMS ± 2%	10 Hz	5 MHz	3	4	0	65	mVpp	29
Q16	Ripple, total on -15V, OVL	Item P102	115 VRMS ± 2%	10 Hz	5 MHz	5	6	0	65	mVpp	
Q17	Ripple, total, low input	Item P102	86 VRMS ± 1%	10 Hz	5 MHz	7	8	0	130	mVpp	
Q18	Ripple, total, low input	Item P102		10 Hz	5 MHz	3	4	0	25	mVpp	
Q19	Ripple, total, low input	Item P102	86 VRMS ± 1%	10 Hz	5 MHz	5	6	0	25	mVpp	
Q20	Ripple, total, high input	Item P102	144 VRMS ± 1%	10 Hz	5 MHz	7	8	0	130	mVpp	
Q21	Ripple, total, high input	Item P102		10 Hz	5 MHz	3	4	0	25	mVpp	
Q22	Ripple, total, high input	Item P102	144 VRMS ± 1%	10 Hz	5 MHz	5	6	0	25	mVpp	
Q23	Ripple, total, -5.05V	Item P102	115 VRMS ± 2%	10 Hz	5 MHz	7	8	0	130	mVpp	29
Q24	Spectral line, 5V	NA	115 VRMS ± 2%	50 Hz	10 MHz	7	8	0	10	mVRMS / √Hz	30

1/ When using P101 or P102, use the AC coupling to obtain the 10 Hz lower 3dB point.

2/ Test condition Q7 and Q9 limits will be as specified except between the frequencies of 1 MHz to 3 MHz where the maximum limits will be 0.18 mVRMS/√Hz.

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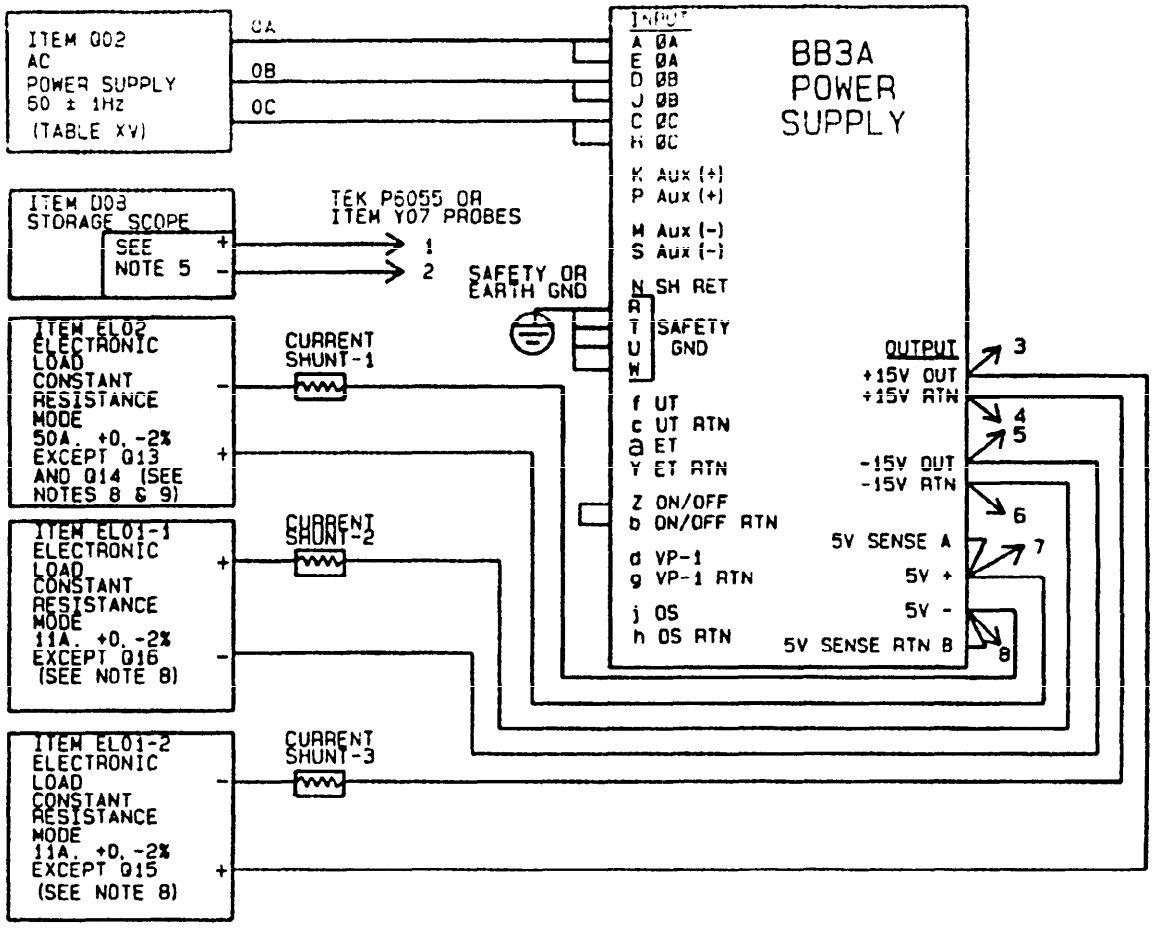


FIGURE 29. Ripple test circuit for table XV.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 1=75A/50mV; 2,3=50A/50mV.

4. Monitor shunt voltages with an item V04 voltmeter to ensure that the load currents are within specification.

5. The smaller box within item D08 indicates a plug-in. See table XV for the plug-ins to be used. Use items PI02 and PI03 in the differential mode.

6. For test conditions Q1-Q5, and Q10-Q23, measure the total peak-to-peak ripple and noise observed on the oscilloscope within the frequency range per table XV (including all spikes, hash periodic and aperiodic components). The intensity must be turned up adequately to catch all the peaks. The scope trigger should be varied to assure observation of both positive and negative peaks.

7. For test conditions Q10-Q12, measurement shall be made with item D08 scope and item PI01 plug-in (with bandwidth set to 20 MHz). The item Y07 probe shall be used so differential measurements can be made.

8. For test conditions Q13, Q15, and Q16, the ripple measurement is made with the current protection circuit energized. The current on the output being measured shall be increased until one of the following things shall happen:

(a) To the point where the current saturates or begins to drop (in the case of current limiting or foldback).

(b) To just prior to the point where the power supply shuts down (if the power supply shuts down on overcurrent).

(c) Until the output voltage decreases to 95 percent of the value at full load current.

The ripple measurement shall be made at the condition which occurs first. In no case shall the current be increased beyond

FIGURE 29. Ripple test circuit for table XV - Continued.

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NOTES: - Continued.

125 percent of rated load current. For test condition Q13, adjust item EL02; for test condition Q15, adjust item EL01-2; and for test condition Q16, adjust item EL01-1.

9. For test condition Q14, adjust item EL02 to 0 amperes.

10. For test condition Q23, connect point 7 to the safety ground contacts on the input connector and to points 4 and 6.

FIGURE 29. Ripple test circuit for table XV - Continued.

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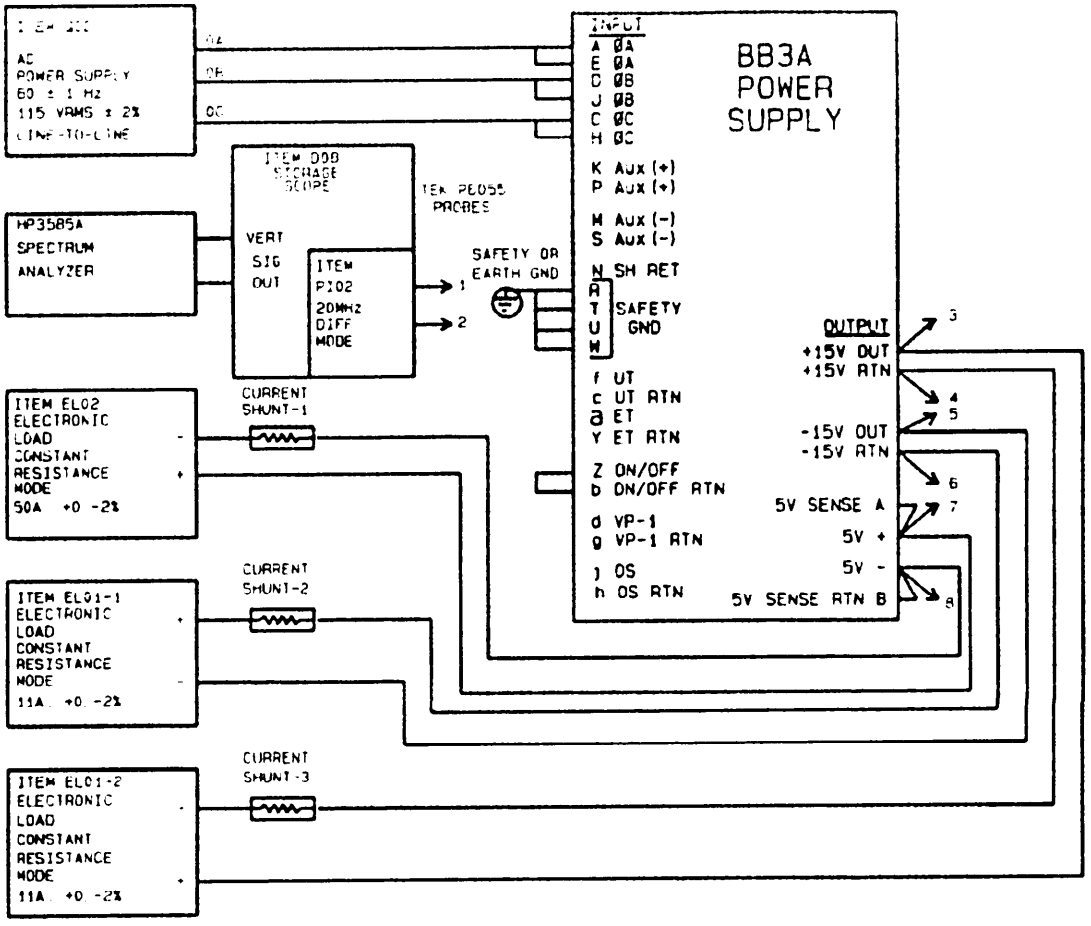


FIGURE 30. Spectral line test circuit for table XV.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 1=75A/50mV; 2,3=50A/50mV.

4. Monitor shunt voltages with an item V04 voltmeter to ensure that the load currents are within specification.

5. For test conditions Q6, Q8, and Q24, sweep the frequency between 50 Hz and 10 kHz using a 30 Hz or 100 Hz bandwidth to find the frequencies of peak spectral lines. Then use a 10 Hz bandwidth to verify that the power supply meets the specified limits.

6. For test conditions Q7, Q9, and Q24, sweep the frequency between 10 kHz and 10 MHz using a 1 kHz or 3 kHz bandwidth to find the frequencies to peak spectral lines. Then use a 30 Hz bandwidth to verify that the power supply meets the specified limits.

7. Do not apply bandwidth correction to the Q6-Q9 and Q24 test limits.

8. To measure the spectral line ripple differentially, item D08 must be used and connected as shown. Item PI02 should be used to provide a voltage gain if required (the vertical signal output is 0.5 volt per division into 1 MOhm and if item PI02/P6055 measurement combination volt/division is set at 50 mV, then the nominal gain would be 0.5/0.05 = 10). The item D08 vertical signal output has a tolerance of ±25 percent and its exact gain at the frequencies of measurement should be checked by using a known voltage amplitude into item PI02. The spectrum analyzer voltage measured shall be corrected by the item PI02 gain before comparing to the table XV limits.

9. Erroneous data may be obtained from 60 Hz power line frequency and its harmonics or nearby spurious frequency sources being measured. A frequency sweep with the power off to the power supply under test (all other applicable equipment on) should be used to identify and exclude all but power supply generated signals.

FIGURE 30. Spectral line test circuit for table XV - Continued.

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TABLE XVI. Dynamic line regulation test.

Test cond	Parameter	Connect 1 to	Connect 2 to	Initial limits 25 ± 5°C mounting surface temperature Min	Initial limits 25 ± 5°C mounting surface temperature Max	Unit	Test fig
R1	Dynamic line regulation on 5V,+21%	7	8	0	152	mV	31,32
R2	Dynamic line ringing on 5V,+21% (2nd half cycle) 1/	7	8	0	76	mV	31,32
R3	Dynamic line regulation on 5V,-21%	7	8	0	152	mV	31,32
R4	Dynamic line ringing on 5V,-21% (2nd half cycle) 1/	7	8	0	76	mV	31,32
R5	Dynamic line regulation on +15V, +21%	3	4	0	450	mV	31,32
R6	Dynamic line ringing on +15V,+21% (2nd half cycle) 2/	3	4	0	225	mV	31,32
R7	Dynamic line regulation on +15V, -21%	3	4	0	450	mV	31,32
R8	Dynamic line ringing on +15V,-21% (2nd half cycle) 2/	3	4	0	225	mV	31,32
R9	Dynamic line regulation on -15V, +21%	5	6	0	450	mV	31,32
R10	Dynamic line ringing on -15V,+21% (2nd half cycle) 3/	5	6	0	225	mV	31,32
R11	Dynamic line regulation on -15V, -21%	5	6	0	450	mV	31,32
R12	Dynamic line ringing on -15V,-21% (2nd half cycle) 3/	5	6	0	225	mV	31,32

1/ Any additional ringing beyond the 2nd half cycle shall be within the regulation limits of test condition D1 and the total ripple envelope of test condition Q1 peak-to-peak limits.
2/ Any additional ringing beyond the 2nd half cycle shall be within the regulation limits of test condition D5 and the total ripple envelope of test condition Q2 peak-to-peak limits.
3/ Any additional ringing beyond the 2nd half cycle shall be within the regulation limits of test condition D9 and the total ripple envelope of test condition Q3 peak-to-peak limits.
4/ +21%--> from 115 VRMS to 139 VRMS: -21%--> from 115 VRMS to 91 VRMS, see fig.32.

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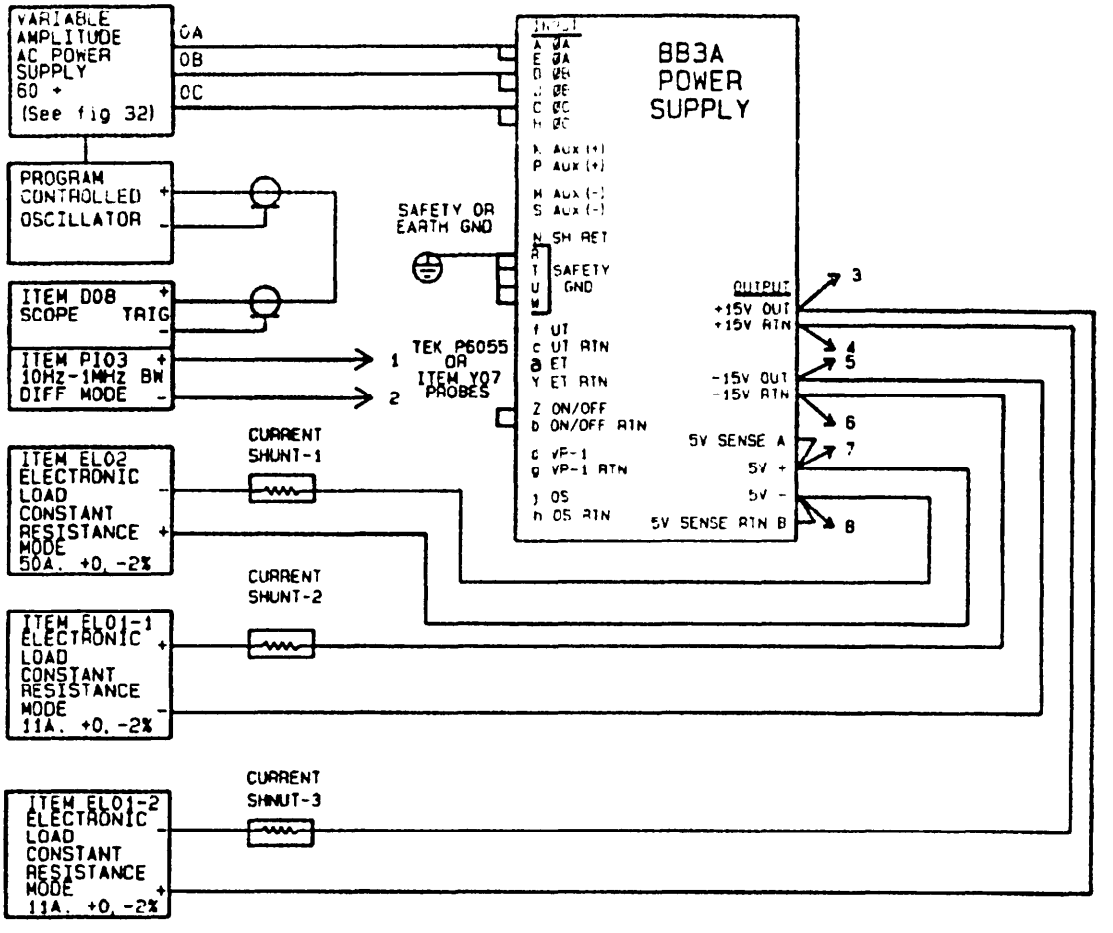


FIGURE 31. Dynamic line regulation test circuit for table XVI.

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NOTES:

1. See table II for output connector contact assignments.

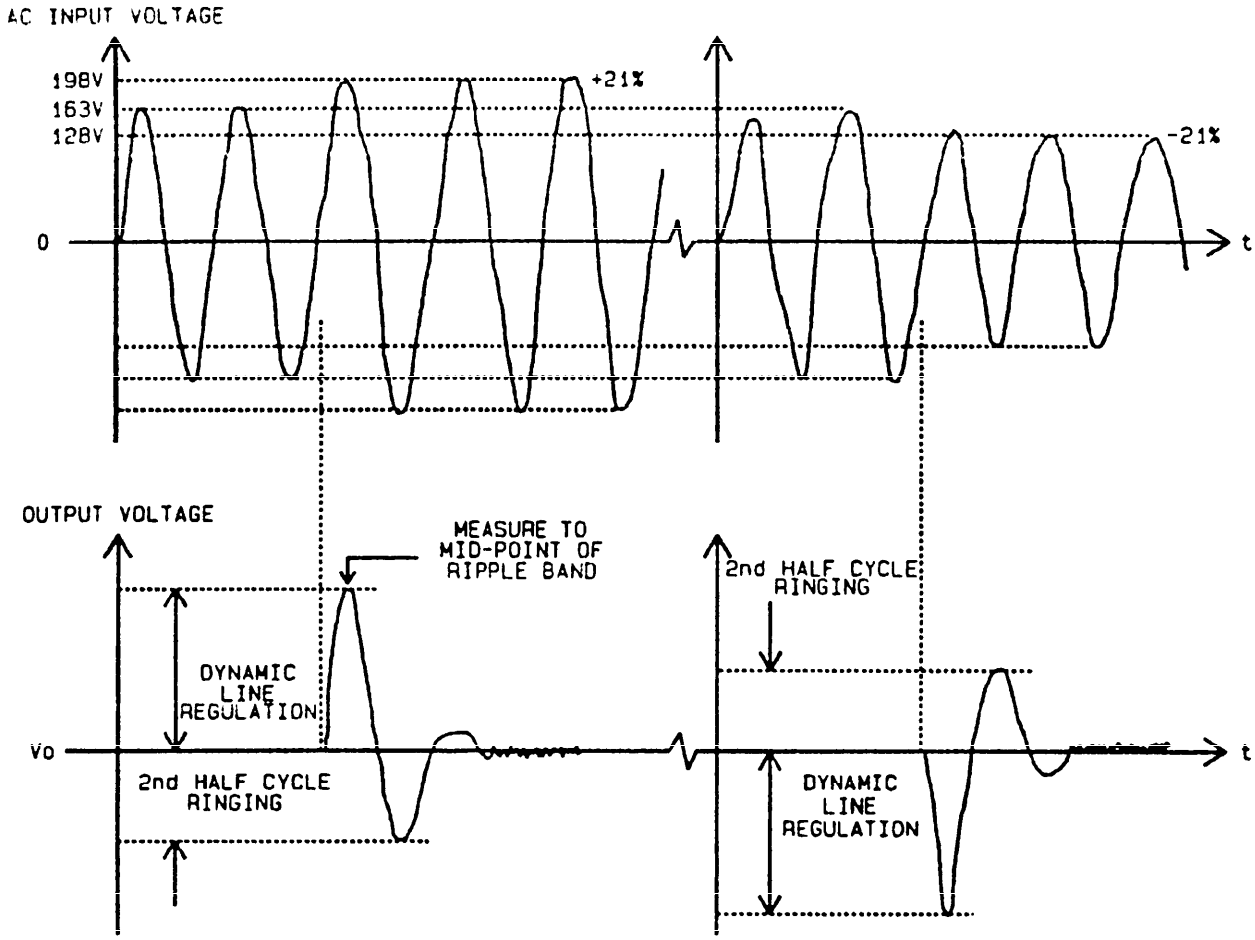
2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 1=75A/50mV; 2,3=50A/50mV.

4. Monitor shunt voltages with an item V04 voltmeter to ensure that load currents are within specification.

FIGURE 31. Dynamic line regulation test circuit for table XVI - Continued.

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NOTES:

1. The initial overshoot may be positive or negative.

2. Transition time for the AC amplitude shall be between 50 and 500 microseconds.

3. The AC input amplitude shall remain at each amplitude for a minimum of 2 seconds.

4. Nominal voltage is 115 VRMS, +21 percent=139 VRMS, -21 percent=91 VRMS.

FIGURE 32. Dynamic line regulation for table XVI.

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TABLE XVII. Dynamic load regulation and capacitive load turn on test.

Test cond	Parameter 4/	Connect		Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		1 to	2 to	Min	Max		
S1	80% to 90%, 5.05V, regulation	7	8	0	152	mV	33,36
S2	80% to 90%, 5.08V, ringing 1/	7	8	0	76	mV	33,36
S3	90% to 80%, 5.05V, regulation	7	8	0	152	mV	33,36
S4	90% to 80%, 5.05V, ringing 1/	7	8	0	76	mV	33,36
S5	80% to 90%, 5.05V, full capacitor load, regulation	7	8	0	152	mV	33,36
S6	80% to 90%, 5.05V, full capacitor load, ringing 1/	7	8	0	76	mV	33,36
S7	90% to 80%, 5.05V, full capacitor load, regulation	7	8	0	152	mV	33,36
S8	90% to 80%, 5.05V, full capacitor load, ringing 1/	7	8	0	76	mV	33,36
S9	80% to 90%, 5.25V, max sense lead, regulation	n/a	n/a	0	162	mV	34,36
S10	80% to 90%, 5.25V, max sense lead, ringing 1/	n/a	n/a	0	81	mV	34,36
S11	90% to 80%, 5.25V, max sense lead, regulation	n/a	n/a	0	162	mV	34,36
S12	90% to 80%, 5.25V, max sense lead, ringing 1/	n/a	n/a	0	81	mV	34,36
S13	80% to 90%, +15V, regulation	3	4	0	450	mV	33,36
S14	80% to 90%, +15V, ringing 2/	3	4	0	225	mV	33,36
S15	90% to 80%, +15V, regulation	3	4	0	450	mV	33,36

- See footnotes at end of table.

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TABLE XVII. Dynamic load regulation and capacitive load turn on test - Continued.

Test cond	Parameter 4/	Connect 1 to	Connect 2 to	Initial limits 25 ± 5°C mounting surface temperature Min	Initial limits 25 ± 5°C mounting surface temperature Max	Unit	Test fig
S16	90% to 80%, +15V, ringing 2/	3	4	0	225	mV	33,36
S17	80% to 90%, -15V, regulation	6	5	0	450	mV	33,36
S18	80% to 90%, -15V, ringing 3/	6	5	0	225	mV	33,36
S19	90% to 80%, -15V, regulation	6	5	0	450	mV	33,36
S20	90% to 80%, -15V, ringing 3/	6	5	0	225	mV	33,36
S21	60% to 90%, 5.05V, regulation	7	8	0	500	mV	33,36
S22	60% to 90%, 5.05V, ringing 1/	7	8	0	250	mV	33,36
S23	90% to 60%, 5.05V, regulation	7	8	0	500	mV	33,36
S24	90% to 60%, 5.05V, ringing 1/	7	8	0	250	mV	33,36
S25	40% to 90%, +15V, regulation	3	4	0	1200	mV	33,36
S26	40% to 90%, +15V, ringing 2/	3	4	0	600	mV	33,36
S27	90% to 40%, +15V, regulation	3	4	0	1200	mV	33,36
S28	90% to 40%, +15V, ringing 2/	3	4	0	600	mV	33,36
S29	40% to 90%, -15V, regulation	6	5	0	1200	mV	33,36
S30	40% to 90%, -15V, ringing 3/	6	5	0	600	mV	33,36
S31	90% to 40%, -15V, regulation	6	5	0	1200	mV	33,36
S32	90% to 40%, -15V, ringing 3/	6	5	0	600	mV	33,36
T1	Turn on time into capacitive load 4/	--	--	15	250	ms	35
T2	Turn off into capacitive load	--	--	14.85	15.15	VDC	35
T3	Turn off into capacitive load	--	--	-15.15	-14.85	VDC	35

- See footnotes on next page.

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1/ Additional ringing beyond the 2nd half cycle shall fall within the regulation limits of test condition C1 and the peak-to-peak noise envelope of test condition Q1.

2/ Additional ringing beyond the 2nd half cycle shall fall within the regulation limits of test condition C7 and the peak-to-peak noise envelope of test condition Q2.

3/ Additional ringing beyond the 2nd half cycle shall fall within the regulation limits of test condition C11 and the peak-to-peak noise envelope of test condition Q3.

4/ At the option of the testing facility, an AC input of 3-phase, 115 VRMS ± 2 percent line-to-line at 60 Hz can be applied instead of using the DC source specified on figures 33,34, and 35.

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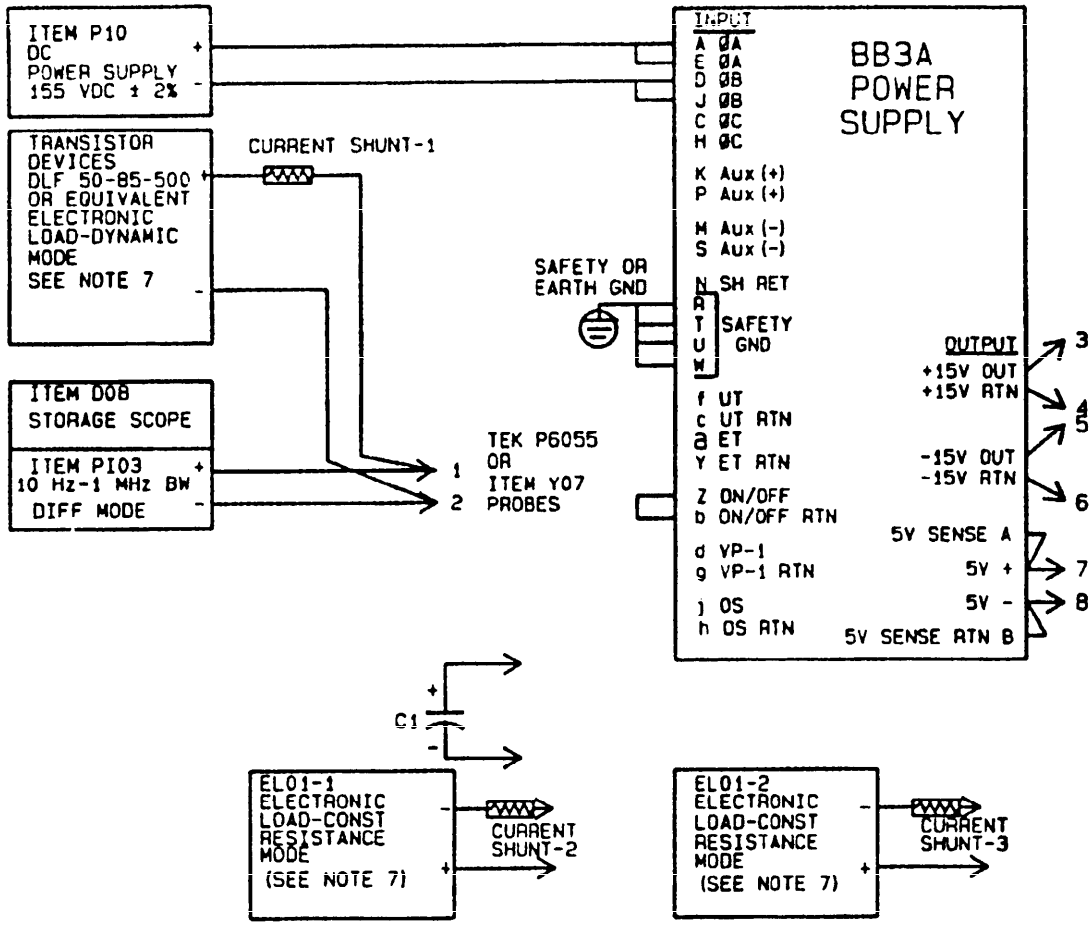


FIGURE 33. Dynamic load regulation test circuit for table XVII.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 1=75A/50mV; 2,3=50A/50mV.

4. Monitor shunt voltages with an item V04 voltmeter to ensure that the load currents are within specification.

5. C1 = 4000 uF ± 10 percent, 10 VDC, connected + to 7 and - to 8 for test conditions S5-S8.

6. Keep the leads between the power supply under test and the electronic load as short as possible.

7. The dynamic load, transistor devices DLF 50-85-500 or equivalent, shall be set to switch between the following transitions:

TEST CONDITION	FROM A CURRENT LEVEL OF	TO A CURRENT LEVEL OF
S1,S2	40 A	45 A
S3,S4	45 A	40 A
S5,S6	40 A	45 A
S7,S8	45 A	40 A
S13,S14	8.8 A	9.9 A
S15,S16	9.9 A	8.8 A
S17,S18	8.8 A	9.9 A
S19,S20	9.9 A	8.8 A
S21,S22	30 A	45 A
S23,S24	45 A	30 A
S25,S26	4.4 A	9.9 A
S27,S28	9.9 A	4.4 A
S29,S30	4.4 A	9.9 A
S31,S32	9.9 A	4.4 A

The current step accuracy shall be ± 5 percent. The outputs not being tested shall be loaded with a static load equal to 50 percent full load ± 10 percent using item EL01-1 and EL01-2.

FIGURE 33. Dynamic load regulation test circuit for table XVII - Continued.

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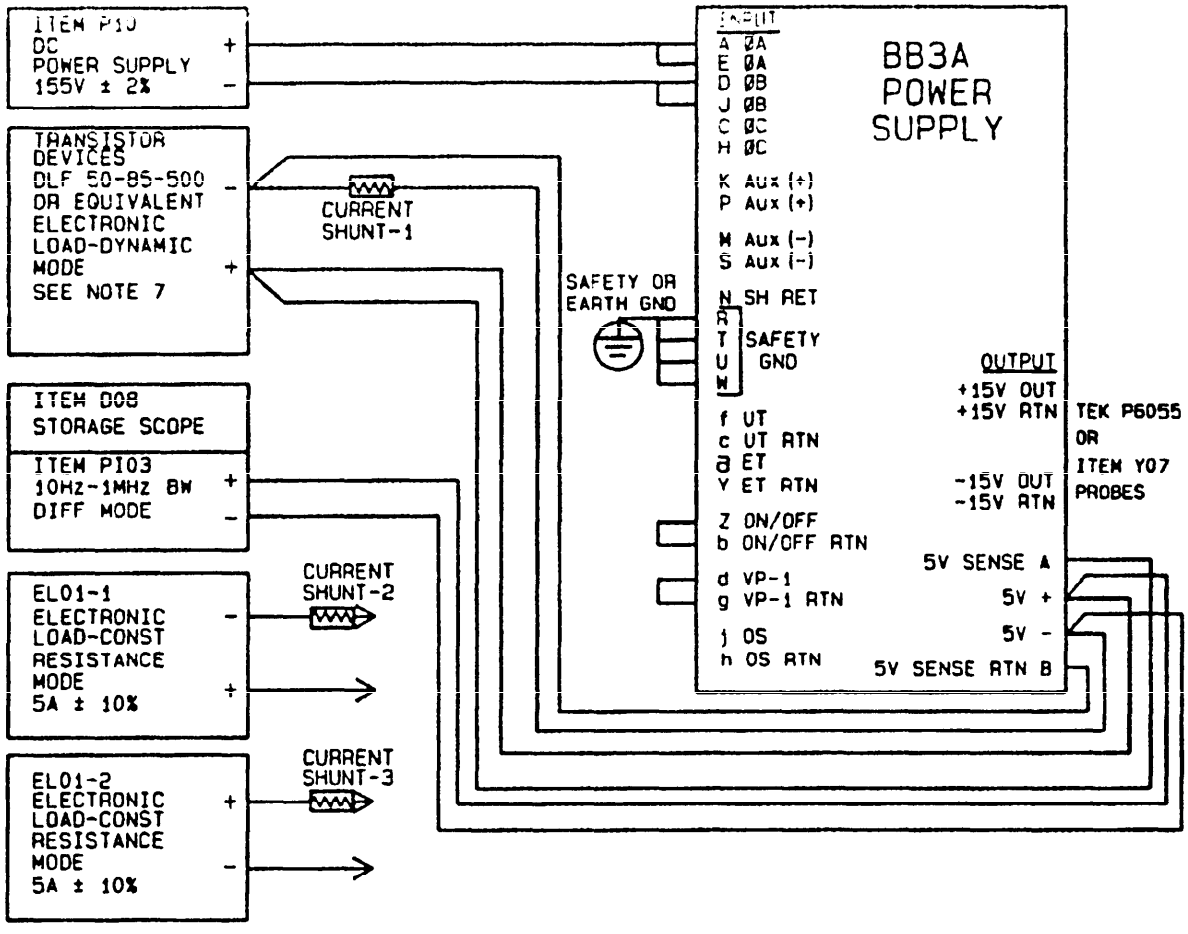


FIGURE 34. Dynamic load regulation (maximum sense leads) test circuit for table XVII.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 1=75A/50mV; 2,3=50A/50mV.

4. Monitor shunt voltages with an item V04 voltmeter to ensure that the load currents are within specification.

5. Sense leads shall be as close to 10 inches long as practical. The two sense leads shall be twisted together.

6. The load leads shall be 10 ± 2 inches long.

7. The electronic load, transistor devices DLF 50-85-500, or equivalent shall be in the dynamic mode, switching between 40 amperes (80 percent full load) and 45 amperes (90 percent full load).

8. The electronic loads, items EL01-1 and EL01-2, shall be connected to the +15V and -15V outputs respectively.

9. The maximum load lead voltage drop (difference between the voltage directly at the BB3A output voltage contacts and the sense lead connection point) shall be 0.188 VDC. This shall apply under steady state load current conditions with the current at 90 percent full load.

FIGURE 34. Dynamic load regulation (maximum sense leads) test circuit for table XVII - Continued.

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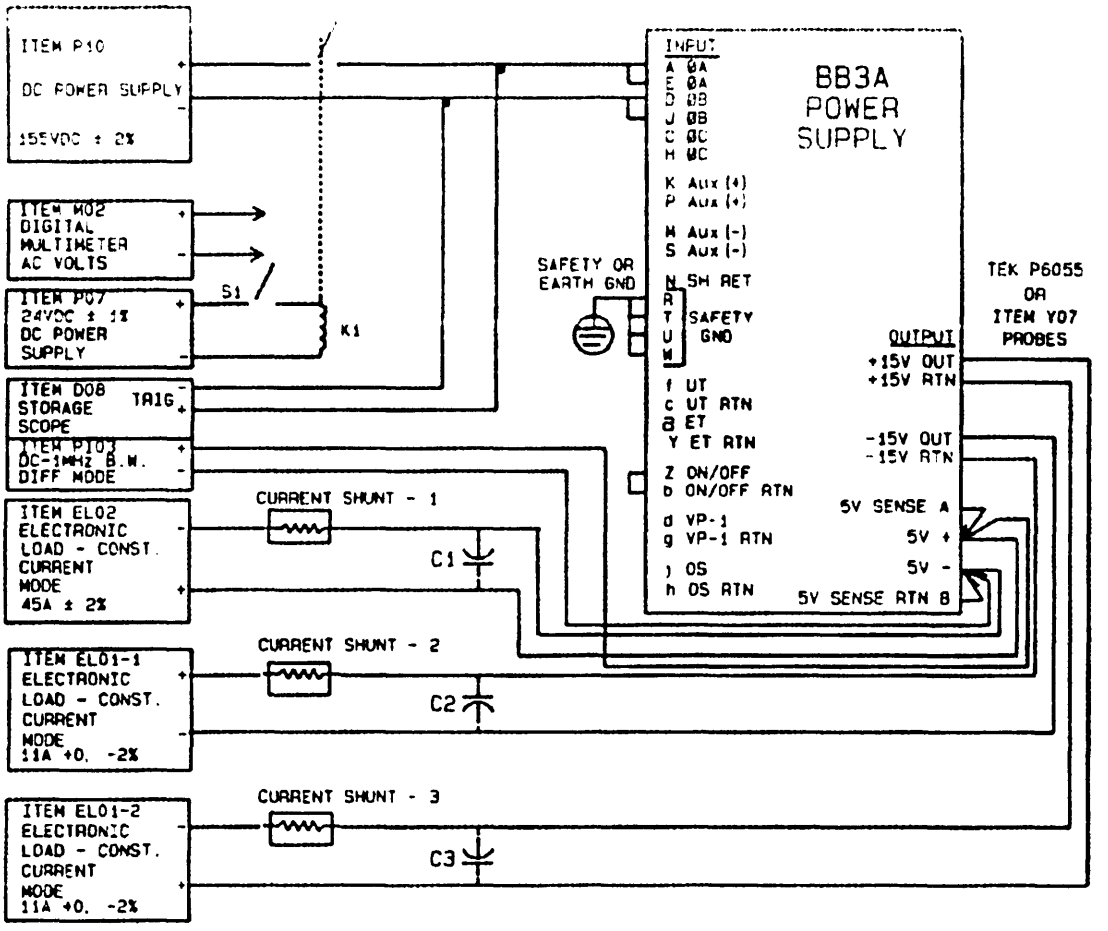


FIGURE 35. Turn on time into capacitive load test circuit for table XVII.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 1 = 75A/50mV; 2,3 = 50A/50mV.

4. Monitor shunt voltages with an item VO4 voltmeter to ensure that the load currents are within specification.

5. C1 = 4000 uF, ± 10 percent, 10 VDC capacitor; C2,C3 = 660 uF ± 10 percent, 20 VDC capacitor.

6. S1 = spst switch.

7. K1 = spst mercury relay, Magnecraft WM35A-24D or equivalent.

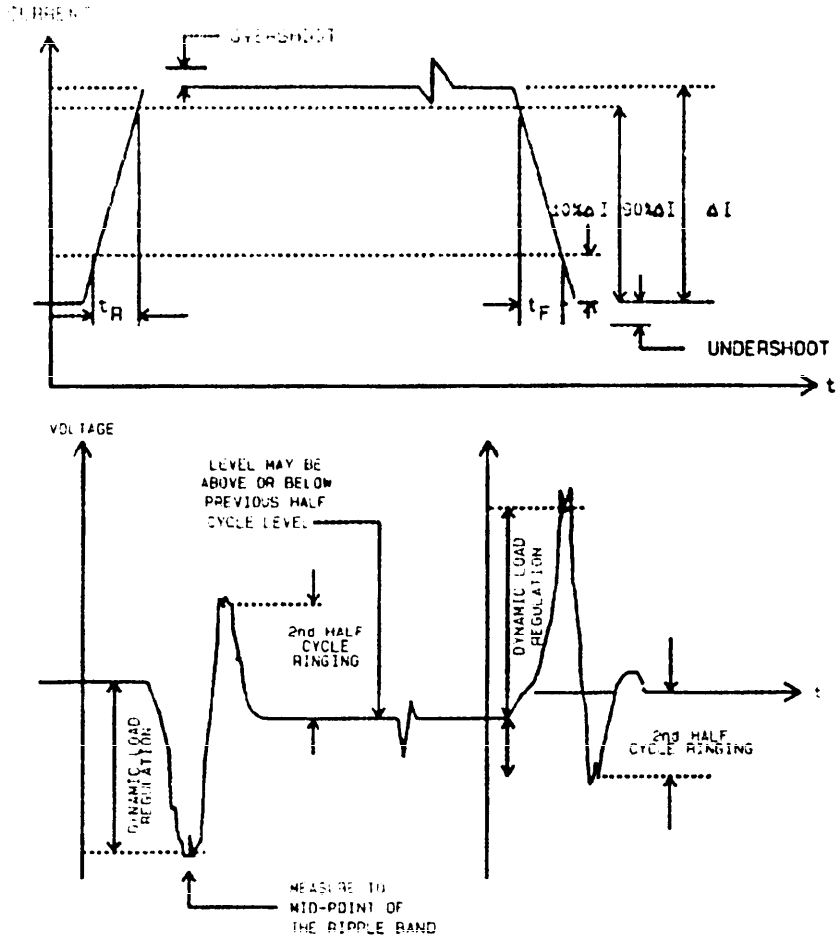
8. To perform test T1, close switch S1 and check turn on time. Turn on time is defined as the time from when power is applied to the input, until the output reaches and remains within regulation limits (test condition C1).

9. For test condition T1, item EL02 shall act as a constant current load while the 5V output is rising. Therefore, above a power supply output terminal voltage of 2.0 volt, the load current shall be rising to its set value at a rate two times, or greater, than the rate of the voltage.

10. For test conditions T2 and T3, remove the electronic loads EL01-1, EL01-2 and EL02 from the circuit and apply power to the power supply under test by closing switch S1. Wait a minimum of two seconds, then open switch S1 for a minimum of two seconds. Repeat 5 times. On the last cycle, attach item M02 (DC mode) to the +15V output. The voltage reading shall fall within table XVII limits. Connect item M02 (DC mode) to the -15V output. The voltage reading shall fall within table XVII limits.

FIGURE 35. Turn on time into capacitive load test circuit for table XVII - Continued.

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NOTES:

1. Rise time, tR, and fall time, tF, both shall be 60 us maximum.

2. The dynamic load may be a single pulse or repetitive plus and minus transitions with each state being maintained for no less than 1 millisecond.

3. The maximum load current overshoot or undershoot shall be minimized to the greatest extent possible.

FIGURE 36. Dynamic load regulation and ringing test definitions for table XVII.

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TABLE XVIII. Undervoltage and overvoltage output status test.

Test cond	Parameter	Connect 1 to	Connect 2 to	Initial limits 25 ± 5°C mounting surface temperature Min	Initial limits 25 ± 5°C mounting surface temperature Max	Unit	Test fig
U1	5V Undervoltage-output status LED off	7	8	4.39	4.80	VDC	37
U2	5V Overvoltage-output status LED off	7	8	5.57	6.45		38
U3	+15V Undervoltage-output status LED off	3	4	13.05	14.25		37
U4	+15V Overvoltage-output status LED off	3	4	15.90	17.25		38
U5	-15V Undervoltage-output status LED off	5	6	-14.25	-13.05		37
U6	-15V Overvoltage-output status LED off	6	5	-17.25	-15.90		37
U7	Status LED check	7	8	---Status LED on----			37
V1	Output status signal logic low output (5 mA)	--	--	0	0.4		37
V2	Output status signal logic high output	--	--	29.0	30.5		37
V3	5V Undervoltage-output status	7	8	4.39	4.80		37
V4	5V Overvoltage-output status	7	8	5.57	6.45		38
V5	+15V Undervoltage-output status	3	4	13.05	14.25		38
V6	+15V Overvoltage-output status	3	4	15.90	17.25		38
V7	-15V Undervoltage-output status	5	6	-14.25	-13.05		37
V8	-15V Overvoltage-output status	6	5	-17.25	-15.90	VDC	38
V9	Output status signal leakage current	--	--	0	4.4	uADC	37
V10	Output status signal settling time (high to low)	--	--		2.0	ms	39,40
V11	Output status signal settling time (low to high)	--	--		2.0	ms	39,40
V12	Output status signal, logic low (0.5 mA)	--	--	0	0.4	VDC	39

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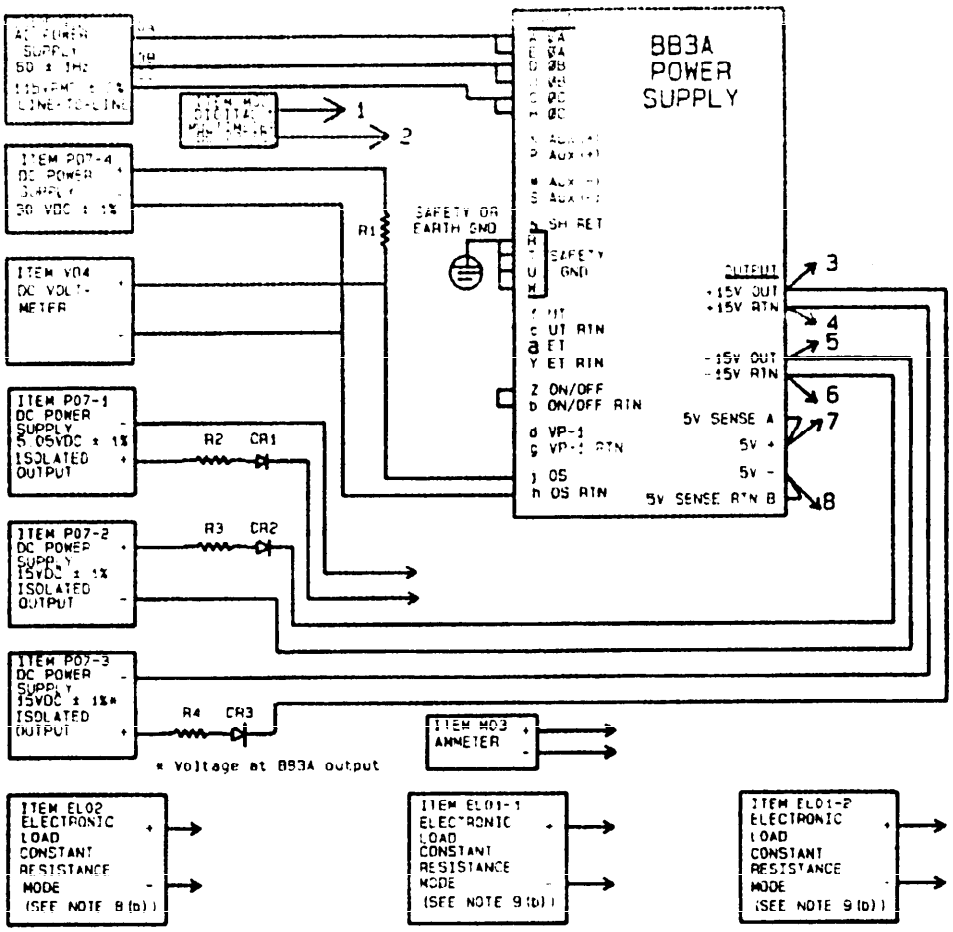


FIGURE 37. Undervoltage output status test circuit for table XVIII.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. R1 = 5.9 kOhm ± 1 percent, 1/8 watt resistor.

4. R2 = 5 Ohm ± 1 percent, 5 watt resistor.

5. R3 = R4 = 15 Ohm ± 1 percent, 15 watt resistor.

6. CR1 = CR2 = CR3 = 1N5616 diode.

7. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

8. The 5V output voltage can be reduced for undervoltage testing in one of two ways (the method to use will depend upon the design of the power supply being tested):

(a) Set item P07-1 voltage to 0 volts. Connect item P07-1 plus terminal to input contact d and minus terminal to input contact g (CR1 and R2 shall not be used). Apply input power with item P10. Gradually increase the item P07-1 voltage until the output is 5.05 ± 0.05 volts. Further increase of the item P07-1 voltage will cause the output to drop as required. Items P07-2, P07-3, R3-R4 and CR2-CR3 are not required.

(b) Connect item EL02 plus terminal to point 7 and minus terminal to point 8. With the BB3A power supply powered up and running, begin slowly increasing the load current from 50 amperes and the output voltage shall gradually drop as required for testing. Do not exceed 75 amperes load current. Items P07-1 through P07-3, R2-R4, and CR1-CR3 are not required.

9. The 15V outputs can be reduced for output voltage testing in one of two ways (the method to use will depend upon the design of the power supply being tested):

(a) Initially disconnect items P07-1, P07-2, P07-3 and associated components. Beginning with item P10 set to 155 VDC, begin reducing this voltage until the + and - output voltages drop and remain at less than (toward zero) +12.00 volts

FIGURE 37. Undervoltage test circuit for table XVIII - Continued.

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NOTES: - Continued.

and -12.00 volts. The internal housekeeping supply should still be working at its nominal voltage. If required to maintain the 5V output at 5.05 ± 0.13 volts, connect item P07-1 with the minus terminal to point 8 and the CR1 cathode to point 7. Connect items P07-2, P07-3 and their associated components as shown. Adjust items P07-1 through P07-3 until the nominal output voltages are applied at the associated BB3A output contacts (as measured with the item M02 voltmeter). Item P07-2 or P07-3, as applicable, can be reduced to give the undervoltage for the output being tested.

(b) Connect item EL01-1 plus terminal to point 3 and minus terminal to point 4 and connect item EL01-2 plus terminal to point 6 and minus terminal to point 5. With the BB3A power supply powered up and running, begin slowly increasing the load current from 11 amperes (on the output being tested) and the output voltage shall gradually drop as required for testing. Do not exceed 15 amperes load current. Items P07-1 through P07-3, R2-R4, and CR1-CR3 are not required.

10. Perform test conditions U7 and V1 before performing the undervoltage and overvoltage tests. For U7 and V1, connect the EL02 plus terminal to point 7 and minus terminal to point 8, connect the EL01-1 plus terminal to point 3 and minus terminal to point 4, and connect the EL01-2 plus terminal to point 6 and minus terminal to point 5. Adjust EL02 to 50, +0, -2 amperes, EL01-1 to 11, +0, -1 ampere, and the EL01-2 to 11, +0, -1 ampere. Using item M02, ensure that all outputs are within regulation limits for the applicable temperature (test conditions C4, C10, and C13 for the 5V output, +15V output, and -15V output respectively). For test condition U7, verify that the output status LED is on. For test condition V1, the item V04 reading shall be within the table XVIII limits.

11. Perform the test condition U undervoltage tests as follows:

(a) For U1, with applicable connections as in note 7, begin reducing the 5V output voltage as measured with item M02. Record the output voltage at which the output status LED goes off.

(b) For U3, with applicable connections as on note 8 (all output voltages within specification), begin reducing the +15V output voltage. Record the output voltage at which the output status LED goes off. Return the output voltage to 15 volt ± 1 percent.

FIGURE 37. Undervoltage test circuit for table XVIII - Continued.

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NOTES: - Continued.

(c) For U5, repeat the note 9(b) procedure for the -15V output.

12. Perform the test condition V undervoltage tests as follows:

(a) For V3, with the applicable connections as in note 7, begin reducing the 5V output voltage until the output status signal, as read by item V04, goes from a low voltage to a high voltage (within the test condition V2 limit). Record this high voltage level for test condition V2. When the output status signal goes within the V2 limits, read the output voltage from item M02 and compare to V3 limits.

(b) For V5, with the connections as in note 8 (all output voltages within specification), reduce the +15V output voltage until the output status signal, as read by item V04, goes from a low voltage to a high voltage (within test condition V2 limits). When the output status signal goes within the V2 limits, read the output voltage from item M02 and compare to the V5 limit. Return the output voltage to within specification.

(c) For V7, repeat the note 11(c) procedure for the -15V output.

(d) For V9, use the setup for V7 and reduce the output voltage until the status signal goes to a high voltage (within the test condition V2 limits). Connect the item M03 between resistor R1 and item P07-4 and the measured current shall be within the V9 limits. Disconnect item V04 while making the current reading.

FIGURE 37. Undervoltage test circuit for table XVIII - Continued.

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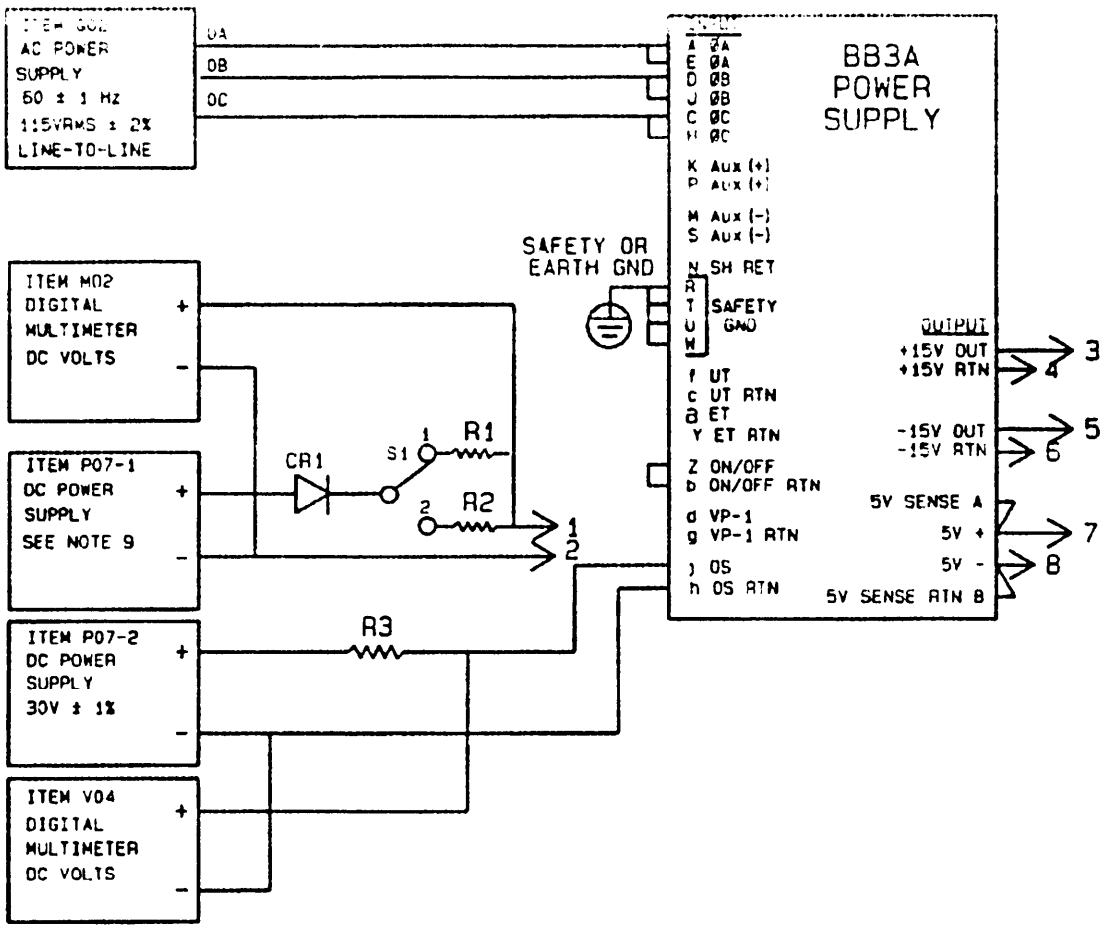


FIGURE 38. Overvoltage output status test circuit for table XVIII.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. R1=5 Ohm ± 5 percent, 8 watt resistor.

4. R2=15 Ohm ± 5 percent, 15 watt resistor.

5. R3=5.9 kOhm ± 1 percent, 1/8 watt resistor.

6. S1=spdt switch.

7. CR1 = 1N5616 or equivalent diode.

8. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

9. Before connecting points 1 and 2 to the power supply under test, adjust item P07-1 initially to: +5V ± 1 percent for test conditions (T.C.) U2 and V4; to +15V ± 1 percent for T.C. U4 and V6; and -15V ± 1 percent for T.C. U6 and V8.

10. Test conditions U2, U4, U6, V4, V6, and V8 verify that the output status LED and output status signal detect an overvoltage condition. Additionally, the power supply vendor shall verify by 100 percent (X) testing during power supply manufacturing, that the output voltage is limited as specified.

11. Test conditions U2 and V4. Connect per table XVIII and switch S1 to position 1. Slowly increase item P07-1 voltage until the LED goes off. The item M02 measured voltage when this occurs shall be recorded for T.C. U2. For T.C. V4, repeat the above procedure except record the item M02 measured voltage when the item V04 voltage (output status signal) goes greater than 29.0 volts (T.C. V2 min. limit). Do not increase the voltage at points 1 and 2 (as measured with item M02) above 6.6 volts.

12. Test conditions U4 and V6. Connect per table XVIII and switch S1 to position 2. Slowly increase item P07-1 voltage until the LED goes off. The item M02 measured voltage when this occurs shall be recorded for T.C. U4. For T.C. V6, repeat the

FIGURE 38. Overvoltage output status test circuit for table XVIII - Continued.

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NOTES: - Continued.

above procedure except record the item M02 measured voltage when the item V04 voltage (output status signal) goes greater than 29.0 volts (T.C. V2 min. limit). Do not increase the voltage at points 1 and 2 (as measured with item M02) above 20 volts.

13. Test conditions U6 and V8. Connect per table XVIII and switch S1 to position 2. Slowly increase item P07-1 voltage until the LED goes off. The item M02 measured voltage when this occurs shall be recorded for T.C. U6. For T.C. V8, repeat the above procedure except record the item M02 measured voltage when the item V04 voltage (output status signal) goes greater than 29.0 volts (T.C. V2 min. limit). Do not increase the voltage at points 1 and 2 (as measured with item M02) above 20 volts.

14. It may be necessary to recycle the input to 0 volts to allow for recovery of the overvoltage condition.

FIGURE 38. Overvoltage output status test circuit for table XVIII - Continued.

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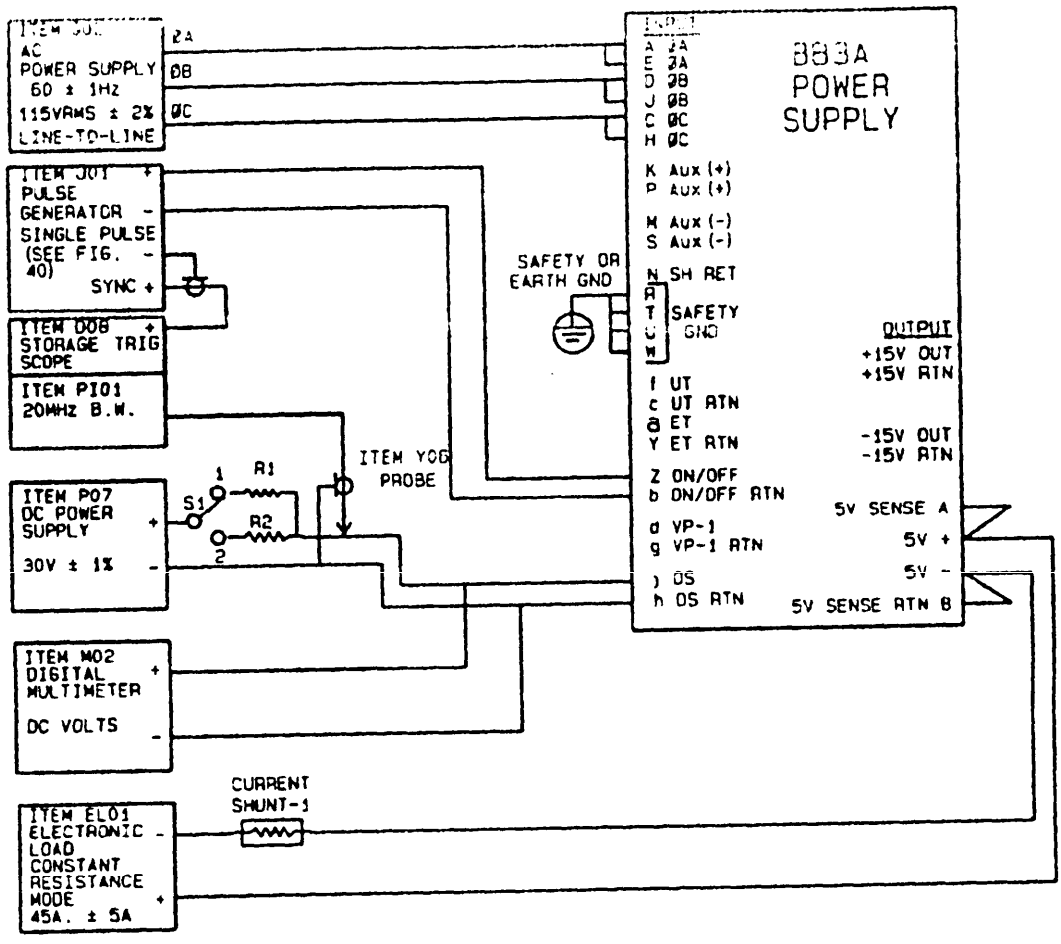


FIGURE 39. Output status signal settling time and logic low test circuit for table XVIII.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 1=75A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. R1 = 5.9 kOhm ± 1 percent, 1/4 watt resistor.

6. R2 = 60.4 kOhm ± 1 percent, 1/10 watt resistor.

7. S1 = spdt switch.

8. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used in place of the AC input specified.

9. For test conditions V10 and V11, with switch S1 in position 1, set up the power supply input voltage and load current with item J01 disconnected and input contacts Z and b shorted (power supply on). Disconnect the short after setup.

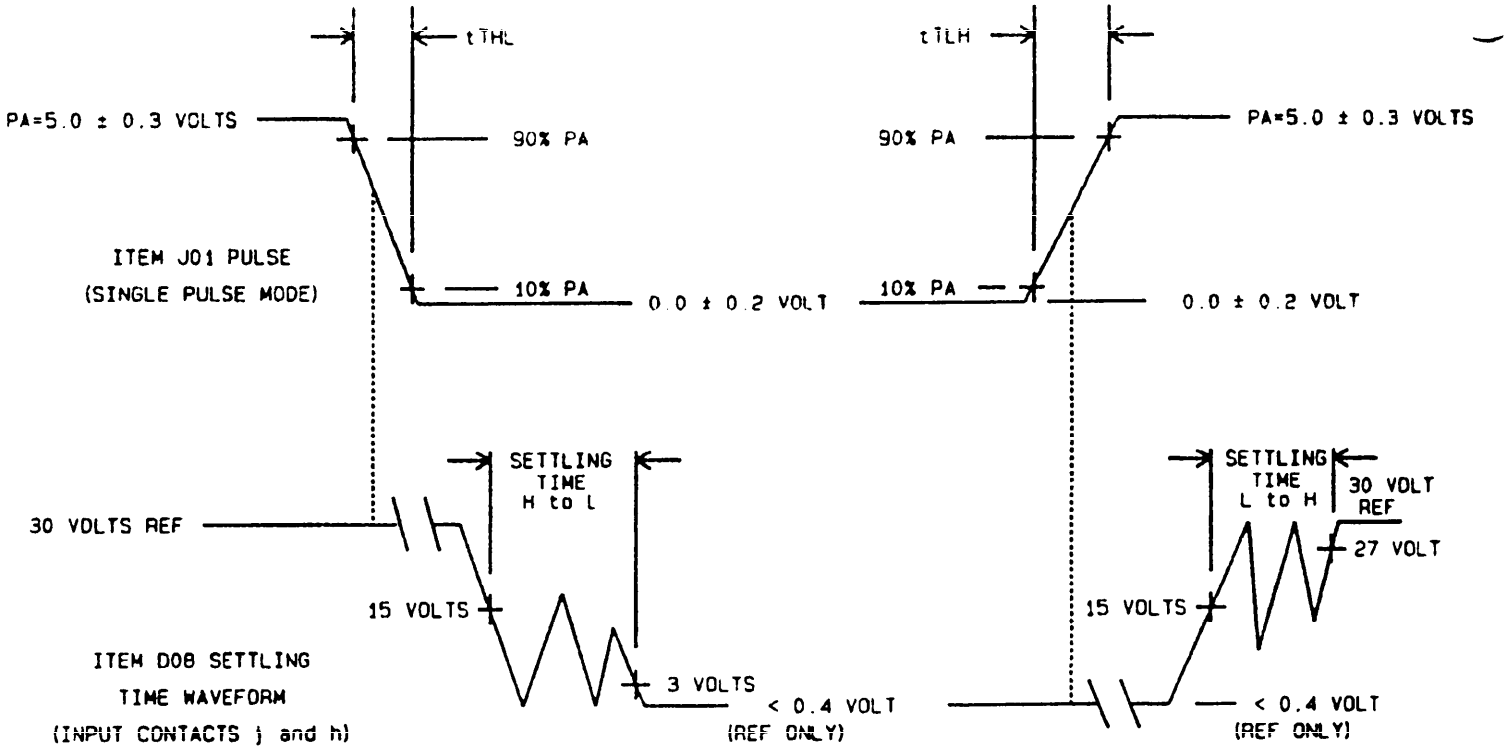
(a) For test condition V10, set up item J01 for a single high to low level pulse in accordance with figure 40. Connect item J01 as shown and trigger item J01 for a single pulse. Using the storage mode of item D08, scope measure the settling time (see figure 40).

(b) For test condition V11, repeat 9(a) above except for a single low to high level pulse from item J01.

10. For test condition V12, set switch S1 to position 2, disconnect item D08 and item J01 and short input contact Z to b. Connect OS RTN (input contact h) to SAFETY GND (input contact R). Using item M02, measure the voltage across input contacts j and h.

FIGURE 39. Output status signal settling time and logic low test circuit for table XVIII - Continued.

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NOTE:

1. Item J01 pulse parameters:
   tTHL = tTLH = 10 ± 5 us.

FIGURE 40. Output status settling time test circuit waveforms for table XVIII.

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TABLE XIX. Excessive and unsafe temperature indication test.

Test cond	Parameter	Initial limits elevated ambient temperature		Unit	Test fig
		Min	Max		
W1	Excessive temperature indication (5 mA current) 2/	0	0.4	VDC	41
W2	Excessive temperature indication (500 mA current) 2/	"	"	"	"
W3	Unsafe temperature indication (5 mA current) 2/	"	"	"	"
W4	Unsafe temperature indication (500 mA current) 2/	0	0.4	VDC	"
W5	Unsafe-excessive differential (5mA current) 1/	1.0	25	C	"
W6	Indicator reset (5mA current)	60	--	C	"
W7	Unsafe-excessive differential (500mA current) 1/	1.0	25	C	"
W8	Indicator reset (500mA current)	60	--	C	41

1/ Test conditions W5 and W7 are calculations. Subtract the trip point temperature measured in test condition W1 (W2) from the trip point temperature measured in test condition W3 (W4).

2/ Test conditions W2 and W4 verify that the closed contact resistance is a maximum of 0.8 Ohms with a contact current of 500 mA. Test conditions W1 and W3 verify that the closed contact resistance is a maximum of 80 Ohms with a contact current of 5 mA.

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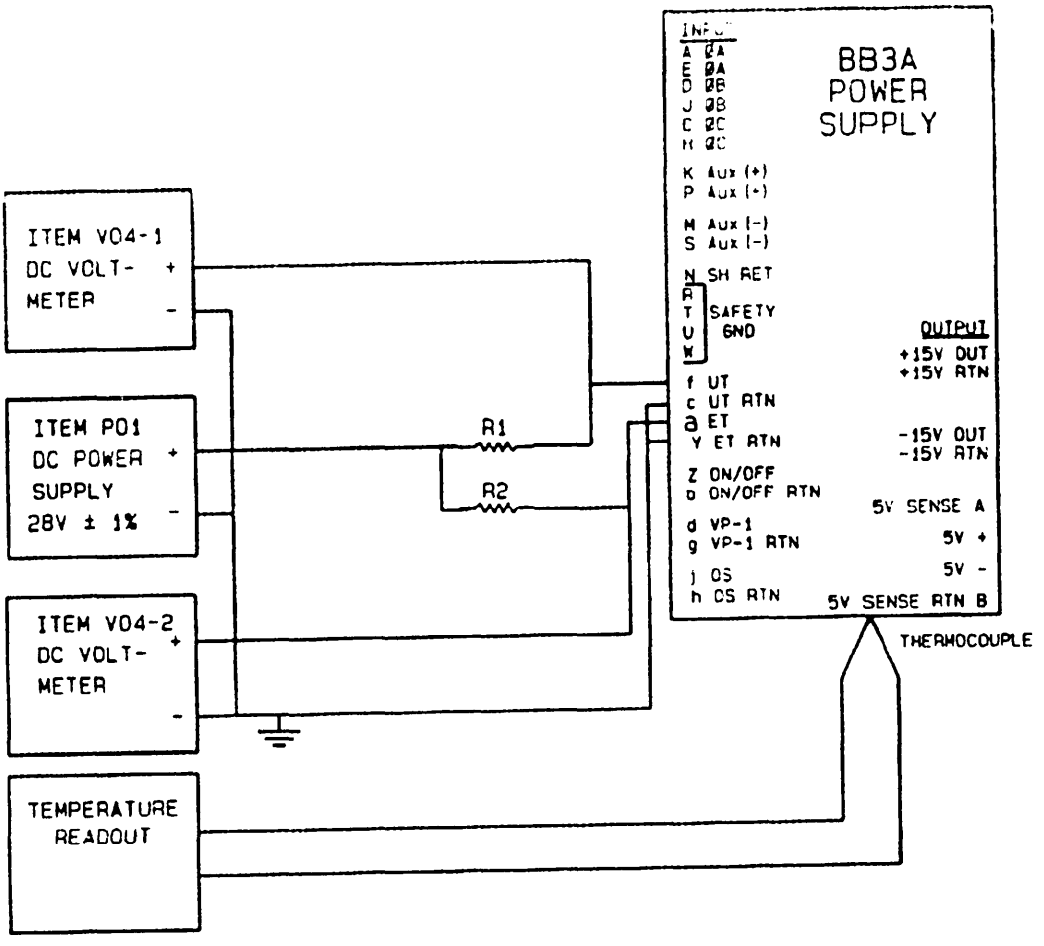


FIGURE 41. Excessive and unsafe temperature indication test circuit for table XIX.

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NOTES:

1. Perform tests W1-W8 only after all other electrical and environmental tests have been performed.

2. R1 = R2 = 56 Ohm ± 1 percent, 15 watt resistor, for test conditions W2, W4, and W8. R1 = R2 = 5.62 kOhm ± 1 percent, 1/4 watt resistor, for test conditions W1, W3, and W6.

3. The power supply shall be unenergized during this test.

4. The temperature shall be monitored with a thermocouple connected to the power supply frame. Do not attach the thermocouple to the mounting surface.

5. Test conditions W1 and W3 shall be performed as follows and shall be performed prior to performing test conditions W2 and W4:

(a) Place the power supply in an oven and raise the temperature to 65 ± 2°C at a rate not to exceed 1°C per minute. Let the power supply soak for a minimum of 1 hour at this temperature. Verify that items VO4-1 and VO4-2 both read 28.0 ± 0.5 volts (thermal switches open).

(b) Begin increasing the oven temperature, at a rate not to exceed 1°C per 15 minutes, until VO4-2 voltage switches to a low level (excessive temperature thermal switch closed). This low level voltage shall meet the test condition W1 limits. Record the power supply frame temperature at which the thermal switch closed for use in test condition W5 calculation. Do not exceed 140°C.

(c) Continue increasing the oven temperature, as noted in (b) above, until item VO4-1 switches to a low level (unsafe temperature thermal switch closed). This low level voltage shall meet test condition W3 limits. Record the power supply frame temperature at which the thermal switch closed for use in test condition W5 calculation. Do not exceed 140°C.

(d) Begin decreasing the oven temperature at a rate not to exceed 1°C per 15 minutes, until VO4-1 and VO4-2 both read 28.0 ± 0.5 volts (thermal switches open). The power supply frame temperature at which both thermal switches are open shall meet the limits of test condition W6.

6. Repeat note 5 procedure for test conditions W2, W4, W7 and W8.

FIGURE 41. Excessive and unsafe temperature indication test circuit for table XIX - Continued.

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TABLE XX. Output voltage hold-up time.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
X	Output voltage hold-up time	50	-	us	42 43

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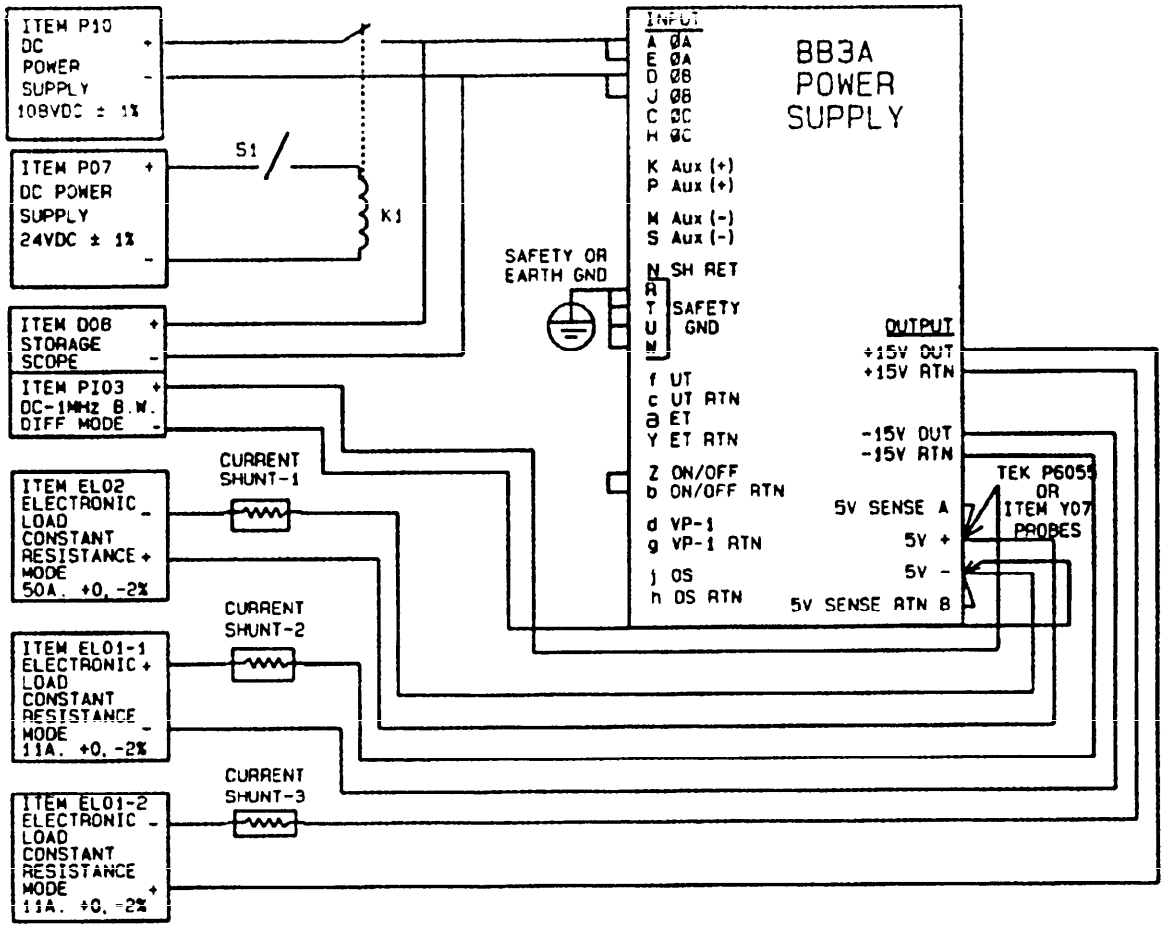


FIGURE 42. Output voltage hold-up time test circuit for table XX.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 1=75A/50mV; 2,3=50A/50mV.

4. Monitor shunt voltages with an item VO4 voltmeter to ensure that the load currents are within specification.

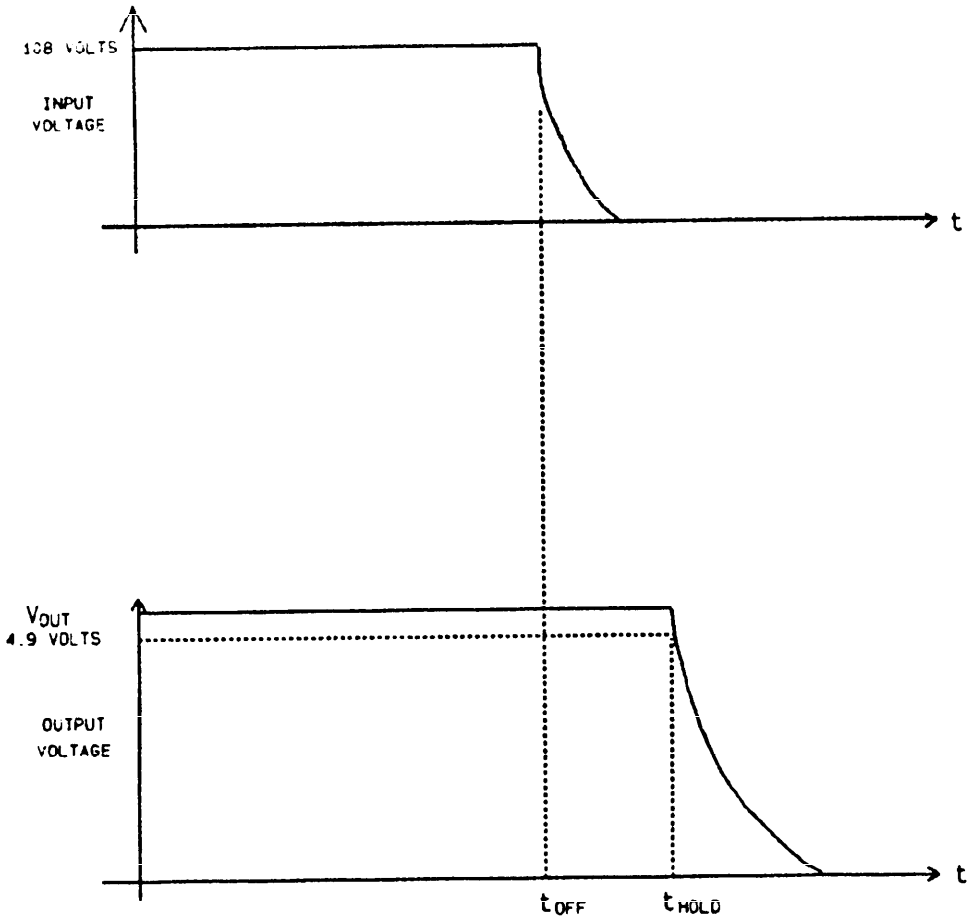
5. S1 = spst switch.

6. K1 = spst mercury relay, Magnecraft WM35A-24D or equivalent.

7. For test condition X, connect the circuit as shown. Use switch S1 to apply power and then disconnect power and measure hold-up time.

FIGURE 42. Output voltage hold-up time test circuit for table XX - Continued.

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NOTE:

1. Hold-up time is defined as tHOLD minus tOFF.

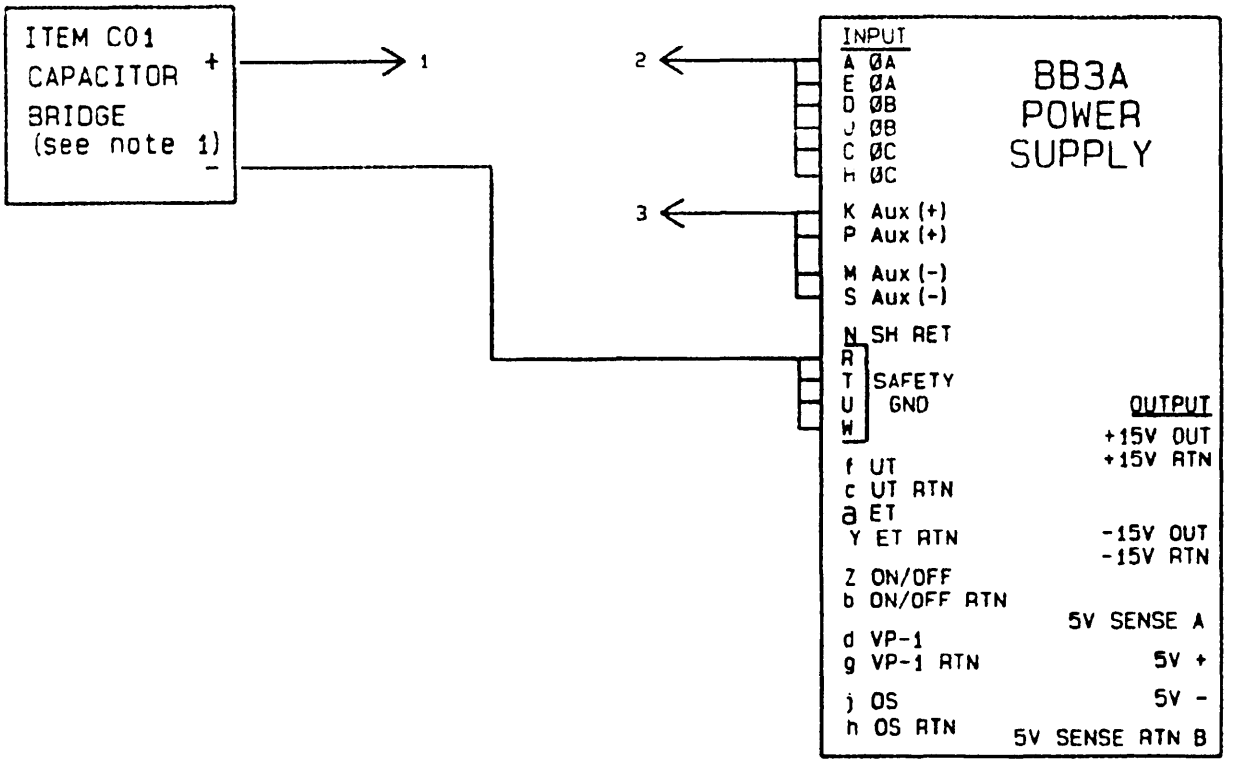
FIGURE 43. Output voltage hold-up time for table XX.

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TABLE XXI. Input to chassis capacitance test.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
Y1	AC input to chassis capacitance	0	0.03	uF	41
Y2	Auxiliary input to chassis capacitance	0	0.05	uF	44

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NOTES:

1. Item C01 setup frequency = 1 kHz ± 5 percent, Vpk = 0.3 volts peak maximum.

2. For test condition Y1, connect point 1 to point 2. The item C01 capacitance reading shall meet the test condition Y1 limits.

3. For test condition Y2, connect point 1 to point 3. The item C01 capacitance reading shall meet the test condition Y2 limits.

FIGURE 44. Input to chassis capacitance test circuit for table XXI.

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TABLE XXII. Output impedance test. 1/

Test cond	Parameter	R2 Value (including lead resistance ±5%, 250 watt)	Connect		Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
			1 to point	2 to point	Min	Max		
Z1	Output impedance, +15V	2.0 Ohm	3	4	0	6.91	mV	45
Z2	Output impedance, -15V	2.0 Ohm	5	6	0	5.91	mV	45
Z3	Output impedance, 5.05V	0.133 Ohm	7	8	0	4.90	mV	45

1/ This test verifies a maximum output impedance of 30 mOhm. The mV limits in this table do not apply if the power supply under test (PSUT) load resistance is not exactly as specified or varies with increasing frequency. The Zo relationship in the figure 45 notes shall be used to calculate the PSUT maximum impedance.

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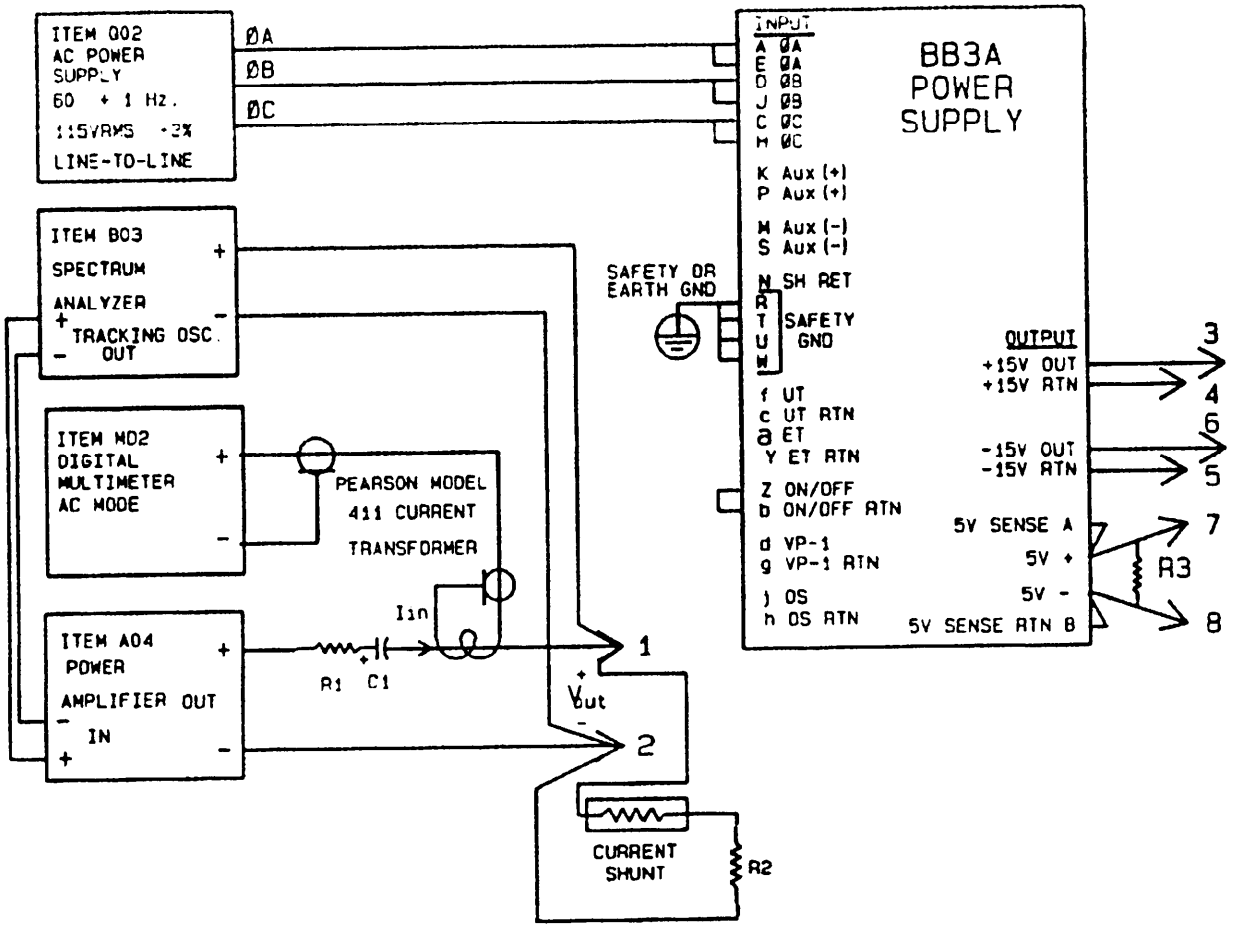


FIGURE 45. Output impedance test circuit for table XXII.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor the shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. R1 = 10 Ohm ± 5 percent, 0.5 watt, non-inductive.

6. R2 = value per table XXII, non-inductive.

7. R3 = 0.8 ± 0.2 Ohm, 50 watt (remove during Z3).

8. C1 = 6000 uF +75 percent, -10 percent, 25VDC.

9. At the option of the testing facility, 155 VDC ±2 percent can be applied between any two AC inputs, instead of the AC source specified on figure 45.

10. The output impedance test shall be performed as follows:

(a) Make the required connections for test condition Z1 with the item A01 amplitude adjusted to 0V and apply input power to the BB3A power supply. The DC load current on the power supply will be 7.5 ampere ± 10 percent for test conditions Z1 and Z2 and 37.5 ampere ± 10 percent for test condition Z3. A load will only be applied to the output being tested (except the 5.05V output should be loaded with R3 during test conditions Z1 and Z2).

(b) AC current adjustment-set the spectrum analyzer to the manual mode and at a fixed frequency between 100 Hz and 1 kHz. Use the model 411 current transformer and item M02 to monitor the AC current (conversion factor=0.1 volts per ampere). Use the item B03 tracking oscillator level control and the item A04 amplitude control to increase the AC current into the power supply output to 200 mA ± 3 percent.

FIGURE 45. Output impedance test circuit for table XXII - Continued.

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NOTES: - Continued.

(c) Set the item B03 spectrum analyzer to the sweep mode and find maximum Vo (as measured by item B03) over the frequency range of 30 Hz to 25 kHz. Sweep the frequency over a minimum of two bands: 30 Hz to 5 kHz, resolution bandwidth = 10 Hz; 5 kHz to 25 kHz, resolution bandwidth = 30 Hz. The circuit is set up so the current into the power supply output remains constant over the sweep range. The frequency of maximum Vo is the frequency of maximum output impedance. Read the maximum value displayed on item B03. Record the frequency of maximum Vo.

(d) Repeat the above procedure for test conditions Z2 and Z3. Remove resistor R3 for test condition Z3.

(e) To calculate the maximum output impedance from the maximum Vo:

$$Zo = \frac{\text{item B03 voltage}}{Iin - \frac{\text{item B03 voltage}}{R2}} = \frac{\text{item B03 voltage}}{0.2 \text{ ampere} - \frac{\text{item B03 voltage}}{R2}}$$

where R2 includes all impedance (may be frequency dependent) from point 1 through the current shunt, through R2 and to point 2.

FIGURE 45. Output impedance test circuit for table XXII - Continued.

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TABLE XXIII. Open load lead and return protection test.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
AA1	Open 5.05V load lead protection	5.00	5.10	VDC	46
AA2	Open 5.05V RTN load lead protection	5.00	5.10	VDC	46

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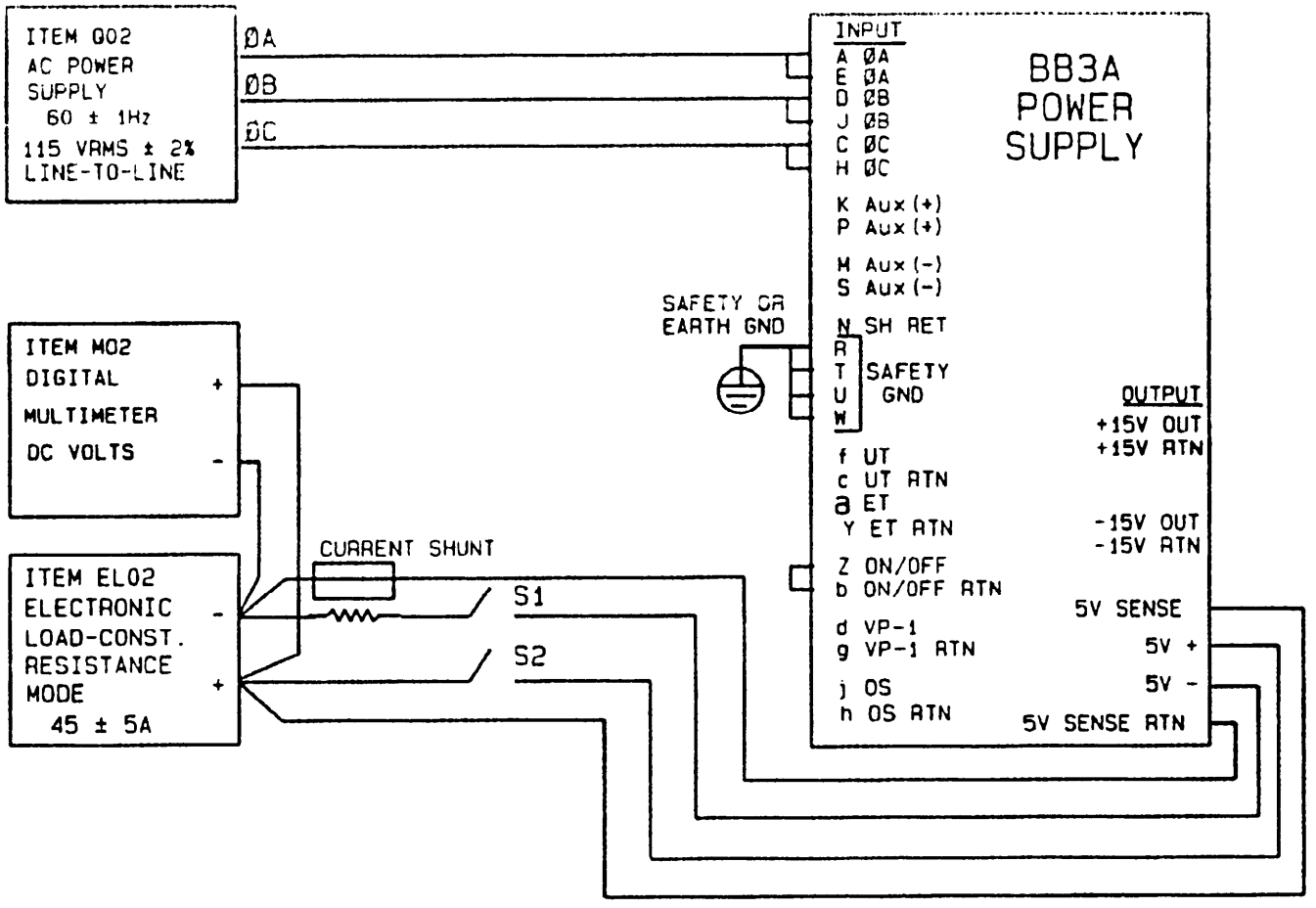


FIGURE 46. Open load lead and return protection test circuit for table XXIII.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 75A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. S1 = S2 = spst switch.

6. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

7. Sense leads shall be as close to 10 inches in length as practical. The two sense leads shall be twisted together.

8. The maximum load lead voltage drop to the sense lead connection point (with S1 and S2 closed) shall be 0.358 VDC.

9. The test shall be performed as follows:

(a) Connect the circuit as shown with both S1 and S2 closed.

(b) For test condition AA1, open the +5V load lead (represented by opening S2) for a minimum of 5 seconds. Reconnect the load lead and measure the output voltage with item M02. The item M02 reading meeting the table XXIII limits verifies the power supply has not been damaged by opening the load lead.

(c) For test condition AA2, repeat (b) above but by opening the RTN load lead (represented by opening S1) for a minimum of 5 seconds.

10. During the open load lead condition, the power supply may go into an overvoltage state. Recycling of the input to 0 volts may be necessary for full recovery of the output voltage.

FIGURE 46. Open load lead and return protection test circuit for table XXIII - Continued.

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TABLE XXIV. Spike voltage protection test.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
BB1	Spike voltage protection, AC input, initial transient (during first 20 us)	-253	253	mV	47, 48
BB2	Spike voltage protection, AC input, secondary transient	-126	126	mV	47, 48
BB3	Spike voltage protection, DC input, initial transient (during first 20 us)	-253	253	mV	47, 48
BB4	Spike voltage protection, DC input, secondary transient	-126	126	mV	47, 48

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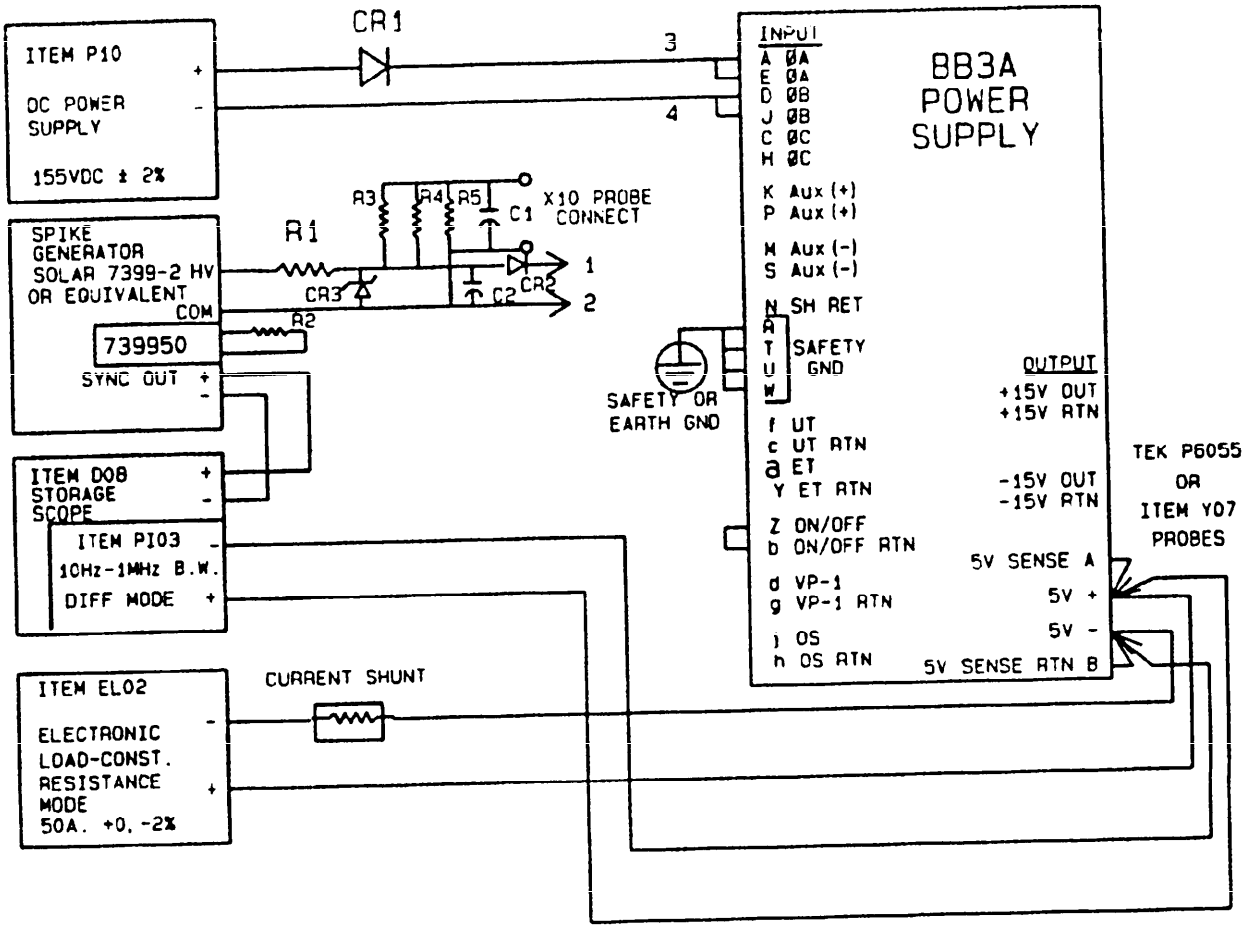


FIGURE 47. Spike voltage protection (AC input) test circuit for table XXIV.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 75A/50mV.

4. Monitor shunt voltage with an item VO4 voltmeter to ensure that the load current is within specification.

5. CR1 = CR2 = International Rectifier 12FL100S05 or equivalent.

6. CR3 = Transorb, General Semiconductor PIP120 or equivalent (see note 12(b)).

7. R1 = 7 Ohm ± 1 percent,10 watt.

8. R2 = .5 Ohm ± 5 percent, 250 watt (see note 12(b)).

9. R3 = R4 = 180 kOhm ± 5 percent, 1 watt carbon composition resistor. R5 = 10 kOhm ± 5 percent, 1/2 watt carbon composition. C1= 27 pF ± 10 percent, 200 volts.

Select the resistors so:

$$\frac{R3//R4}{R5} = \frac{90k}{10k} = 9.0 \pm 1 \text{ percent}$$

A X100 probe may be used directly in lieu of these components.

10. C2 = Selected (start with approximately 1 uF), 400 volts (see note 12(b)).

11. WARNING: High voltage is present during this test. Exercise caution and observe the procedures given in the spike generator instruction book. The spike generator shall be in the single pulse mode. Allow a minimum of 30 seconds between spikes while adjusting the waveform.

12. The input voltage waveform shown on figure 48 shall be obtained at points 1 and 2 (as measured with an oscilloscope connected to the X10 PROBE CONNECT points) prior to connection to the power supply under test:

FIGURE 47. Spike voltage protection (AC input) test circuit for table XXIV - Continued.

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NOTES: - Continued.

(a) Adjust the spike generator to charge its internal capacitors to approximately 1500 volts (using the spike generator meter) prior to discharging to produce the spike.

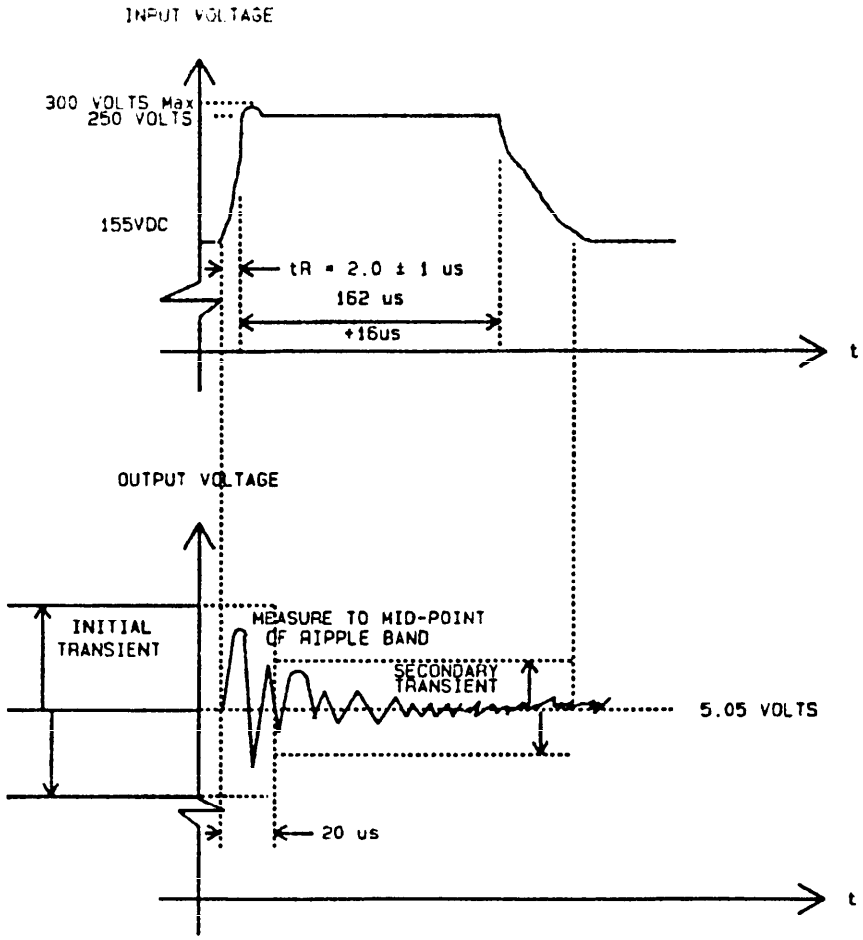
(b) Trigger the spike generator and observe that the input voltage waveform as shown on figure 48 (except the waveform will start at 0 volts and decay to 0 volts) is obtained. Select C2 to give the specified rise time. Increasing or decreasing resistor R2 will increase or decrease the pulse width. Selection of CR3 (or paralleling of more than one transorb) will set the 250 volts peak level.

13. After the input waveform has been adjusted, connect point 1 to point 3 and point 2 to point 4. Apply a single voltage spike to the power supply under test and observe that the output waveform meets the test limits. Do not reapply the input spike voltage to the power supply at intervals of less than one minute (the spike voltage and current limiting resistor R1 have been scaled down, but simulate a 2500 volt spike clipped to 250 volts).

14. Perform test conditions BB3 and BB4 (after the input waveform has been adjusted according to note 12) as follows: Connect the circuit as shown except point 3 shall be connected to input contacts K and P and point 4 shall be connected to input contacts M and S. Repeat the procedure of note 13.

FIGURE 47. Spike voltage protection (AC input) test circuit for table XXIV - Continued.

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NOTES:

1. The initial transient shall be defined as the highest output transient (plus or minus) occurring during the first 20 us after application of the input spike voltage.

2. Secondary transient shall be defined as the highest output transient (plus or minus) occurring between 20 us after the spike application and the end of the input spike.

FIGURE 48. Spike input voltage test for table XXIV.

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TABLE XXV. Reverse voltage protection, ON/OFF parallel outputs, and reverse bias turn on test.

Test cond	Parameter	Item P01 voltage ±1%	Elec-tronic load curr. +0, -2%	Connect		Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
				1 to Point	2 to Point	Min	Max		
CC	Reverse voltage protection, aux input	-	fig 50	7	8	5.00	5.10	VDC	49
DD1	Reverse voltage protection,5V out	5V	-	7	8	-1.0	0		50
DD2	Reverse voltage bias, turn-on, 5V out	5V	5A	7	8	5.00	5.10		50
DD3	Reverse voltage protection,+15V out	15V	-	3	4	-1.0	0		50
DD4	Reverse voltage protection,-15V out	15V	-	6	5	-1.0	0		50
DD5	ON/OFF parallel out,5V out	-	50A.	-	-	5.00	5.10		51
DD6	ON/OFF parallel out,+15V out	-	11A.	-	-	14.85	15.15		
DD7	ON/OFF parallel out,-15V out	-	11A.	-	-	-15.15	-14.85		
DD8	ON/OFF parallel out, OS high	-	-	-	-	2.4	5.5	VDC	
DD9	ON/OFF parallel out, OS LED off	-	-	-	-	O.S.	LED off	-	51
DD10	Positive voltage bias, turn on, 5V out	-	-	-	-	5.00	5.10	VDC	52

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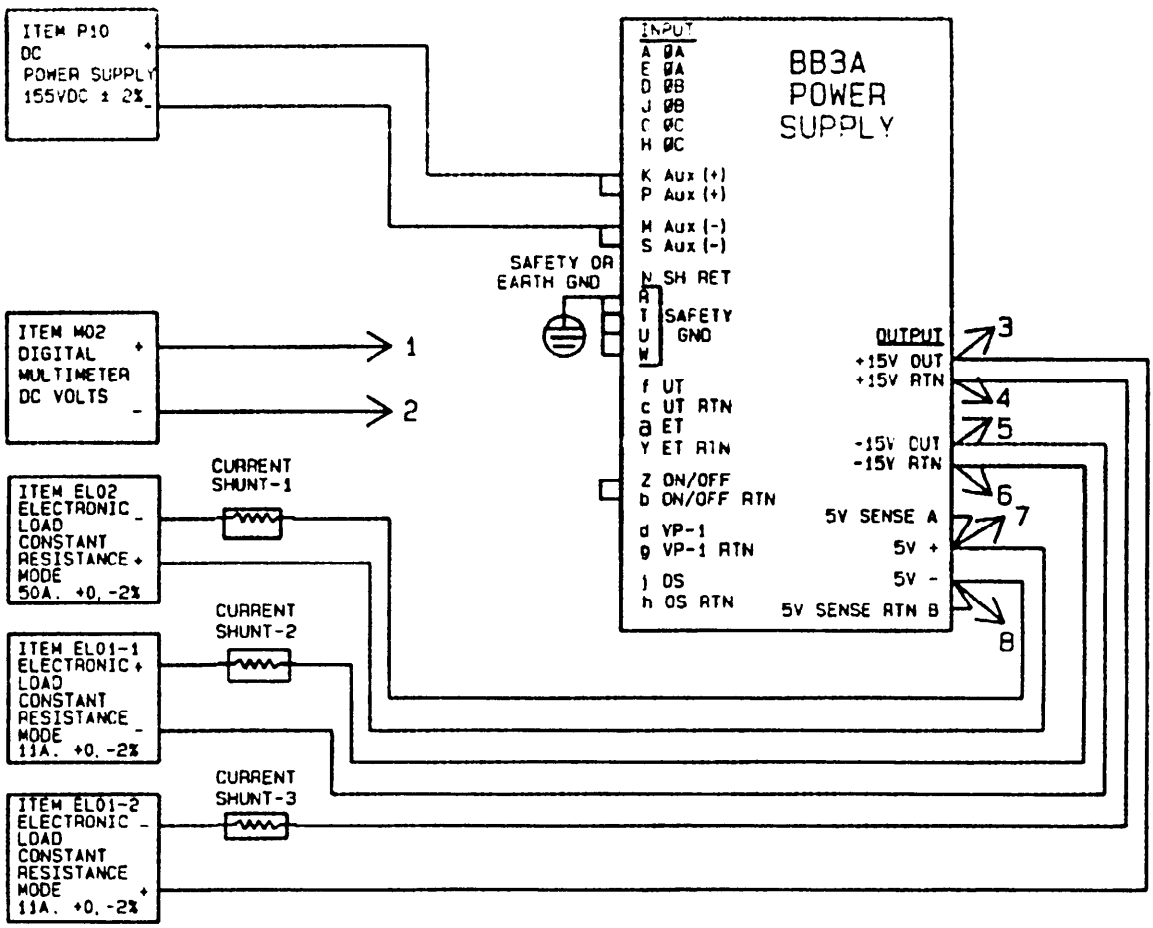


FIGURE 49. Reverse voltage protection on auxiliary input test circuit for table XXV.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 1=75A/50mV; 2,3=50A/50mV.

4. Monitor shunt voltages with an item VO4 voltmeter to ensure that the load currents are within specification.

5. For test condition CC, connect as shown, then reverse the inputs polarity connection for a minimum time of 5 seconds. Then reconnect to the circuit as shown and read the voltage on the item M02 multimeter. This voltage shall be within table XXV limits. Item P10 shall be turned off before reversing the input leads and then turned back on.

FIGURE 49. Reverse voltage protection on auxiliary input test circuit for table XXV - Continued.

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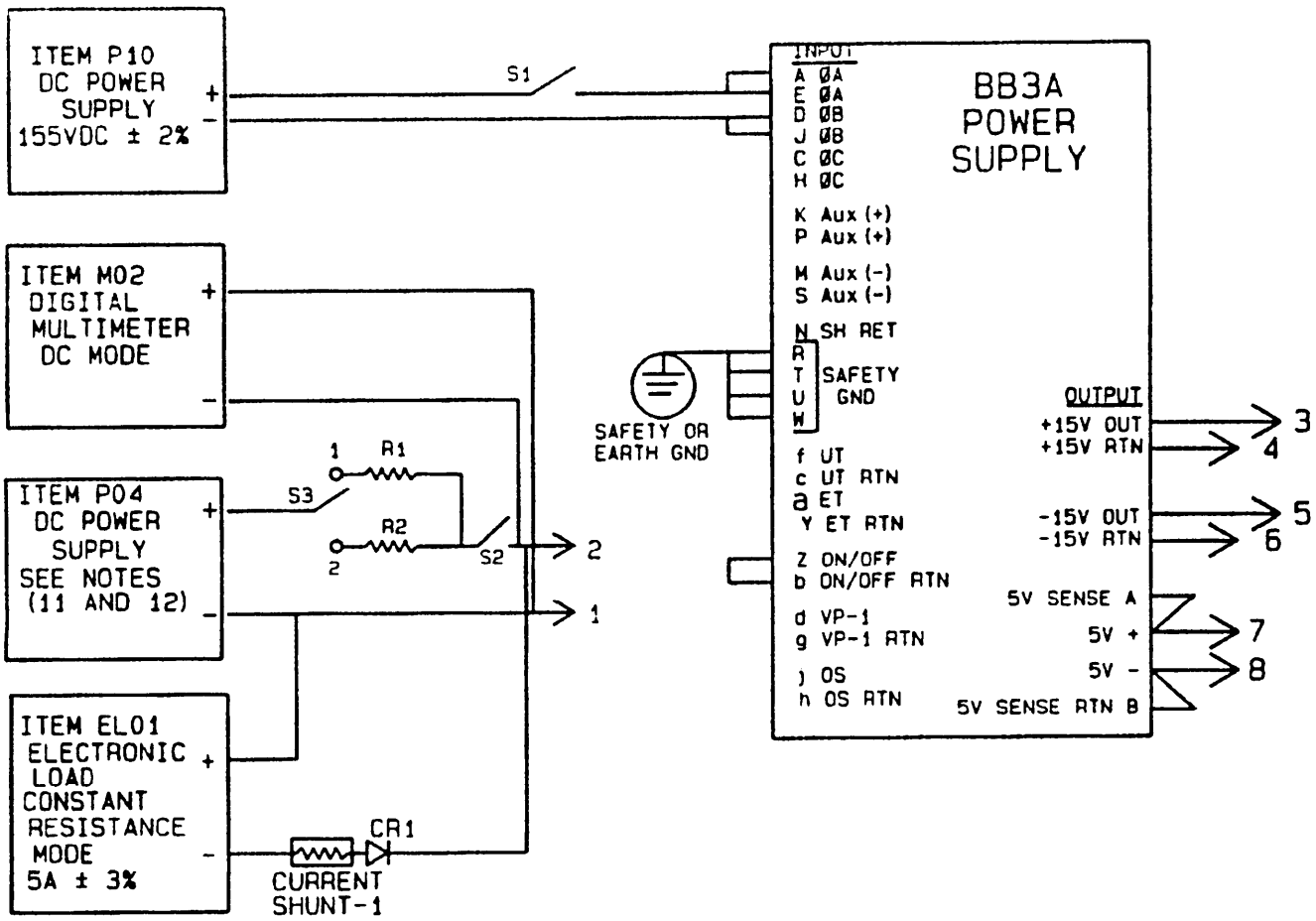


FIGURE 50. Reverse voltage protection and reverse bias test circuit for table XXV - Continued.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50 mV.

4. Monitor the shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. S1 = S2 = spst switch.

6. S3 = spdt switch.

7. R1 = 3 Ohm ± 10 percent, 50 watt resistor.

8. R2 = 14 Ohm ± 10 percent, 25 watt resistor.

9. CR1 = 1N2132 or equivalent (require forward current = 5 amperes and reverse volt = 20 volts).

10. At the option of the testing facility, a 3-phase AC input of 115 VRMS ± 2 percent, 60 ±1 Hz, line-to-line may be used in place of the DC input.

11. Test condition DD1, DD2:

(a) Connect per table XXV and this figure with S1 and S2 both open and S3 in position 1. Adjust item P04 for 5 volts ± 1 percent.

(b) Close S1, adjust item EL01 for 5 amperes and open S1.

(c) Close S2 and read item M02 for test condition DD1 measurement.

(d) Close S1 and read item M02 for test condition DD2 measurement.

12. Test condition DD3, DD4:

(a) Connect per table XXV and this figure with S1 and S2 both open and S3 in position 2. Adjust item P04 for 15 volts ± 1 percent. Items P10 and EL01 are not required.

FIGURE 50. Reverse voltage protection and reverse bias test circuit for table XXV - Continued.

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NOTES:

(b) Close S2 and read item M02 for test condition DD3 measurement.

(c) Repeat (a) and (b) above for test condition DD4.

FIGURE 50. Reverse voltage protection and reverse bias test circuit for table XXV - Continued.

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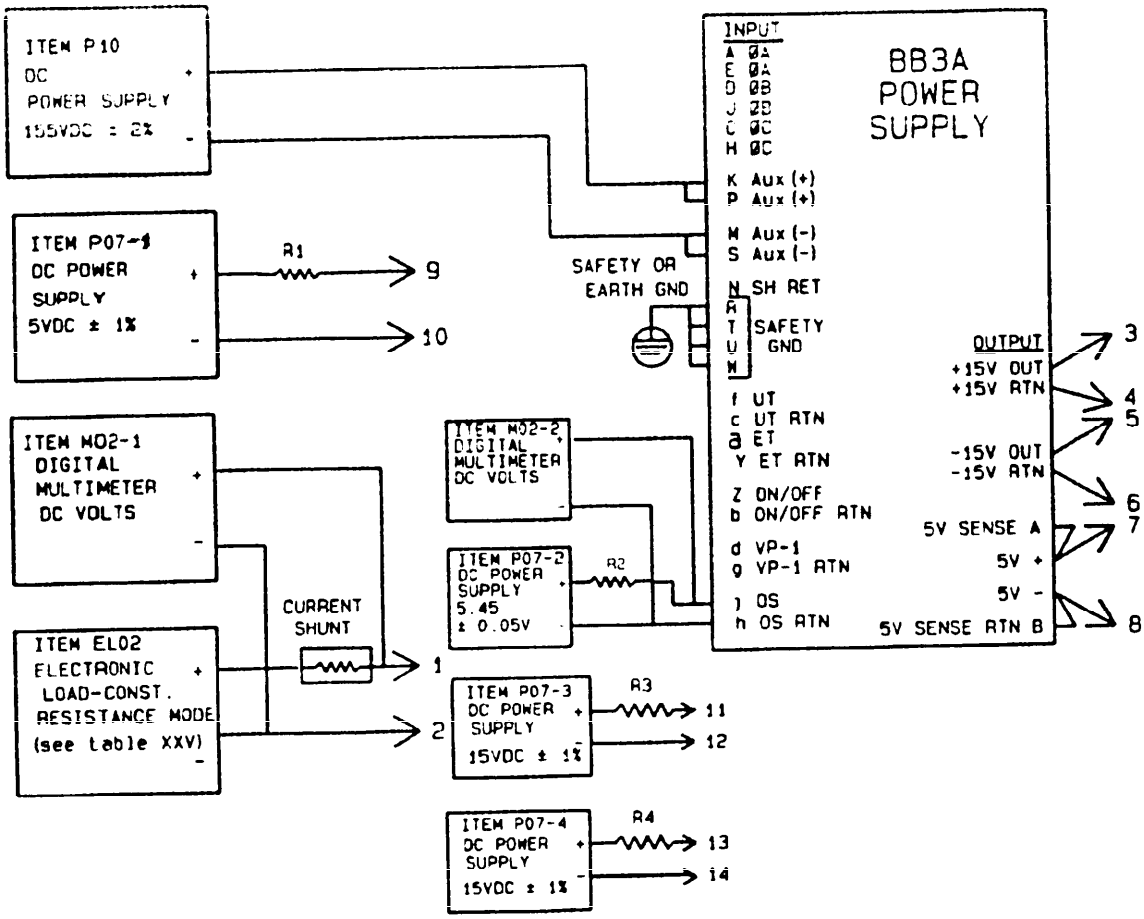


FIGURE 51. ON/OFF parallel outputs test circuit for table XXV.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor the shunt voltage with an item VO4 voltmeter to ensure that the load current is within specification.

5. R1 = 5 Ohm ±5 percent, 1 watt; R3 = R4 = 15 Ohm ±5 percent, 1.5 watt; R2 = 1.69 kOhm, ±1 percent, 1/8 watt resistor.

6. At the option of the testing facility, a 3-phase AC input of 115 VRMS ± 2 percent, 60 ± 1 Hz, line-to-line may be used in place of the DC input.

7. For test conditions DD5-DD7, use the following procedures:

(a) Turn item P10 on. Item EL02 is not connected.

(b) Turn off the power supply under test by opening input contacts Z and b.

(c) Make connection 1 and hold for a minimum of 5 seconds; disconnect and repeat for four more 5 second intervals. Make the connection 2 for five consecutive 5 second intervals. Make the connection 3 for five consecutive 5 second intervals.

			Voltage
connection 1	point 9 to 7	and point 10 to 8	5 volts
connection 2	point 11 to 3	and point 12 to 4	15 volts
connection 3	point 13 to 6	and point 14 to 5	-15 volts

(d) Short input contacts Z and b.

(e) Verify that the BB3A power supply was not damaged by making the following measurements with item EL02 connected:

Test condition	Connect Point 1	Point 2
DD5	7	8
DD6	3	4
DD7	6	5

For load currents, refer to table XXV.

FIGURE 51. ON/OFF parallel cutputs test circuit for table XXV - Continued.

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NOTES: - Continued.

8. The purpose is to simulate the connection of two or more power supplies in parallel where only one power supply is to be on at any one time.

9. For test conditions DD8 and DD9, use the following procedure:

(a) Open the connection between contacts Z and b.

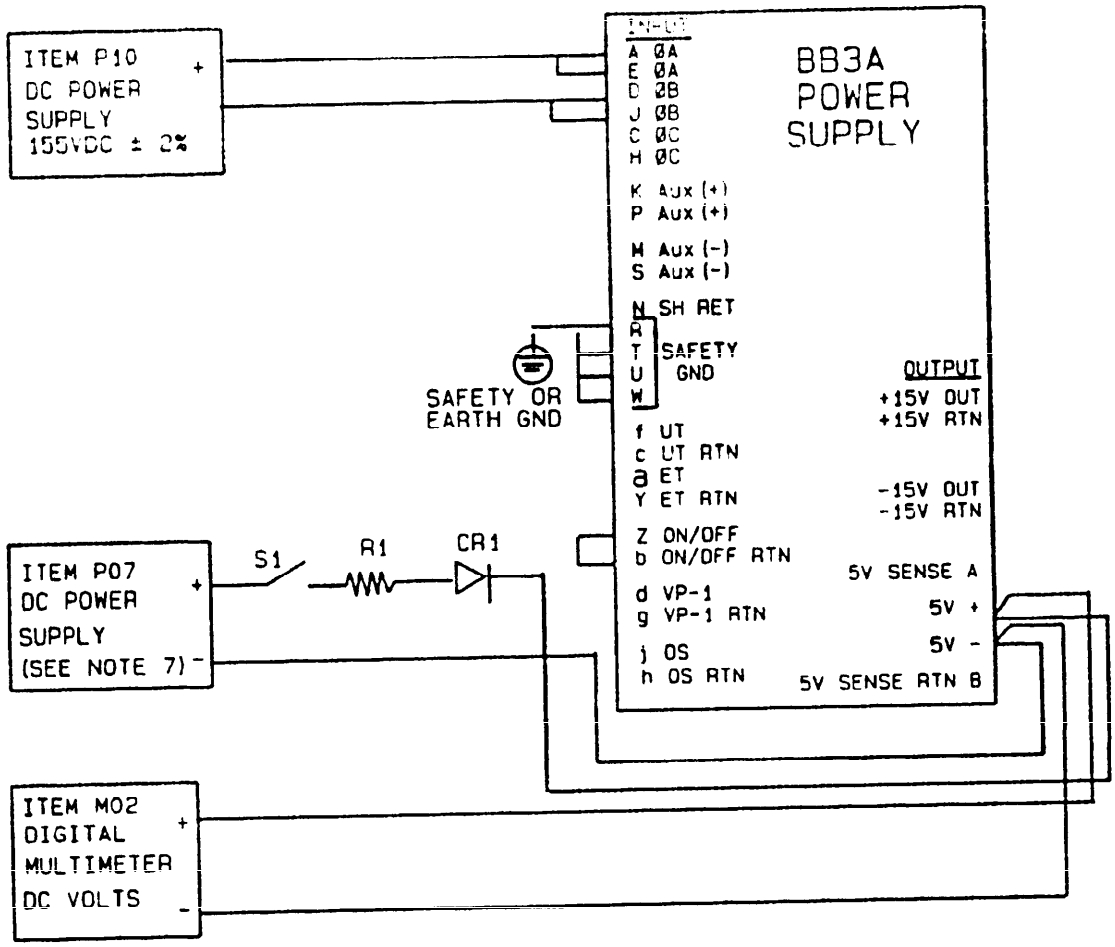
(b) Connect point 9 to 7, point 10 to 8, point 11 to 3, point 12 to 4, point 13 to 6, and point 14 to 5.

(c) Verify that the output status signal as read with item M02-2 meets the DD8 limits.

(d) For test condition DD9, verify that the output status LED is off when input contacts Z and b are open.

FIGURE 51. ON/OFF parallel outputs test circuit for table XXV - Continued.

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NOTES:

1. See table II for output connector contact assignments.
2. For cooling requirements, refer to 3.5.6.
3. R1 = 5 Ohm ± 5 percent, 8 watt resistor.
4. S1 = spst switch.
5. CR1 = 1N5616 or equivalent.
6. At the option of the testing facility, a 3-phase AC input of 115 VRMS ± 2 percent, 60 ± 1 Hz, line-to-line may be used in place of the DC input.

FIGURE 52. Positive voltage bias test circuit for table XXV.

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NOTES: - Continued.

7. For test condition DD10:

(a) Connect per this figure with S1 closed and item P10 off. Adjust item P07 until item M02 reads 5.30 VDC ± 1 percent.

(b) Turn item P10 on. Open input contacts Z and b (remote ON/OFF). Short input contacts Z and b. Open switch S1. The item M02 voltage reading shall meet the table XXV limits.

FIGURE 52. Positive voltage bias test circuit for table XXV - Continued.

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TABLE XXVI. Common mode output current test.

Test cond	Parameter	Connect point 1 to point:	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
			Min	Max		
EE1	Common mode current broadband, 5V, 50A	3	0	200	mAp-p	54
EE2	Common mode current broadband, 5V, 25A	3	0	200	mAp-p	
EE3	Common mode current broadband, ±15V, 11A	3	0	200	mAp-p	
EE4	Common mode current broadband, ±15V, 5A	3	0	200	mAp-p	
EE5	Common mode current spectral line, 5V, 40 Hz - 40kHz 2/ , 4/	2	-INF	-80 1/	$\frac{dBV}{\sqrt{Hz}}$	
EE6	Common mode current spectral line, 5V, 40kHz - 350kHz 2/ , 4/	2	-INF	-56.5 1/		
EE7	Common mode current spectral line, 5V 350kHz - 5MHz 2/ , 4/	2	See figure 54			
EE8	Common mode current spectral line, 5V, 5MHz - 10MHz 2/ , 4/	2	-INF	-80 1/		
EE9	Common mode current spectral line, ±15V, 40Hz - 40kHz 3/ , 4/	2	-INF	-80 1/		
EE10	Common mode current spectral line, ±15V, 40kHz - 350kHz 3/ , 4/	2	-INF	-56.5 1/		
EE11	Common mode current spectral line, ±15V, 350kHz - 5MHz 3/ , 4/	2	See figure 54			
EE12	Common mode current spectral line, ±15V, 5MHz - 10MHz 3/ , 4/	2	-INF	-80 1/	$\frac{dBV}{\sqrt{Hz}}$	54

- See footnotes on next page.

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NOTES:

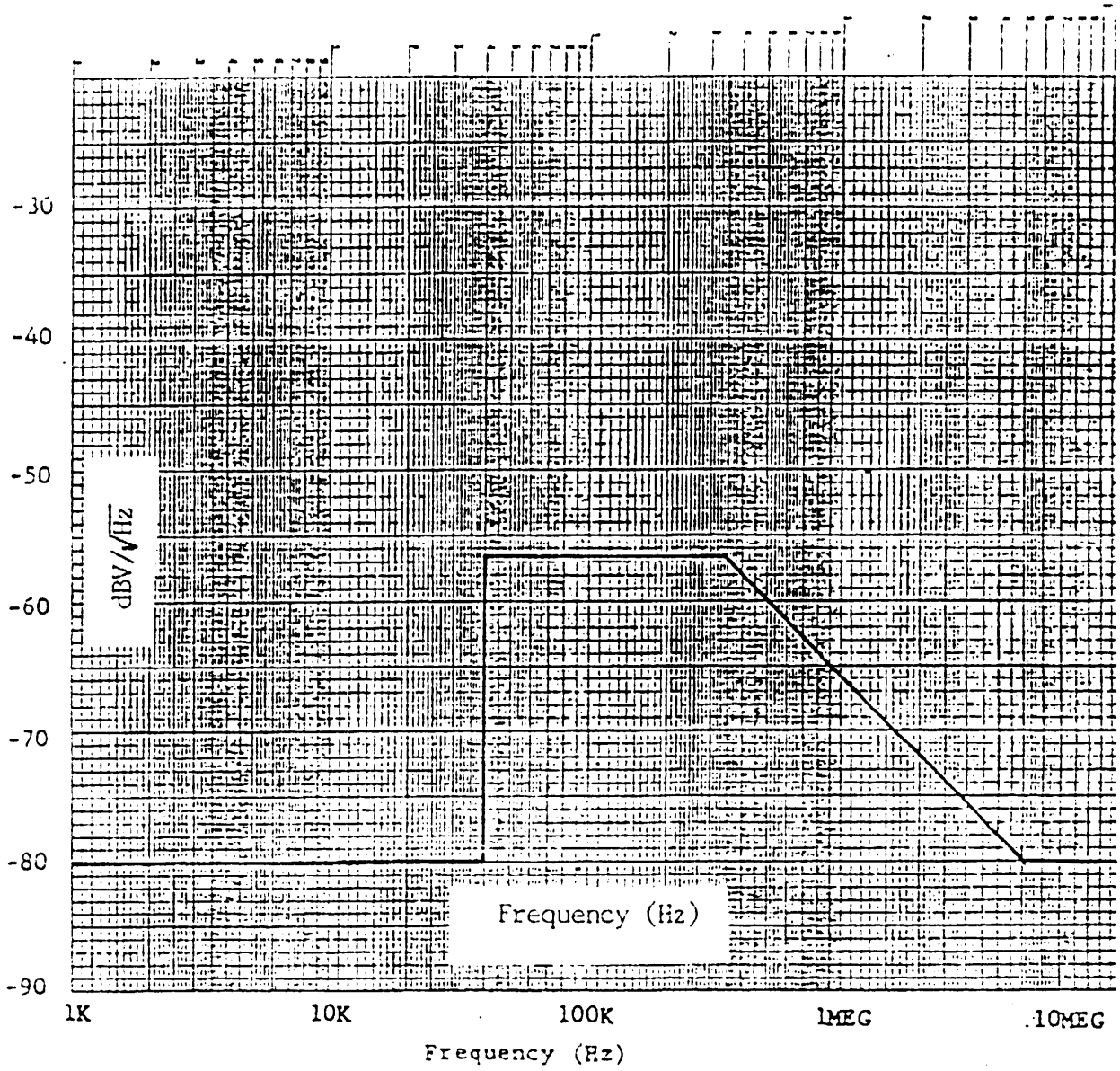
1/ Common mode current (dBA) =
(item B05 reading in dBV) - 20 LOG(Z)
= (item B05 reading in dBV) + 20dB.
where Z = current transformer transfer function of mV/mA (shown calculated above for Z=0.1), dBA = decibel (ref 1 ampere), and dBV = decibel (ref 1 volt). The table XXVI limits for EE5-EE12 will change in accordance with the above relationship if Z is not equal to 0.1. Refer to table I for dBA/\|Hz limits.

2/ Test conditions EE5-EE8 shall be run using the setup from EE1 or EE2 which gives the largest amplitude signal.

3/ Test conditions EE9-EE12 shall be run using the setup from EE3 or EE4 which gives the largest amplitude signal.

4/ Do not apply bandwidth correction to the dBV per \|$\overline{\text{Hz}}$ limits.

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NOTE:

1. Add 20dB for conversion to dBA (see note 1 to table XXVI).

FIGURE 53. Common mode output current spectral line limits for table XXVI.

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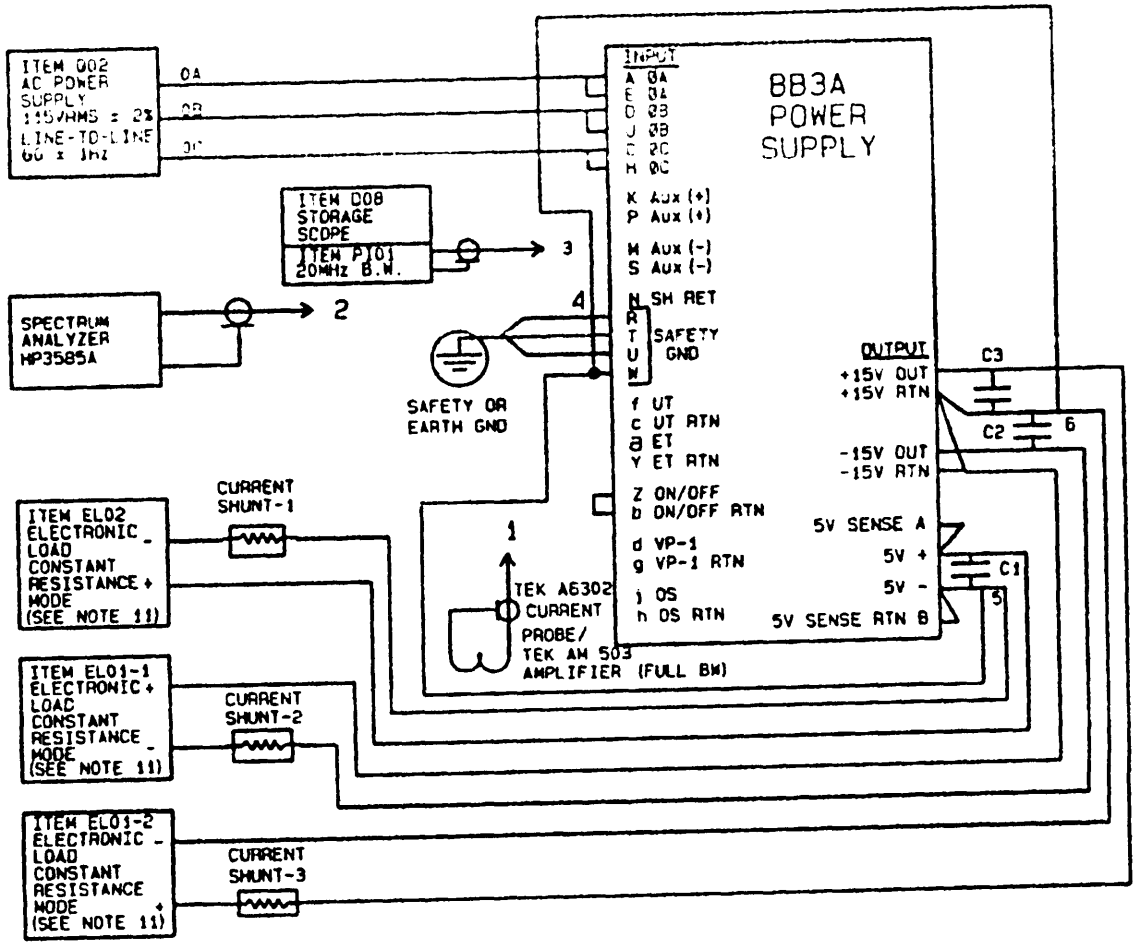


FIGURE 54. Common mode output current test circuit for table XXVI.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 1=75A/50mV, 2,3=50A/50mV.

4. Monitor shunt voltages with an item VO4 voltmeter to ensure that load currents are within specification.

5. C1 = C2 = C3 = 10 uF ± 5 percent, 50 VDC, MIL-C-83421/01-359, or equivalent capacitor.

6. The connection from point 4 (safety gnd) to the output returns (at points 5 and 6) shall be made using a 4 ± 0.125 inch piece of copper, stranded, number 16 gauge wire (length measurements shall be made from the mating connector solder cups). The connections from the power supply output contacts to the C1-C3 capacitors and points 5 and 6 shall be kept as short as possible. The wires from input connector contacts R, T, and U to the common tie point shall each be a 4.0 ± 0.5 inch piece of number 16 gauge wire.

7. For test conditions EE1 and EE2, place the current probe around the wire going from point 4 to point 5. Measure the total peak-to-peak current observed on item D08 oscilloscope (including all spikes, hash, periodic and aperiodic components). Use the single sweep storage mode.

8. For test conditions EE3 and EE4, place the current probe around the wire going from point 4 to point 6. Measure the total peak-to-peak current observed on item D08 oscilloscope (including all spikes, hash, periodic and aperiodic components). Use the single sweep storage mode.

9. For test conditions EE5 and EE9, sweep the spectrum analyzer frequency between 40 Hz and 40 kHz. Use a 10 Hz bandwidth and 5 Hz step size to make final measurements verifying the power supply meets the specified limits. See notes 2 and 3 of table XXVI to determine the setup to be used.

10. For test conditions EE6-EE8, EE10 and EE12, sweep the spectrum analyzer frequency between 40 kHz and 10 MHz. Use a 30 Hz bandwidth and a 15 Hz step size to make final measurements verifying the power supply meets the specified limits.

FIGURE 54. Common mode output current test circuit for table XXVI - Continued.

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NOTES: - Continued.

11. The electronic loads shall be set as follows:

Test Condition	EL02 +0,-2%	EL01-1 +0,-2%	EL01-2 +0,-2%
EE1	50 amperes	11 amperes	11 amperes
EE2	25 amperes	11 amperes	11 amperes
EE3	50 amperes	11 amperes	11 amperes
EE4	50 amperes	5 amperes	5 amperes

The electronic loads connection to the power supply shall be isolated from chassis ground so the common mode current will flow to point 4.

12. When performing the tests of table XXVI, ensure that the current being measured is generated by the power supply under test and is not spurious noise from external sources.

FIGURE 54. Common mode output current test circuit for table XXVI - Continued.

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TABLE XXVII. Voltage programming contact signal rejection test.

Test cond	Parameter	Spectrum analyzer frequency ±1%	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
			Min	Max		
FF1	Rejection at 100 Hz	100 Hz	15	-	dBV	55
FF2	Rejection at 5 kHz	5 kHz	15	-	dBV	55
FF3	Rejection at 50 kHz	50 kHz	15	-	dBV	55

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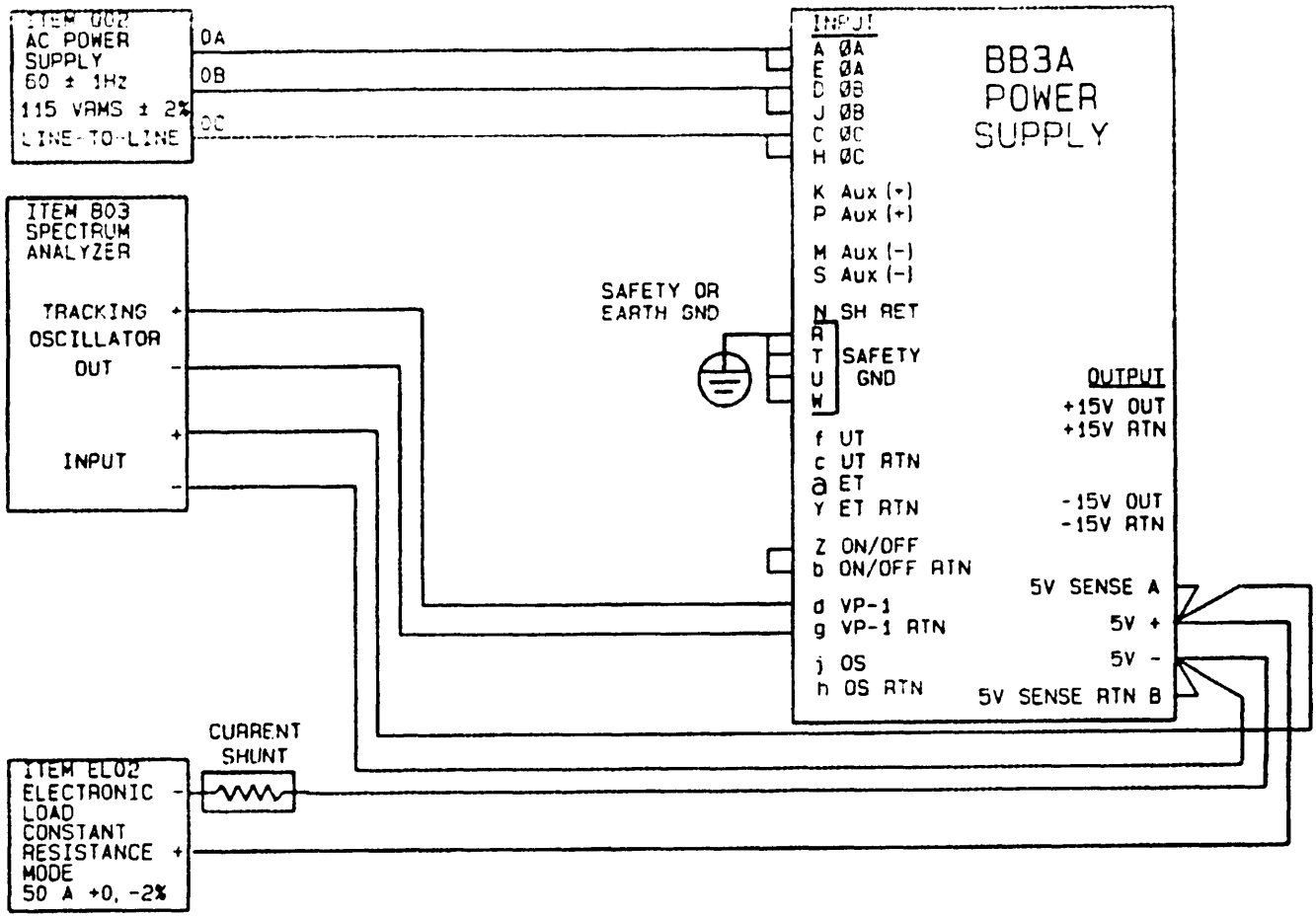


FIGURE 55. Voltage programming contact signal rejection test figure for table XXVII.

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NOTES:

1. See table II for output connector contact assignments.
2. For cooling requirements, refer to 3.5.6.
3. Current shunt: 75A/50mV.
4. Monitor shunt voltage with an item VO4 voltmeter to ensure that the load current is within specification.
5. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two of the AC inputs may be used in place of the AC input specified.
6. For each test condition set the tracking oscillator out to input 0.1 VRMS ±5 percent (-20 ±0.5 dBV), frequency per table XXVII, at input contact d. Measure the AC voltage on the power supply 5V output with the spectrum analyzer. The spectrum analyzer measurement shall be 15 dB or greater below the -20 dBV input signal (-20 dBV-measurement level $\geq$ 15 dB). (If the tracking oscillator cannot input as high as 0.1 VRMS to the unit under test, set it to the highest value obtainable. The spectrum analyzer measurement shall still be 15 dB or greater below the input signal.)

FIGURE 55. Voltage programming contact signal rejection test figure for table XXVII - Continued.

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TABLE XXVIII. Input voltage and current characteristics test.

Test cond	Parameter	Switch S1	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
			Min	Max		
GG1	Input voltage turn on threshold	Closed	100	135	VDC	56
GG2	Input voltage turn off threshold		-	100		56
GG3	Input voltage turn on/turn off hysteresis 1/		18	-		-
GG4	Input voltage turn on threshold, auxiliary input		100	135		56
GG5	Input voltage turn off threshold, auxiliary input		-	100		56
GG6	Input voltage turn on/turn off hysteresis, auxiliary input 1/		18	-	VDC	-
GG7	Maximum steady state input current		-	11.9	ADC	56
GG8	Slowly rising input peak current			11.9	Apk	
GG9	Slowly rising input rate of current rise	Closed		9.4	A/ms	
GG10	Input peak current, remote turn-on	See fig. 57, note 11		11.9	Apk	
GG11	Input rate of current rise, remote turn-on	See fig. 57, note 11	-	9.4	A/ms	56

1/ Hysteresis = turn on threshold minus turn off threshold.

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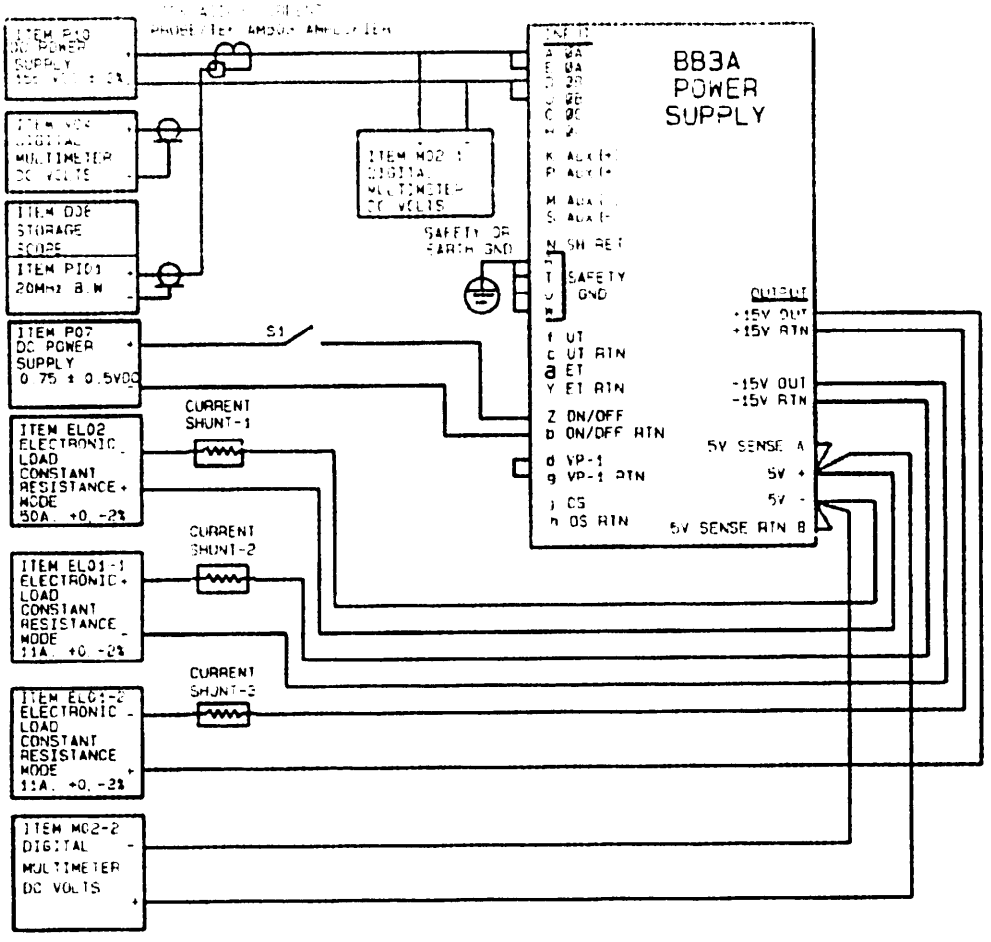


FIGURE 56. Input voltage and current characteristics test circuit for table XXVIII.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 1=75A/50mV; 2,3=50A/50mV.

4. Monitor shunt voltages with an item V04 voltmeter to ensure that load currents are within specification.

5. S1=spst switch.

6. For test conditions GG1, GG2, and GG7 through GG11, connect item P10 power supply to the AC inputs as shown. For test conditions GG4 and GG5, connect item P10 and item M02 to the auxiliary inputs: + to input contacts K and P, - to input contacts M and S.

7. Test conditions GG1 and GG4 determine the AC/DC input and auxiliary input voltage threshold for power supply turn-on. Perform the test by slowly increasing the item P10 voltage (1 volt/sec. or less) from 0 volts until the output voltage, as measured with the item M02-2, just equals or exceeds 5.00 VDC. The item M02-1 voltage at this point shall meet the test condition GG1 or GG4 limits.

8. Test conditions GG2 and GG5 determine the AC/DC input and auxiliary input voltage threshold for power supply turn-off. Perform the test by slowly decreasing the item P10 voltage (1 volt/sec. or less) from 155 VDC ± 2 percent until the output voltage, as measured with the item M02-2, just equals or drops below 5.00 VDC. The item M02-1 voltage at this point shall meet the test condition GG2 or GG5 limits.

9. For test condition GG7, slowly (1 V/sec. or less) sweep the item P10 voltage from 0 volts to 202 VDC ± 2 percent and back to 0 volts. The maximum steady state input current observed as measured with the current probe and item V04 shall meet the GG7 limits.

10. For test conditions GG8 and GG9, increase the item P10 voltage from 0 volts (below the turn on threshold and output voltage off) to 140 VDC ± 2 percent (above the turn-on threshold) at a rate of 1 volt/millisecond or less. Use the

FIGURE 56. Input voltage and current characteristics circuit for table XXVIII - Continued.

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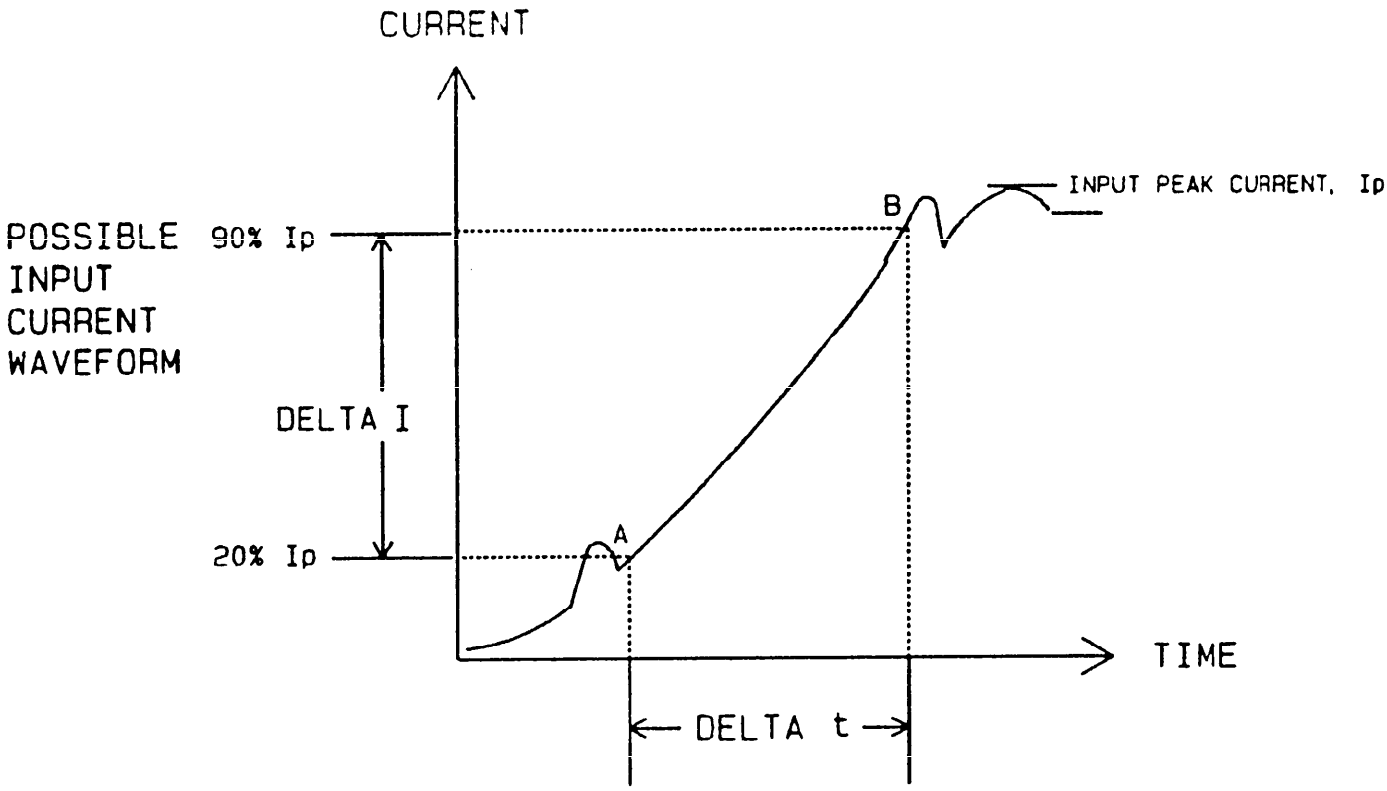
NOTES: - Continued.

current probe and item D08 to store the input current waveform as the power supply turns on. The input current observed shall meet the GG8 and GG9 limits (see figure 57). The test shall be performed 5 times and the highest values recorded.

11. For test conditions GG10 and GG11, close switch S1 and adjust item P10 to 155 VDC ± 2 percent. Open switch S1. Use the current probe and item D08 to store the input current waveform as S1 is closed and the power supply turns on. The input current observed shall meet the GG10 and GG11 limits (see figure 58). The test shall be performed 5 times and the highest values recorded.

FIGURE 56. Input voltage and current characteristics circuit for table XXVIII - Continued.

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NOTE:

1. The rate of current rise to be recorded for test condition GG9 or GG11 is the slope between points A and B on the rising input current vs time waveform. If the curve crosses the value of current represented by 20 percent Ip and/or 90 percent Ip more than once, the time between the last time it crosses 20 percent Ip and the first time it crosses 90 percent Ip shall be taken as the recorded DELTA t. The slope is defined as

rate of current rise = DELTA I in Amperes/DELTA t in ms.

(The figure is for illustrative purposes only and the input current waveform may look different).

FIGURE 57. Input current waveform for table XXVIII.

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TABLE XXIX. Emergency condition input voltage protection test.

Test cond	Parameter	Connect		Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		1 to	2 to	Min	Max		
HH1	Emergency condition input voltage protection (during overvoltage), 5.25V	3	4	4.99	5.51	VDC	58
HH2	Emergency condition input voltage protection, 5.25V	3	4	5.20	5.30	VDC	58
HH3	Emergency condition input voltage protection (during overvoltage), +15V	7	8	14.25	15.75	VDC	58
HH4	Emergency condition input voltage protection, +15V	7	8	14.85	15.15	VDC	58
HH5	Emergency condition input voltage protection (during overvoltage), -15V	6	5	-15.75	-14.25	VDC	58
HH6	Emergency condition input voltage protection, -15V	6	5	-15.15	-14.85	VDC	58

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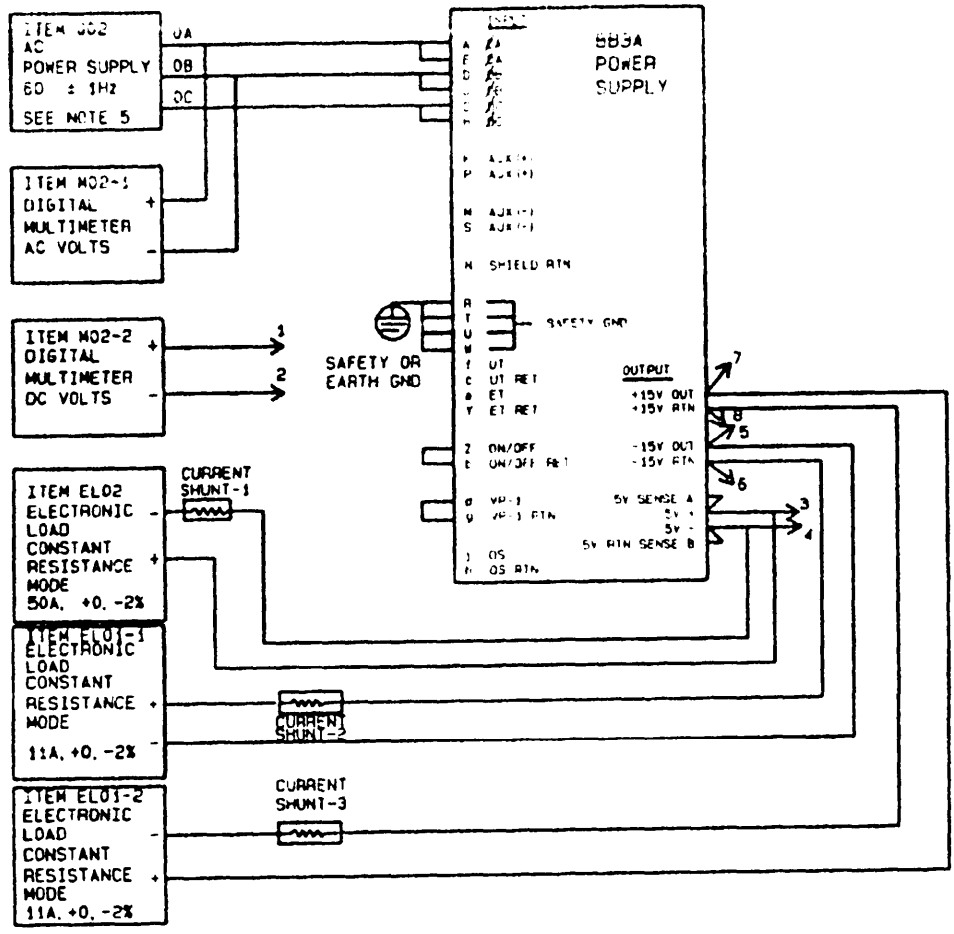


FIGURE 58. Emergency condition input voltage protection test circuit for table XXIX.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 1 = 75A/50mV, 2,3 = 50A/50mV.

4. Monitor shunt voltages with an item V04 voltmeter to ensure that load currents are within specification.

5. Connect as shown and apply power to the PSUT. Increase item Q02 voltage to 155 VRMS ± 2 percent line-to-line. Leave the 155 VRMS applied for a minimum of two minutes and monitor item M02-2 to verify that the voltage remains within the test condition HH1, HH3, and HH5 limits (record the highest voltage observed). Reduce item Q02 voltage to 115 VRMS line-to-line. Item M02-2 voltage shall be within the test condition HH2, HH4, and HH6 limits.

FIGURE 58. Emergency condition input voltage protection test circuit for table XXIX - Continued.

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TABLE XXX. Output status noise test.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
JJ1	Output status noise, positive	0	500	mVpk	59
JJ2	Output status noise, negative	-500	0	mVpk	59

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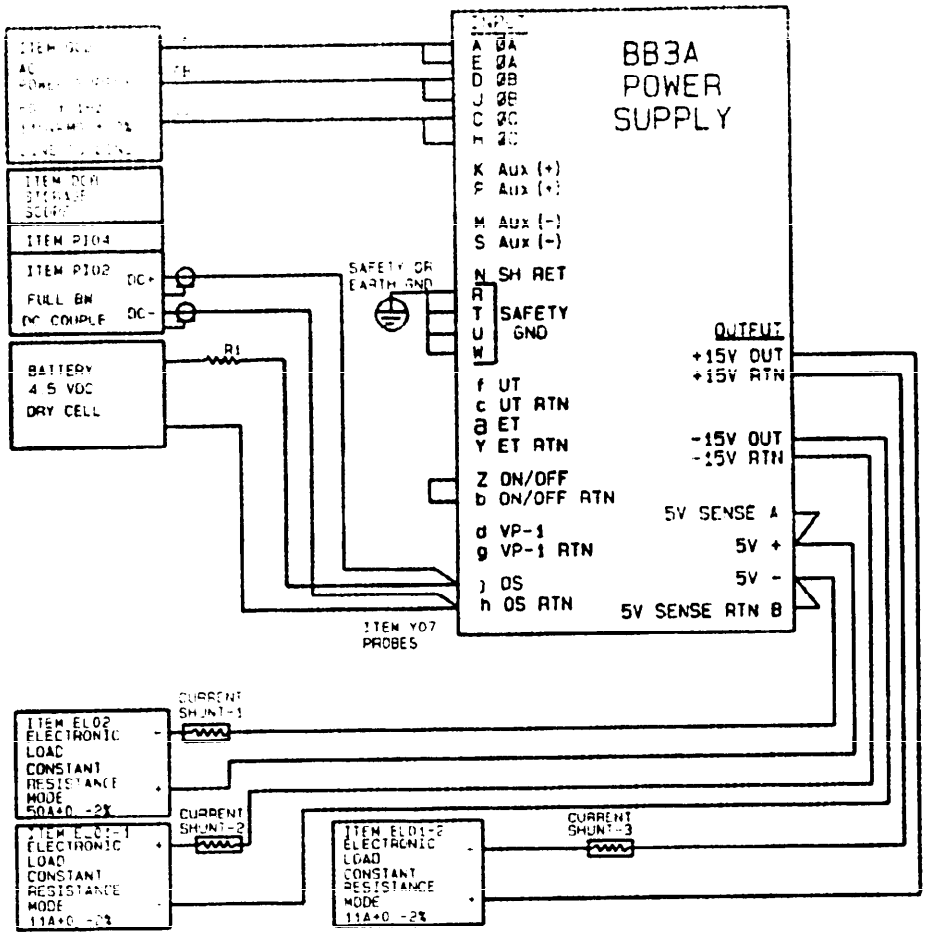


FIGURE 59. Output status noise test for table XXX.

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NOTES:

1. See table II for output connector assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 1=75A/50mV; 2,3=50A/50mV.

4. Monitor the shunt voltages with an item V04 voltmeter to ensure that the load currents are as specified.

5. At the option of the test facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two of the AC inputs may be used instead of the AC specified.

6. R1 = R2 = 5.9 kOhm, ±1 percent, 1/8 watt resistor.

7. The leads between the battery terminals and the output status contacts of the PSUT shall be a tightly twisted pair, with a maximum length of 10 inches.

8. The millivolt peak (mVpk) measurements for test conditions KK1 and KK2 shall be referenced to input contact h. The measurement shall include all spikes, hash, periodic and aperiodic components.

FIGURE 59. Output status noise test for table XXX - Continued.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - SH
Air Force - 99

Review activity:
DLA - GS

Preparing activity:
Navy - NW

(Project 6130-0344)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-24764/2	31 December 1992

3. DOCUMENT TITLE
Power Supplies, Shipboard, Electronic, 5 Volts and ± 15 Volts, Key Code BB3A

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contact (TPOC): Mr. Mike Roach (Code 6023) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial TPOC: 812-854-5106	(2) AUTOVON 88-482-5106
c. ADDRESS (Include Zip Code) Commander Code 6023, Bldg. 2917 (Mike Roach) NAVSURFWARCENDIV, 300 Highway 361 Crane, Indiana 47522-5001	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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